

Physics Publications with Major Personal Contributions

Oliver Gutsche

May 10, 2026

1. Search for a new heavy scalar resonance decaying into the Higgs boson and a new scalar particle in the $b\bar{b}b\bar{b}$ final state using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.02848](#) | 2026-05-04

2. Observation of the rare decay $\eta \rightarrow \mu^+ \mu^- e^+ e^-$

Hayrapetyan, Aram et al.

[arXiv:2605.00615](#) | 2026-05-01

3. Search for light charged Higgs bosons decaying to charm and strange quarks in $t\bar{t}$ events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.27795](#) | 2026-05-01

4. Improved results on Higgs boson pair production in the $4b$ final state

Hayrapetyan, Aram et al.

[arXiv:2604.27044](#) | 2026-04-29

5. Search for electroweakinos in compressed-spectrum scenarios with low-momentum isolated tracks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.25604](#) | 2026-04-28

6. Simultaneous measurements of N-subjettiness observables in jets from gluons and light-flavour quarks, and in decays of boosted W bosons and top quarks

Hayrapetyan, Aram et al.

[arXiv:2604.25538](#) | 2026-04-28

7. Measurement of the $Z \rightarrow \mu^+ \mu^-$ angular coefficients in pp collisions at $\sqrt{s} = 13$ TeV as functions of transverse momentum and rapidity

Gevorgyan, Arzunik et al.

[arXiv:2604.25678](#) | 2026-04-28

8. Search for associated production of a Higgs boson and two vector bosons via vector boson scattering at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.24531](#) | 2026-04-27

9. Search for dark matter produced in association with a dark Higgs boson decaying into a bottom quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.20999](#) | 2026-04-22

10. Search for the single production of vector-like quarks decaying into a W boson and a b quark using single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.17564](#) | 2026-04-19

11. Highly boosted dielectron identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.13320](#) | 2026-04-14

12. Search for heavy resonances decaying into four-lepton final states via light bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.14236](#) | 2026-04-14

13. A search for microscopic black holes, string balls, and sphalerons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10732](#) | 2026-04-12

14. Search for a new heavy resonance decaying to a top quark and a neutral scalar boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10729](#) | 2026-04-12

15. Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons

Hayrapetyan, Aram et al.

[arXiv:2604.09809](#) | 2026-04-10

16. Observation of nuclear suppression in coherent $\Upsilon(1S)$ photoproduction off heavy nuclei at the LHC

Belyaev, Andrey et al.

[arXiv:2604.05814](#) | 2026-04-07

17. Search for soft unclustered energy patterns produced in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.05996](#) | 2026-04-07

18. Evidence of $ZZ\gamma$ production and observation of $4\ell\gamma$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.02594](#) | 2026-04-03

19. Search for Higgs boson pair production in the $b\bar{b}WW$ decay channel with two leptons in the final state using proton-proton collision data at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2604.02127](#) | 2026-04-02

20. Measurement of dijet angular distributions and search for beyond the standard model physics in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2603.25458](#) | 2026-03-26

21. Search for new particles decaying into top quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2603.23454](#) | 2026-03-24

22. Search for low-mass vector and scalar resonances decaying into a quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.21965](#) | 2026-03-23

23. Search for Higgs boson production at high transverse momentum in the WW decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.22233](#) | 2026-03-23

24. Measurement of the jet mass in hadronic decays of boosted W bosons at 13 TeV and extraction of the W boson mass

Hayrapetyan, Aram et al.

[arXiv:2603.19963](#) | 2026-03-20

25. Jet peak shapes based on two-particle angular correlations in lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.

[arXiv:2603.14385](#) | 2026-03-15

26. Measurement of the t-channel single top quark cross section in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.13592](#) | 2026-03-13

27. Search for Z' bosons decaying into charginos in final states with two oppositely charged leptons and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.11035](#) | 2026-03-11

28. Measurement of angular correlations inside jets induced by gluon polarization in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.03689](#) | 2026-03-04

29. Search for a narrow resonance with a mass between 10 and 70 GeV decaying to a pair of photons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.03250](#) | 2026-03-03

30. Combination of ATLAS and CMS searches for Higgs boson pair production at $\sqrt{s} = 13$ TeV

Aad, Georges et al.

[arXiv:2602.23991](#) | 2026-02-27

31. System-size dependence of charged-particle suppression in ultrarelativistic nucleus-nucleus collisions

Belyaev, Andrey et al.

[arXiv:2602.21325](#) | 2026-02-24

32. Search for pair production of heavy resonances in final states with a photon and large-radius jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.20477](#) | 2026-02-24

33. Observation of the jet diffusion wake using dijets in heavy ion collisions

Hayrapetyan, Aram et al.

[arXiv:2602.19431](#) | 2026-02-23

34. Search for a new resonance decaying to a Higgs boson and a scalar boson in events with two b jets and two Z bosons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18223](#) | 2026-02-20

35. Combined measurements and interpretations of Higgs boson production and decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18611](#) | 2026-02-20

36. First measurement of jet axis decorrelation with photon-tagged jets in pp and PbPb collisions at 5.02 TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18279](#) | 2026-02-20

37. Measurement of event shape variables using charged particles inside jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.17509](#) | 2026-02-19

38. Influence of strangeness on the anisotropic flow of prompt $D^{\pm}_s$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.

[arXiv:2602.14221](#) | 2026-02-15

39. High-level hadronic tau lepton triggers of the CMS experiment in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[JINST 21 \(2026\) P04002](#) | [arXiv:2602.11359](#) | [DOI:10.1088/1748-0221/21/04/P04002](#) | 2026-02-11

40. Measurement of B meson production fraction ratios in proton-proton collisions at $\sqrt{s} = 13$ TeV using open-charm and charmonium decays

Hayrapetyan, Aram et al.

[arXiv:2602.10270](#) | 2026-02-10

41. Probing early parton emissions in heavy ion collisions using the Lund jet plane

Belyaev, Andrey et al.

[arXiv:2602.09271](#) | 2026-02-09

42. Observation of $\Upsilon(1S) + Z$ associated production and measurement of the effective double-parton scattering cross section in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.02770](#) | 2026-02-02

43. Observation of a family of all-charm tetraquarks

Hayrapetyan, Aram et al.

[arXiv:2602.02252](#) | 2026-02-02

44. Search for heavy scalar resonances decaying to Lorentz-boosted Higgs and Higgs-like bosons in the $b\bar{b}4q$ final state at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.00273](#) | 2026-01-30

45. Measurement and effective field theory interpretation of the photon-fusion production cross section of a pair of W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.21574](#) | 2026-01-29

46. Measurement of the $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ differential cross sections in pp collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20023](#) | 2026-01-27

47. Search for heavy resonances decaying into two Higgs bosons in the $b\bar{b}\tau^+\tau^-$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20011](#) | 2026-01-27

48. Search for heavy long-lived charged particles with level-1 trigger scouting data from proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20063](#) | 2026-01-27

49. Strategy and performance of the CMS long-lived particle trigger program in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.17544](#) | 2026-01-24

50. Search for the pair production of long-lived supersymmetric partners of the tau lepton in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.17576](#) | 2026-01-24

51. Full event interpretation with machine-learning-based particle-flow reconstruction in the CMS detector

Hayrapetyan, Aram et al.

[arXiv:2601.17554](#) | 2026-01-24

52. Measurement of the $Z\gamma$ production cross section and search for anomalous neutral triple gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.14102](#) | 2026-01-20

53. Combination of searches for heavy vector boson resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.12583](#) | 2026-01-18

54. Search for dark matter produced in association with a Higgs boson decaying to bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.11330](#) | 2026-01-16

55. Search for a boosted Higgs boson decaying to bottom quark pairs in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.05362](#) | 2026-01-08

56. Measurement of the Higgs boson total decay width using the $H \rightarrow WW \rightarrow e\nu\mu\nu$ decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.05168](#) | 2026-01-08

57. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A}$ with $\mathcal{A}\mathcal{A} \rightarrow \gamma\gamma$ in events with a semi-merged topology in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.00183](#) | 2026-01-01

58. Search for charged Higgs bosons decaying into top and bottom quarks in lepton+jets final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2512.24471](#) | 2025-12-30

59. Machine-learning techniques for model-independent searches in dijet final states

Hayrapetyan, Aram et al.

[arXiv:2512.20395](#) | [DOI:10.5281/zenodo.16656501](#) | 2025-12-23

60. Characterization of the quantum state of top quark pairs produced in proton-proton collisions at $\sqrt{s}=13$ TeV using the beam and helicity bases

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) L051101](#) | [arXiv:2512.17557](#) | [DOI:10.1103/hkgw-dztk](#) | 2025-12-19

61. Measurement of $Z\gamma$ production in proton-proton collisions at $\sqrt{s} = 13.6$ TeV and constraints on neutral triple gauge couplings

Hayrapetyan, Aram et al.

[arXiv:2512.08582](#) | 2025-12-09

62. Measurements of electroweak production of a photon in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2512.00502](#) | 2025-11-29

63. Search for light pseudoscalar bosons, pair-produced in Higgs boson decays in the four-electron final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2511.19563](#) | 2025-11-24

64. Search for new physics in the final state with a single photon and large missing transverse momentum in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 072020](#) | [arXiv:2511.17310](#) | [DOI:10.1103/f9r8-styj](#) | 2025-11-21

65. Search for Higgsinos in final states with low-momentum lepton-track pairs at 13 TeV

Hayrapetyan, Aram et al.

[arXiv:2511.16394](#) | 2025-11-20

66. Search for heavy $H\gamma$ and $Z\gamma$ resonances with a bottom quark-antiquark pair in the final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2511.14583](#) | 2025-11-18

67. Search for low-mass hidden-valley dark showers with non-prompt muon pairs in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 189](#) | [arXiv:2511.11888](#) | [DOI:10.1007/JHEP03\(2026\)189](#) | 2025-11-14

68. Search for a new neutral gauge boson produced in association with one or two b jets and decaying into a pair of muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

[arXiv:2511.11853](#) | 2025-11-14

69. Search for long-lived particles using displaced vertices with low-momentum tracks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2511.08212](#) | 2025-11-11

70. Identification of tau leptons using a convolutional neural network with domain adaptation

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P12032](#) | [arXiv:2511.05468](#) | [DOI:10.1088/1748-0221/20/12/P12032](#) | 2025-11-07

71. Jet fragmentation function and groomed substructure of bottom quark jets in proton-proton collisions at 5.02 TeV

Chekhovsky, Vladimir et al.

[JHEP 04 \(2026\) 147](#) | [arXiv:2511.10666](#) | [DOI:10.1007/JHEP04\(2026\)147](#) | 2025-11-06

72. Inclusive and differential measurements of the $t\bar{t}\gamma$ cross section and the $t\bar{t}\gamma / t\bar{t}$ cross section ratio in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2026\) 086](#) | [arXiv:2511.01995](#) | [DOI:10.1007/JHEP04\(2026\)086](#) | 2025-11-03

73. Characterizing the initial state and dynamical evolution in XeXe and PbPb collisions using multiparticle cumulants

Hayrapetyan, Aram et al.

[Phys.Lett.B 876 \(2026\) 140359](#) | [arXiv:2510.26766](#) | [DOI:10.1016/j.physletb.2026.140359](#) | 2025-10-30

74. Search for single production of a vector-like T quark decaying to a top quark and a neutral scalar boson in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2510.25874](#) | 2025-10-29

75. Measurement of the ratio of the $B_c^+ \rightarrow J/\psi \tau^+ \nu_\tau$ and $B_c^+ \rightarrow J/\psi \mu^+ \nu_\mu$ branching fractions using three-prong τ lepton decays

Chekhovsky, Vladimir et al.

[arXiv:2510.21559](#) | 2025-10-24

76. Search for dijet resonances with data scouting in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2026\) 133](#) | [arXiv:2510.21641](#) | [DOI:10.1007/JHEP04\(2026\)133](#) | 2025-10-24

77. Observation of tWZ Production at the CMS Experiment

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 081802](#) | [arXiv:2510.19080](#) | [DOI:10.1103/rk6w-1pcl](#) | 2025-10-21

78. Search for dark matter production in association with bottom quarks and a lepton pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

[arXiv:2510.12396](#) | 2025-10-14

79. Performance of heavy-flavour jet identification in Lorentz-boosted topologies in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P11006](#) | [arXiv:2510.10228](#) | [DOI:10.1088/1748-0221/20/11/P11006](#) | 2025-10-11

80. Observation of Suppressed Charged-Particle Production in Ultrarelativistic Oxygen-Oxygen Collisions

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 162301](#) | [arXiv:2510.09864](#) | [DOI:10.1103/89sf-9t1x](#) | 2025-10-10

81. Combination of searches for nonresonant Higgs boson pair production in proton-proton collisions at $\sqrt{s}= 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2510.07527](#) | 2025-10-08

82. Observation of long-range collective flow in OO and NeNe collisions and implications for nuclear structure studies

Hayrapetyan, Aram et al.
[arXiv:2510.02580](#) | 2025-10-02

83. Wasserstein normalized autoencoder for anomaly detection

Hayrapetyan, Aram et al.
[arXiv:2510.02168](#) | 2025-10-02

84. Vector boson scattering and anomalous quartic couplings in final states with $\ell\nu qq$ or $\ell\ell qq$ plus jets using proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.
[JHEP 03 \(2026\) 022](#) | [arXiv:2510.00118](#) | [DOI:10.1007/JHEP03\(2026\)022](#) | 2025-09-30

85. Measurement of the dineutrino system kinematic variables in dileptonic top quark pair production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.
[arXiv:2510.00160](#) | 2025-09-30

86. Simultaneous Probe of the Charm and Bottom Quark Yukawa Couplings Using $t\bar{t}H$ Events

Hayrapetyan, Aram et al.
[Phys.Rev.Lett. 136 \(2026\) 011801](#) | [arXiv:2509.22535](#) | [DOI:10.1103/9nwb-splk](#) | 2025-09-26

87. Search for resonances decaying to an anomalous jet and a Higgs boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.
[Eur.Phys.J.C 86 \(2026\) 137](#) | [arXiv:2509.13635](#) | [DOI:10.1140/epjc/s10052-025-15176-9](#) | 2025-09-17

88. Measurements of $t\bar{t}W$ differential cross sections and the leptonic charge asymmetry at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.
[JHEP 03 \(2026\) 083](#) | [arXiv:2509.13512](#) | [DOI:10.1007/JHEP03\(2026\)083](#) | 2025-09-16

89. Improving missing transverse momentum estimation with a deep neural network

Hayrapetyan, Aram et al.
[Phys.Rev.D 113 \(2026\) 072010](#) | [arXiv:2509.12012](#) | [DOI:10.1103/c4z7-tqvc](#) | 2025-09-15

90. Measurement of D^0 Meson Photoproduction in Ultraperipheral Heavy Ion Collisions

Chekhovsky, Vladimir et al.
[Phys.Rev.Lett. 136 \(2026\) 122303](#) | [arXiv:2509.08626](#) | [DOI:10.1103/lckg-sdh9](#) | 2025-09-10

91. Model-independent measurement of the Higgs boson associated production with two jets and decaying to a pair of W bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.
[JHEP 03 \(2026\) 073](#) | [arXiv:2509.07958](#) | [DOI:10.1007/JHEP03\(2026\)073](#) | 2025-09-09

92. Search for a Higgs boson produced in association with a charm quark and decaying to a W boson pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2508.14988](#) | 2025-08-20

93. General search for supersymmetric particles in scenarios with compressed mass spectra using proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[Phys.Rev.D 112 \(2025\) 112023](#) | [arXiv:2508.13900](#) | [DOI:10.1103/b26z-zmpy](#) | 2025-08-19

94. Search for a new scalar resonance decaying to a Higgs boson and another new scalar particle in the final state with two bottom quarks and two photons in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 12 \(2025\) 178](#) | [arXiv:2508.11494](#) | [DOI:10.1007/JHEP12\(2025\)178](#) | 2025-08-15

95. Search for charged lepton flavor violating Z and Z' boson decays in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 112 \(2025\) 112011](#) | [arXiv:2508.07512](#) | [DOI:10.1103/8lw4-6lyk](#) | 2025-08-11

96. Search for light pseudoscalar boson pairs produced from Higgs boson decays using the 4τ and $2\mu 2\tau$ final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 04 \(2026\) 132](#) | [arXiv:2508.06947](#) | [DOI:10.1007/JHEP04\(2026\)132](#) | 2025-08-09

97. Search for b-hadron decays to long-lived particles in the CMS endcap muon detectors

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 012009](#) | [arXiv:2508.06363](#) | [DOI:10.1103/7ldn-snzn](#) | 2025-08-08

98. First Exclusive Reconstruction of the B^{*+} , B^{*0} , and B_s^{*0} Mesons and Precise Measurement of Their Masses

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 031902](#) | [arXiv:2508.05820](#) | [DOI:10.1103/njf9-4zfv](#) | 2025-08-07

99. Search for resonant production of pairs of dijet resonances through broad mediators in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2507.17884](#) | 2025-07-23

100. Probing the flavour structure of dimension-6 EFT operators in multilepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 140](#) | [arXiv:2507.17498](#) | [DOI:10.1007/JHEP03\(2026\)140](#) | 2025-07-23

101. Exploring small-angle emissions in charm quark jets in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Chekhovsky, Vladimir et al.

[arXiv:2507.13469](#) | 2025-07-17

102. Evidence of medium response to hard probes using correlations of Z bosons with hadrons in heavy ion collisions

Chekhovsky, Vladimir et al.

[Phys.Lett.B 874 \(2026\) 140120](#) | [arXiv:2507.09307](#) | [DOI:10.1016/j.physletb.2025.140120](#) | 2025-07-12

103. A method for correcting the substructure of multiprong jets using the Lund jet plane

Hayrapetyan, Aram et al.

[JHEP 11 \(2025\) 038](#) | [arXiv:2507.07775](#) | [DOI:10.1007/JHEP11\(2025\)038](#) | 2025-07-10

104. Search for heavy pseudoscalar and scalar bosons decaying to a top quark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 88 \(2025\) 127801](#) | [arXiv:2507.05119](#) | [DOI:10.1088/1361-6633/ae2207](#) | 2025-07-07

105. Search for the nonresonant and resonant production of a Higgs boson in association with an additional scalar boson in the $\gamma\gamma\tau\tau$ final state in proton-proton collisions at $\sqrt{s} =$

13 TeV

Hayrapetyan, Aram et al.

[arXiv:2506.23012](#) | 2025-06-28

106. Operation and performance of the CMS silicon strip tracker with proton-proton collisions at the CERN LHC

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P08027](#) | [arXiv:2506.17195](#) | [DOI:10.1088/1748-0221/20/08/P08027](#) | 2025-06-20

107. Search for nonresonant new physics signals in high-mass dilepton events produced in association with b-tagged jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 02 \(2026\) 141](#) | [arXiv:2506.13565](#) | [DOI:10.1007/JHEP02\(2026\)141](#) | 2025-06-16

108. Search for top squarks in final states with many light-flavor jets and 0, 1, or 2 charged leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 10 \(2025\) 236](#) | [arXiv:2506.08825](#) | [DOI:10.1007/JHEP10\(2025\)236](#) | 2025-06-10

109. Machine learning method for enforcing variable independence in background estimation with LHC data: ABCDisCoTEC

Hayrapetyan, Aram et al.

[arXiv:2506.08826](#) | 2025-06-10

110. Determination of the spin and parity of all-charm tetraquarks

Hayrapetyan, Aram et al.

[Nature 648 \(2025\) 58](#) | [arXiv:2506.07944](#) | [DOI:10.1038/s41586-025-09711-7](#) | 2025-06-09

111. Search for the Rare Decay $D^0 \rightarrow \mu^+ \mu^-$ in Proton-Proton Collisions at $\sqrt{s}=13.6$ TeV

Chekhovsky, Vladimir et al.

[Phys.Rev.Lett. 135 \(2025\) 151803](#) | [arXiv:2506.06152](#) | [DOI:10.1103/zc76-rgcp](#) | 2025-06-06

112. Search for dark matter produced in association with a Higgs boson decaying to a τ lepton pair in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 10 \(2025\) 170](#) | [arXiv:2506.04431](#) | [DOI:10.1007/JHEP10\(2025\)170](#) | 2025-06-04

113. Search for CP violation in events with top quarks and Z bosons at $\sqrt{s} = 13$ and 13.6 TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 869 \(2025\) 139857](#) | [arXiv:2505.21206](#) | [DOI:10.1016/j.physletb.2025.139857](#) | 2025-05-27

114. Measurement of WWZ and ZH Production Cross Sections at $\sqrt{s}=13$ and 13.6 TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 135 \(2025\) 091802](#) | [arXiv:2505.20483](#) | [DOI:10.1103/6z3d-zjw4](#) | 2025-05-26

115. Measurement of event shapes in minimum-bias events from proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[Phys.Rev.D 112 \(2025\) 112006](#) | [arXiv:2505.17850](#) | [DOI:10.1103/rqvb-gfb7](#) | 2025-05-23

116. Search for dark matter produced in association with one or two top quarks in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 08 \(2025\) 085](#) | [arXiv:2505.05300](#) | [DOI:10.1007/JHEP08\(2025\)085](#) | 2025-05-08

117. Measurements of inclusive and differential Higgs boson production cross sections at $\sqrt{s}=13.6$ TeV in the $H \rightarrow \gamma\gamma$ decay channel

Hayrapetyan, Aram et al.

[JHEP 09 \(2025\) 070](#) | [arXiv:2504.17755](#) | [DOI:10.1007/JHEP09\(2025\)070](#) | 2025-04-24

118. Combination and interpretation of differential Higgs boson production cross sections in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 04 \(2026\) 093](#) | [arXiv:2504.13081](#) | [DOI:10.1007/JHEP04\(2026\)093](#) | 2025-04-17

119. Search for jet quenching with dijets from high-multiplicity pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Chekhovsky, Vladimir et al.

[JHEP 07 \(2025\) 118](#) | [arXiv:2504.08507](#) | [DOI:10.1007/JHEP07\(2025\)118](#) | 2025-04-11

120. Search for charged-lepton flavour violation in top quark interactions with an up-type quark, a muon, and a τ lepton in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 12 \(2025\) 065](#) | [arXiv:2504.08532](#) | [DOI:10.1007/JHEP12\(2025\)065](#) | 2025-04-11

121. Observation of Coherent $\phi(1020)$ Meson Photoproduction in Ultraperipheral PbPb Collisions at $\sqrt{s_{NN}}=5.36$ TeV

Chekhovsky, Vladimir et al.

[Phys.Rev.Lett. 135 \(2025\) 262301](#) | [arXiv:2504.05193](#) | [DOI:10.1103/2ssw-wwyy](#) | 2025-04-07

122. Combined effective field theory interpretation of Higgs boson, electroweak vector boson, top quark, and multijet measurements

Chekhovsky, Vladimir et al.

[Eur.Phys.J.C 86 \(2026\) 331](#) | [arXiv:2504.02958](#) | [DOI:10.1140/epjc/s10052-025-14997-y](#) | 2025-04-03

123. Highlights of the HL-LHC physics projections by ATLAS and CMS

Aad, Georges et al.

[arXiv:2504.00672](#) | 2025-04-01

124. Observation of a pseudoscalar excess at the top quark pair production threshold

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 88 \(2025\) 087801](#) | [arXiv:2503.22382](#) | [DOI:10.1088/1361-6633/adf7d3](#) | 2025-03-28

125. Observation of $WZ\gamma$ production and constraints on new physics scenarios in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, V. et al.

[Phys.Rev.D 112 \(2025\) 012009](#) | [arXiv:2503.21977](#) | [DOI:10.1103/cm24-665b](#) | 2025-03-27

126. Search for t-channel scalar and vector leptoquark exchange in the high-mass dimuon and dielectron spectra in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, A. et al.

[JHEP 12 \(2025\) 052](#) | [arXiv:2503.20023](#) | [DOI:10.1007/JHEP12\(2025\)052](#) | 2025-03-25

127. Search for dark matter production in association with a single top quark in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 09 \(2025\) 141](#) | [arXiv:2503.20033](#) | [DOI:10.1007/JHEP09\(2025\)141](#) | 2025-03-25

128. Observation of nuclear modification of energy-energy correlators inside jets in heavy ion collisions

Chekhovsky, Vladimir et al.

[Phys.Lett.B 866 \(2025\) 139556](#) | [arXiv:2503.19993](#) | [DOI:10.1016/j.physletb.2025.139556](#) | 2025-03-25

129. Search for vector-like leptons with long-lived particle decays in the CMS muon system in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 08 \(2025\) 156](#) | [arXiv:2503.16699](#) | [DOI:10.1007/JHEP08\(2025\)156](#) | 2025-03-20

130. Measurements of the inclusive W and Z boson production cross sections and their ratios in proton-proton collisions at $\sqrt{s}=13.6$ TeV

Chekhovsky, Vladimir et al.

[JHEP 01 \(2026\) 047](#) | [arXiv:2503.09742](#) | [DOI:10.1007/JHEP01\(2026\)047](#) | 2025-03-12

131. Probing Gluon Fluctuations in Nuclei with the First Energy-Dependent Measurement of Incoherent J/ψ Photoproduction in Ultraperipheral PbPb Collisions

Chekhovsky, Vladimir et al.

[Phys.Rev.Lett. 135 \(2025\) 112301](#) | [arXiv:2503.08903](#) | [DOI:10.1103/w9kp-f8xr](#) | 2025-03-11

132. Search for a $c\bar{h}$ signal in the associated production of at least one charm quark with a Higgs boson in the diphoton decay channel in pp collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 11 \(2025\) 060](#) | [arXiv:2503.08797](#) | [DOI:10.1007/JHEP11\(2025\)060](#) | 2025-03-11

133. Search for New Physics in Jet Multiplicity Patterns of Multilepton Events at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 135 \(2025\) 231804](#) | [arXiv:2503.06726](#) | [DOI:10.1103/51fw-klz3](#) | 2025-03-09

134. Luminosity measurement for lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV in 2015 and 2018 at CMS

Chekhovsky, Vladimir et al.

[arXiv:2503.03946](#) | 2025-03-05

135. Observation of the Charged-Particle Multiplicity Dependence of $\sigma_{\psi(2S)}/\sigma_{J/\psi}$ in p-Pb Collisions at 8.16 TeV

Chekhovsky, Vladimir et al.

[Phys.Rev.Lett. 135 \(2025\) 092301](#) | [arXiv:2503.02139](#) | [DOI:10.1103/c9wp-5tq3](#) | 2025-03-03

136. Development of systematic uncertainty-aware neural network trainings for binned-likelihood analyses at the LHC

Chekhovsky, Vladimir et al.

[Eur.Phys.J.C 85 \(2025\) 1360](#) | [arXiv:2502.13047](#) | [DOI:10.1140/epjc/s10052-025-14713-w](#) | 2025-02-18

137. Search for medium effects using jet axis decorrelation in inclusive jets from PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Chekhovsky, Vladimir et al.

[JHEP 06 \(2025\) 120](#) | [arXiv:2502.13020](#) | [DOI:10.1007/JHEP06\(2025\)120](#) | 2025-02-18

138. Evidence for Similar Collectivity of High Transverse-Momentum Particles in p-Pb and Pb-Pb Collisions

Chekhovsky, Vladimir et al.

[Phys.Rev.Lett. 135 \(2025\) 071903](#) | [arXiv:2502.07525](#) | [DOI:10.1103/t5kp-vsv7](#) | 2025-02-11

139. Observation of Λ Hyperon Local Polarization in p-Pb Collisions at $\sqrt{s_{NN}}=8.16$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 135 \(2025\) 132301](#) | [arXiv:2502.07898](#) | [DOI:10.1103/6ywq-gm61](#) | 2025-02-11

140. Search for bosons of an extended Higgs sector in b quark final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 06 \(2025\) 144](#) | [arXiv:2502.06568](#) | [DOI:10.1007/JHEP06\(2025\)144](#) | 2025-02-10

141. Search for γH production and constraints on the Yukawa couplings of light quarks to the Higgs boson

Chekhovsky, Vladimir et al.

[Phys.Rev.D 112 \(2025\) 112001](#) | [arXiv:2502.05665](#) | [DOI:10.1103/6s8n-pvmy](#) | 2025-02-08

142. Measurements of the Higgs boson production cross section in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Chekhovsky, Vladimir et al.

[JHEP 05 \(2025\) 079](#) | [arXiv:2501.14849](#) | [DOI:10.1007/JHEP05\(2025\)079](#) | 2025-01-24

143. Search for a heavy pseudoscalar Higgs boson decaying to a 125 GeV Higgs boson and a Z boson in final states with two tau and two light leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 10 \(2025\) 074](#) | [arXiv:2501.14825](#) | [DOI:10.1007/JHEP10\(2025\)074](#) | 2025-01-21

144. Evidence for CP violation and measurement of CP-violating parameters in $B^0_s \rightarrow J/\psi \phi(1020)$ decays in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2412.19952](#) | 2024-12-27

145. Search for a Neutral Gauge Boson with Nonuniversal Fermion Couplings in Vector Boson Fusion Processes in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 135 \(2025\) 061803](#) | [arXiv:2412.19261](#) | [DOI:10.1103/srvm-f1h3](#) | 2024-12-26

146. Identification of low-momentum muons in the CMS detector using multivariate techniques in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Chekhovsky, Vladimir et al.

[JINST 20 \(2025\) P04021](#) | [arXiv:2412.17590](#) | [DOI:10.1088/1748-0221/20/04/P04021](#) | 2024-12-23

147. Determination of the strong coupling and its running from measurements of inclusive jet production

Chekhovsky, Vladimir et al.

[Phys.Lett.B 868 \(2025\) 139651](#) | [arXiv:2412.16665](#) | [DOI:10.1016/j.physletb.2025.139651](#) | 2024-12-21

148. Measurement of the W boson decay branching fraction ratio $\mathcal{B}(W \rightarrow cq)/\mathcal{B}(W \rightarrow q\bar{q}')$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 868 \(2025\) 139754](#) | [arXiv:2412.16296](#) | [DOI:10.1016/j.physletb.2025.139754](#) | 2024-12-20

149. Measurement of light-by-light scattering and the Breit-Wheeler process, and search for axion-like particles in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Hayrapetyan, Aram et al.

[JHEP 08 \(2025\) 006](#) | [arXiv:2412.15413](#) | [DOI:10.1007/JHEP08\(2025\)006](#) | 2024-12-19

150. High-precision measurement of the W boson mass with the CMS experiment

Chekhovsky, Vladimir et al.

[Nature 652 \(2026\) 321](#) | [arXiv:2412.13872](#) | [DOI:10.1038/s41586-026-10168-5](#) | 2024-12-18

151. Search for heavy neutral resonances decaying to tau lepton pairs in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 111 \(2025\) 112004](#) | [arXiv:2412.04357](#) | [DOI:10.1103/PhysRevD.111.112004](#) | 2024-12-05

152. Model-agnostic search for dijet resonances with anomalous jet substructure in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

[Rept.Prog.Phys. 88 \(2025\) 067802](#) | [arXiv:2412.03747](#) | [DOI:10.1088/1361-6633/add762](#) | 2024-12-04

153. Measurement of the inclusive WZ production cross section in pp collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2025\) 115](#) | [arXiv:2412.02477](#) | [DOI:10.1007/JHEP04\(2025\)115](#) | 2024-12-03

154. Search for heavy neutral Higgs bosons A and H in the $t\bar{t}Z$ channel in proton-proton collisions at 13 TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 866 \(2025\) 139568](#) | [arXiv:2412.00570](#) | [DOI:10.1016/j.physletb.2025.139568](#) | 2024-11-30

155. Proton reconstruction with the TOTEM Roman pot detectors for high- β^* LHC data

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P04012](#) | [arXiv:2411.19749](#) | [DOI:10.1088/1748-0221/20/04/P04012](#) | 2024-11-29

156. Constraints on standard model effective field theory for a Higgs boson produced in association with W or Z bosons in the $H \rightarrow b\bar{b}$ decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 03 \(2025\) 114](#) | [arXiv:2411.16907](#) | [DOI:10.1007/JHEP03\(2025\)114](#) | 2024-11-25

157. Search for rare decays of the Z and Higgs bosons to a J/ψ or $\psi(2S)$ meson and a photon in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 865 \(2025\) 139462](#) | [arXiv:2411.15000](#) | [DOI:10.1016/j.physletb.2025.139462](#) | 2024-11-22

158. Angular analysis of the $B^0 \rightarrow K^{*(892)^0} \mu^+ \mu^-$ decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 864 \(2025\) 139406](#) | [arXiv:2411.11820](#) | [DOI:10.1016/j.physletb.2025.139406](#) | 2024-11-18

159. Reweighting simulated events using machine-learning techniques in the CMS experiment

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 85 \(2025\) 495](#) | [arXiv:2411.03023](#) | [DOI:10.1140/epjc/s10052-025-14097-x](#) | 2024-11-05

160. Search for a heavy resonance decaying into a Z and a Higgs boson in events with an energetic jet and two electrons, two muons, or missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 089](#) | [arXiv:2411.00202](#) | [DOI:10.1007/JHEP02\(2025\)089](#) | 2024-10-31

161. Measurements of inclusive and differential cross sections for top quark production in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 177](#) | [arXiv:2410.23475](#) | [DOI:10.1007/JHEP02\(2025\)177](#) | 2024-10-30

162. Measurement of the inclusive $t\bar{t}$ cross section in final states with at least one lepton and additional jets with 302 pb^{-1} of pp collisions at $\sqrt{s} = 5.02$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2025\) 099](#) | [arXiv:2410.21631](#) | [DOI:10.1007/JHEP04\(2025\)099](#) | 2024-10-28

163. Search for excited tau leptons in the $\tau\tau\gamma$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

164. Search for pair production of heavy particles decaying to a top quark and a gluon in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 85 \(2025\) 342](#) | [arXiv:2410.20601](#) | [DOI:10.1140/epjc/s10052-024-13729-y](#) | 2024-10-27

165. Search for the Higgs boson decays to a ρ^0 , ϕ , or K^*_0 meson and a photon in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 862 \(2025\) 139296](#) | [arXiv:2410.18289](#) | [DOI:10.1016/j.physletb.2025.139296](#) | 2024-10-23

166. Performance of the CMS high-level trigger during LHC Run 2

Hayrapetyan, A. et al.

[JINST 19 \(2024\) P11021](#) | [arXiv:2410.17038](#) | [DOI:10.1088/1748-0221/19/11/P11021](#) | 2024-10-22

167. Search for high-mass resonances in a final state comprising a gluon and two hadronically decaying W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 199](#) | [arXiv:2410.17303](#) | [DOI:10.1007/JHEP02\(2025\)199](#) | 2024-10-22

168. Search for heavy long-lived charged particles with large ionization energy loss in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2025\) 109](#) | [arXiv:2410.09164](#) | [DOI:10.1007/JHEP04\(2025\)109](#) | 2024-10-11

169. Correlations between azimuthal anisotropy and mean transverse momentum in pp, pPb, and peripheral PbPb collisions

Tumasyan, Armen et al.

[arXiv:2410.04578](#) | 2024-10-06

170. Study of same-sign W boson scattering and anomalous couplings in events with one tau lepton from pp collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2025\) 219](#) | [arXiv:2410.04210](#) | [DOI:10.1007/JHEP10\(2025\)219](#) | 2024-10-05

171. Energy-scaling behavior of intrinsic transverse-momentum parameters in Drell-Yan simulation

Hayrapetyan, Aram et al.

[Phys.Rev.D 111 \(2025\) 072003](#) | [arXiv:2409.17770](#) | [DOI:10.1103/PhysRevD.111.072003](#) | 2024-09-26

172. Measurement of the Higgs boson mass and width using the four-lepton final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 111 \(2025\) 092014](#) | [arXiv:2409.13663](#) | [DOI:10.1103/PhysRevD.111.092014](#) | 2024-09-20

173. Measurements of polarization and spin correlation and observation of entanglement in top quark pairs using lepton+jets events from proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 112016](#) | [arXiv:2409.11067](#) | [DOI:10.1103/PhysRevD.110.112016](#) | 2024-09-17

174. Search for light long-lived particles decaying to displaced jets in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 88 \(2025\) 037801](#) | [arXiv:2409.10806](#) | [DOI:10.1088/1361-6633/adaa13](#) | 2024-09-16

175. Bottom quark energy loss and hadronization with B^+ and B_s^0 nuclear modification factors using pp and PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 195](#) | [arXiv:2409.07258](#) | [DOI:10.1007/JHEP02\(2025\)195](#) | 2024-09-11

176. Measurement of inclusive and differential cross sections of single top quark production in association with a W boson in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[JHEP 01 \(2025\) 107](#) | [arXiv:2409.06444](#) | [DOI:10.1007/JHEP01\(2025\)107](#) | 2024-09-10

177. Pseudorapidity distributions of charged hadrons in lead-lead collisions at $\sqrt{s_{NN}}=5.36$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 861 \(2025\) 139279](#) | [arXiv:2409.00838](#) | [DOI:10.1016/j.physletb.2025.139279](#) | 2024-09-01

178. Search for dark matter produced in association with a pair of bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 050](#) | [arXiv:2408.17336](#) | [DOI:10.1007/JHEP02\(2025\)050](#) | 2024-08-30

179. Measurement of the Drell-Yan forward-backward asymmetry and of the effective leptonic weak mixing angle in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 866 \(2025\) 139526](#) | [arXiv:2408.07622](#) | [DOI:10.1016/j.physletb.2025.139526](#) | 2024-08-14

180. Measurement of the inclusive cross sections for W and Z boson production in proton-proton collisions at $\sqrt{s} = 5.02$ and 13 TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2025\) 162](#) | [arXiv:2408.03744](#) | [DOI:10.1007/JHEP04\(2025\)162](#) | 2024-08-07

181. Search for bottom quark associated production of the standard model Higgs boson in final states with leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 860 \(2025\) 139173](#) | [arXiv:2408.01344](#) | [DOI:10.1016/j.physletb.2024.139173](#) | 2024-08-02

182. Test of lepton flavor universality in semileptonic $B_c^+ +$ meson decays in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 111 \(2025\) L051102](#) | [arXiv:2408.00678](#) | [DOI:10.1103/PhysRevD.111.L051102](#) | 2024-08-01

183. Model-independent search for pair production of new bosons decaying into muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 12 \(2024\) 172](#) | [arXiv:2407.20425](#) | [DOI:10.1007/JHEP12\(2024\)172](#) | 2024-07-29

184. Search for flavor-changing neutral current interactions of the top quark mediated by a Higgs boson in proton-proton collisions at 13 TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 112 \(2025\) 032008](#) | [arXiv:2407.15172](#) | [DOI:10.1103/95q6-vvlp](#) | 2024-07-21

185. Measurement of the $B^0_s \rightarrow J/\psi K^0_S$ effective lifetime from proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2024\) 247](#) | [arXiv:2407.13441](#) | [DOI:10.1007/JHEP10\(2024\)247](#) | 2024-07-18

186. Constraints on the Higgs boson self-coupling from the combination of single and double Higgs boson production in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 861 \(2025\) 139210](#) | [arXiv:2407.13554](#) | [DOI:10.1016/j.physletb.2024.139210](#) | 2024-07-18

187. Search for resonant pair production of Higgs bosons in the bbbb final state using large-area jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2025\) 040](#) | [arXiv:2407.13872](#) | [DOI:10.1007/JHEP02\(2025\)040](#) | 2024-07-18

188. Multiplicity dependence of charm baryon and charm meson production in pPb collisions at $\sqrt{s_{NN}}=8.16$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 868 \(2025\) 139672](#) | [arXiv:2407.13615](#) | [DOI:10.1016/j.physletb.2025.139672](#) | 2024-07-18

189. Measurement of the ttH and tH production rates in the $H \rightarrow b\bar{b}$ decay channel using proton-proton collision data at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 097](#) | [arXiv:2407.10896](#) | [DOI:10.1007/JHEP02\(2025\)097](#) | 2024-07-15

190. Search for long-lived heavy neutral leptons in proton-proton collision events with a lepton-jet pair associated with a secondary vertex at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 036](#) | [arXiv:2407.10717](#) | [DOI:10.1007/JHEP02\(2025\)036](#) | 2024-07-15

191. Measurement of boosted Higgs bosons produced via vector boson fusion or gluon fusion in the $H \rightarrow b\bar{b}$ decay mode using LHC proton-proton collision data at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 12 \(2024\) 035](#) | [arXiv:2407.08012](#) | [DOI:10.1007/JHEP12\(2024\)035](#) | 2024-07-10

192. Search for Nuclear Modifications of $B^+ +$ Meson Production in p-Pb Collisions at $\sqrt{s_{NN}}=8.16$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 134 \(2025\) 111903](#) | [arXiv:2407.05402](#) | [DOI:10.1103/PhysRevLett.134.111903](#) | 2024-07-07

193. Observation of double J/ψ meson production in pPb collisions at $\sqrt{s_{NN}}=8.16$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 092002](#) | [arXiv:2407.03223](#) | [DOI:10.1103/PhysRevD.110.092002](#) | 2024-07-03

194. Measurement of the polarizations of prompt and non-prompt J/ψ and $\psi(2S)$ mesons produced in pp collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 858 \(2024\) 139044](#) | [arXiv:2406.14409](#) | [DOI:10.1016/j.physletb.2024.139044](#) | 2024-06-20

195. Search for a resonance decaying to a W boson and a photon in proton-proton collisions at $\sqrt{s} = 13$ TeV using leptonic W boson decays

Hayrapetyan, Aram et al.

[JHEP 09 \(2024\) 186](#) | [arXiv:2406.05737](#) | [DOI:10.1007/JHEP09\(2024\)186](#) | 2024-06-09

196. Measurement of inclusive and differential cross sections for W^+W^- production in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 861 \(2025\) 139231](#) | [arXiv:2406.05101](#) | [DOI:10.1016/j.physletb.2024.139231](#) | 2024-06-07

197. Observation of quantum entanglement in top quark pair production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 87 \(2024\) 117801](#) | [arXiv:2406.03976](#) | [DOI:10.1088/1361-6633/ad7e4d](#) | 2024-06-06

198. Observation of $\gamma\gamma \rightarrow \tau\tau$ in proton-proton collisions and limits on the anomalous electromagnetic moments of the τ lepton

Hayrapetyan, Aram et al.

Rept.Prog.Phys. 87 (2024) 107801 | [arXiv:2406.03975](#) | [DOI:10.1088/1361-6633/ad6fcb](#) | 2024-06-06

199. Stairway to discovery: A report on the CMS programme of cross section measurements from millibarns to femtobarns

Hayrapetyan, Aram et al.

Phys.Rept. 1115 (2025) 3 | [arXiv:2405.18661](#) | [DOI:10.1016/j.physrep.2024.11.005](#) | 2024-05-28

200. Search for a standard model-like Higgs boson in the mass range between 70 and 110 GeV in the diphoton final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

Phys.Lett.B 860 (2025) 139067 | [arXiv:2405.18149](#) | [DOI:10.1016/j.physletb.2024.139067](#) | 2024-05-28

201. Review of searches for vector-like quarks, vector-like leptons, and heavy neutral leptons in proton-proton collisions at $\sqrt{s}=13$ TeV at the CMS experiment

Hayrapetyan, Aram et al.

Phys.Rept. 1115 (2025) 2350 | [arXiv:2405.17605](#) | [DOI:10.1016/j.physrep.2024.09.012](#) | 2024-05-27

202. Study of WH production through vector boson scattering and extraction of the relative sign of the W and Z couplings to the Higgs boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

Phys.Lett.B 860 (2025) 139202 | [arXiv:2405.16566](#) | [DOI:10.1016/j.physletb.2024.139202](#) | 2024-05-26

203. Searches for violation of Lorentz invariance in top quark pair production using dilepton events in 13 TeV proton-proton collisions

Hayrapetyan, Aram et al.

Phys.Lett.B 857 (2024) 138979 | [arXiv:2405.14757](#) | [DOI:10.1016/j.physletb.2024.138979](#) | 2024-05-23

204. Dark sector searches with the CMS experiment

Hayrapetyan, Aram et al.

Phys.Rept. 1115 (2025) 2351 | [arXiv:2405.13778](#) | [DOI:10.1016/j.physrep.2024.09.013](#) | 2024-05-22

205. Search for CP violation in $D^0 \rightarrow K^0_{\text{SK}} S$ decays in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

Eur.Phys.J.C 84 (2024) 1264 | [arXiv:2405.11606](#) | [DOI:10.1140/epjc/s10052-024-13244-0](#) | 2024-05-19

206. Overview of high-density QCD studies with the CMS experiment at the LHC

Hayrapetyan, Aram et al.

Phys.Rept. 1115 (2025) 2367 | [arXiv:2405.10785](#) | [DOI:10.1016/j.physrep.2024.11.007](#) | 2024-05-17

207. Search for new physics in high-mass diphoton events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

JHEP 08 (2024) 215 | [arXiv:2405.09320](#) | [DOI:10.1007/JHEP08\(2024\)215](#) | 2024-05-15

208. Search for production of a single vectorlike quark decaying to tH or tZ in the all-hadronic final state in pp collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

Phys.Rev.D 110 (2024) 072012 | [arXiv:2405.05071](#) | [DOI:10.1103/PhysRevD.110.072012](#) | 2024-05-08

209. Girth and groomed radius of jets recoiling against isolated photons in lead-lead and proton-proton collisions at $\sqrt{s_{\text{NN}}}=5.02$ TeV

Hayrapetyan, Aram et al.

Phys.Lett.B 861 (2025) 139088 | [arXiv:2405.02737](#) | [DOI:10.1016/j.physletb.2024.139088](#) | 2024-05-04

210. Search for New Resonances Decaying to Pairs of Merged Diphotons in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

- Phys.Rev.Lett. 134 (2025) 041801 | [arXiv:2405.00834](#) | [DOI:10.1103/PhysRevLett.134.041801](#) | 2024-05-01
211. Search for the Z Boson Decay to $\tau\tau\mu\mu$ in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$
Hayrapetyan, Aram et al.
[Phys.Rev.Lett. 133 \(2024\) 161805](#) | [arXiv:2404.18298](#) | [DOI:10.1103/PhysRevLett.133.161805](#) | 2024-04-28
212. Performance of CMS muon reconstruction from proton-proton to heavy ion collisions
Hayrapetyan, Aram et al.
[JINST 19 \(2024\) P09012](#) | [arXiv:2404.17377](#) | [DOI:10.1088/1748-0221/19/09/P09012](#) | 2024-04-26
213. Measurement of multijet azimuthal correlations and determination of the strong coupling in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$
Hayrapetyan, Aram et al.
[Eur.Phys.J.C 84 \(2024\) 842](#) | [arXiv:2404.16082](#) | [DOI:10.1140/epjc/s10052-024-13116-7](#) | 2024-04-24
214. Search for Higgs boson pair production with one associated vector boson in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$
Hayrapetyan, Aram et al.
[JHEP 10 \(2024\) 061](#) | [arXiv:2404.08462](#) | [DOI:10.1007/JHEP10\(2024\)061](#) | 2024-04-12
215. The CMS Statistical Analysis and Combination Tool: Combine
Hayrapetyan, Aram et al.
[Comput.Softw.Big Sci. 8 \(2024\) 19](#) | [arXiv:2404.06614](#) | [DOI:10.1007/s41781-024-00121-4](#) | 2024-04-09
216. Measurement of differential ZZ + jets production cross sections in pp collisions at $\sqrt{s} = 13\text{ TeV}$
Hayrapetyan, Aram et al.
[JHEP 10 \(2024\) 209](#) | [arXiv:2404.02711](#) | [DOI:10.1007/JHEP10\(2024\)209](#) | 2024-04-03
217. Searches for Pair-Produced Multijet Resonances Using Data Scouting in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$
Hayrapetyan, Aram et al.
[Phys.Rev.Lett. 133 \(2024\) 201803](#) | [arXiv:2404.02992](#) | [DOI:10.1103/PhysRevLett.133.201803](#) | 2024-04-03
218. Measurement of the production cross section of a Higgs boson with large transverse momentum in its decays to a pair of τ leptons in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$
Hayrapetyan, Aram et al.
[Phys.Lett.B 857 \(2024\) 138964](#) | [arXiv:2403.20201](#) | [DOI:10.1016/j.physletb.2024.138964](#) | 2024-03-29
219. Search for ZZ and ZH production in the $b\bar{b}b\bar{b}$ final state using proton-proton collisions at $\sqrt{s}=13\text{ TeV}$
Hayrapetyan, Aram et al.
[Eur.Phys.J.C 84 \(2024\) 712](#) | [arXiv:2403.20241](#) | [DOI:10.1140/epjc/s10052-024-13021-z](#) | 2024-03-29
220. Searches for Higgs boson production through decays of heavy resonances
Hayrapetyan, Aram et al.
[Phys.Rept. 1115 \(2025\) 2342](#) | [arXiv:2403.16926](#) | [DOI:10.1016/j.physrep.2024.09.004](#) | 2024-03-25
221. Enriching the physics program of the CMS experiment via data scouting and data parking
Hayrapetyan, Aram et al.
[Phys.Rept. 1115 \(2025\) 2344](#) | [arXiv:2403.16134](#) | [DOI:10.1016/j.physrep.2024.09.006](#) | 2024-03-24
222. Performance of the CMS electromagnetic calorimeter in pp collisions at $\sqrt{s} = 13\text{ TeV}$
Hayrapetyan, Aram et al.
[JINST 19 \(2024\) P09004](#) | [arXiv:2403.15518](#) | [DOI:10.1088/1748-0221/19/09/P09004](#) | 2024-03-22

223. Observation of the $J/\psi \rightarrow \mu^+ \mu^- \mu^+ \mu^-$ decay in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) L111101](#) | [arXiv:2403.11352](#) | [DOI:10.1103/PhysRevD.109.L111101](#) | 2024-03-17

224. Search for the decay of the Higgs boson to a pair of light pseudoscalar bosons in the final state with four bottom quarks in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 06 \(2024\) 097](#) | [arXiv:2403.10341](#) | [DOI:10.1007/JHEP06\(2024\)097](#) | 2024-03-15

225. Search for Higgs boson pair production in the bbW^+W^- decay mode in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 07 \(2024\) 293](#) | [arXiv:2403.09430](#) | [DOI:10.1007/JHEP07\(2024\)293](#) | 2024-03-14

226. Search for Soft Unclustered Energy Patterns in Proton-Proton Collisions at 13 TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 191902](#) | [arXiv:2403.05311](#) | [DOI:10.1103/PhysRevLett.133.191902](#) | 2024-03-08

227. Search for long-lived heavy neutrinos in the decays of B mesons produced in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 06 \(2024\) 183](#) | [arXiv:2403.04584](#) | [DOI:10.1007/JHEP06\(2024\)183](#) | 2024-03-07

228. Search for dark QCD with emerging jets in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 07 \(2024\) 142](#) | [arXiv:2403.01556](#) | [DOI:10.1007/JHEP07\(2024\)142](#) | 2024-03-03

229. Review of top quark mass measurements in CMS

Hayrapetyan, Aram et al.

[Phys.Rept. 1115 \(2025\) 2369](#) | [arXiv:2403.01313](#) | [DOI:10.1016/j.physrep.2024.12.002](#) | 2024-03-02

230. Constraints on anomalous Higgs boson couplings from its production and decay using the WW channel in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 779](#) | [arXiv:2403.00657](#) | [DOI:10.1140/epjc/s10052-024-12925-0](#) | 2024-03-01

231. Search for heavy neutral leptons in final states with electrons, muons, and hadronically decaying tau leptons in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 06 \(2024\) 123](#) | [arXiv:2403.00100](#) | [DOI:10.1007/JHEP06\(2024\)123](#) | 2024-02-29

232. Search for long-lived heavy neutral leptons decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 012004](#) | [arXiv:2402.18658](#) | [DOI:10.1103/PhysRevD.110.012004](#) | 2024-02-28

233. Search for baryon number violation in top quark production and decay using proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 241802](#) | [arXiv:2402.18461](#) | [DOI:10.1103/PhysRevLett.132.241802](#) | 2024-02-28

234. Observation of the $\Xi_b^- \rightarrow \psi(2S) \Xi^-$ decay and studies of the $\Xi_b(5945)^0$ baryon in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 012002](#) | [arXiv:2402.17738](#) | [DOI:10.1103/PhysRevD.110.012002](#) | 2024-02-27

235. Search for long-lived particles using displaced vertices and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112005](#) | [arXiv:2402.15804](#) | [DOI:10.1103/PhysRevD.109.112005](#) | 2024-02-24

236. Portable Acceleration of CMS Computing Workflows with Coprocessors as a Service

Hayrapetyan, Aram et al.

[Comput.Softw.Big Sci. 8 \(2024\) 17](#) | [arXiv:2402.15366](#) | [DOI:10.1007/s41781-024-00124-1](#) | 2024-02-23

237. Search for long-lived particles decaying to final states with a pair of muons in proton-proton collisions at $\sqrt{s} = 13.6 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 047](#) | [arXiv:2402.14491](#) | [DOI:10.1007/JHEP05\(2024\)047](#) | 2024-02-22

238. Measurement of Energy Correlators inside Jets and Determination of the Strong Coupling $\alpha_S(m_Z)$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 071903](#) | [arXiv:2402.13864](#) | [DOI:10.1103/PhysRevLett.133.071903](#) | 2024-02-21

239. Search for bottom-type vectorlike quark pair production in dileptonic and fully hadronic final states in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 052004](#) | [arXiv:2402.13808](#) | [DOI:10.1103/PhysRevD.110.052004](#) | 2024-02-21

240. Search for exotic decays of the Higgs boson to a pair of pseudoscalars in the $\mu\mu b\bar{b}$ and $\tau\tau b\bar{b}$ final states

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 493](#) | [arXiv:2402.13358](#) | [DOI:10.1140/epjc/s10052-024-12727-4](#) | 2024-02-20

241. Search for a scalar or pseudoscalar dilepton resonance produced in association with a massive vector boson or top quark-antiquark pair in multilepton events at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.D 110 \(2024\) 012013](#) | [arXiv:2402.11098](#) | [DOI:10.1103/PhysRevD.110.012013](#) | 2024-02-16

242. Search for Fractionally Charged Particles in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 134 \(2025\) 131802](#) | [arXiv:2402.09932](#) | [DOI:10.1103/PhysRevLett.134.131802](#) | 2024-02-15

243. Combination of Measurements of the Top Quark Mass from Data Collected by the ATLAS and CMS Experiments at $\sqrt{s}=7$ and 8 TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 261902](#) | [arXiv:2402.08713](#) | [DOI:10.1103/PhysRevLett.132.261902](#) | 2024-02-13

244. Differential cross section measurements for the production of top quark pairs and of additional jets using dilepton events from pp collisions at $\sqrt{s} = 13 \text{ TeV}$

Tumasyan, Armen et al.

[JHEP 02 \(2025\) 064](#) | [arXiv:2402.08486](#) | [DOI:10.1007/JHEP02\(2025\)064](#) | 2024-02-13

245. Search for pair production of scalar and vector leptoquarks decaying to muons and bottom quarks in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112003](#) | [arXiv:2402.08668](#) | [DOI:10.1103/PhysRevD.109.112003](#) | 2024-02-09

246. Search for long-lived particles decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 032007](#) | [arXiv:2402.01898](#) | [DOI:10.1103/PhysRevD.110.032007](#) | 2024-02-02

247. Combined search for electroweak production of winos, binos, higgsinos, and sleptons in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112001](#) | [arXiv:2402.01888](#) | [DOI:10.1103/PhysRevD.109.112001](#) | 2024-02-02

248. Observation of the $\Lambda_b^0 \rightarrow J/\psi \Xi^- K^+$ decay

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 1062](#) | [arXiv:2401.16303](#) | [DOI:10.1140/epjc/s10052-024-13114-9](#) | 2024-01-29

249. Nonresonant central exclusive production of charged-hadron pairs in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112013](#) | [arXiv:2401.14494](#) | [DOI:10.1103/PhysRevD.109.112013](#) | 2024-01-25

250. Measurement of the double-differential inclusive jet cross section in proton-proton collisions at $\sqrt{s} = 5.02 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 01 \(2025\) 011](#) | [arXiv:2401.11355](#) | [DOI:10.1007/JHEP01\(2025\)011](#) | 2024-01-20

251. Test of lepton flavor universality in $B^{\pm} \rightarrow K^{\pm} \mu^+ \mu^-$ and $B^{\pm} \rightarrow K^{\pm} e^+ e^-$ decays in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 87 \(2024\) 077802](#) | [arXiv:2401.07090](#) | [DOI:10.1088/1361-6633/ad4e65](#) | 2024-01-13

252. Extracting the speed of sound in quark-gluon plasma with ultrarelativistic lead-lead collisions at the LHC

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 87 \(2024\) 077801](#) | [arXiv:2401.06896](#) | [DOI:10.1088/1361-6633/ad4b9b](#) | 2024-01-12

253. Observation of Enhanced Long-Range Elliptic Anisotropies Inside High-Multiplicity Jets in pp Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 142301](#) | [arXiv:2312.17103](#) | [DOI:10.1103/PhysRevLett.133.142301](#) | 2023-12-28

254. Elliptic anisotropy measurement of the $f_0(980)$ hadron in proton-lead collisions and evidence for its quark-antiquark composition

Hayrapetyan, Aram et al.

[Nature Commun. 16 \(2025\) 7990](#) | [arXiv:2312.17092](#) | [DOI:10.1038/s41467-025-56200-6](#) | 2023-12-28

255. Measurement of multidifferential cross sections for dijet production in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 85 \(2025\) 72](#) | [arXiv:2312.16669](#) | [DOI:10.1140/epjc/s10052-024-13606-8](#) | 2023-12-27

256. Measurement of the primary Lund jet plane density in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 116](#) | [arXiv:2312.16343](#) | [DOI:10.1007/JHEP05\(2024\)116](#) | 2023-12-26

257. Evidence for tWZ production in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$ in multilepton final states

Hayrapetyan, Aram et al.

[Phys.Lett.B 855 \(2024\) 138815](#) | [arXiv:2312.11668](#) | [DOI:10.1016/j.physletb.2024.138815](#) | 2023-12-18

258. Search for Long-Lived Heavy Neutral Leptons with Lepton Flavour Conserving or Violating Decays to a Jet and a Charged Lepton

Hayrapetyan, Aram et al.

[JHEP 03 \(2024\) 105](#) | [arXiv:2312.07484](#) | [DOI:10.1007/JHEP03\(2024\)105](#) | 2023-12-12

259. Search for flavor changing neutral current interactions of the top quark in final states with a photon and additional jets in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 072004](#) | [arXiv:2312.08229](#) | [DOI:10.1103/PhysRevD.109.072004](#) | 2023-12-11

260. Measurement of simplified template cross sections of the Higgs boson produced in association with W or Z bosons in the $H\rightarrow b\bar{b}$ decay channel in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.D 109 \(2024\) 092011](#) | [arXiv:2312.07562](#) | [DOI:10.1103/PhysRevD.109.092011](#) | 2023-12-09

261. Search for charged-lepton flavor violation in the production and decay of top quarks using trilepton final states in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 111 \(2025\) 012009](#) | [arXiv:2312.03199](#) | [DOI:10.1103/PhysRevD.111.012009](#) | 2023-12-05

262. Search for the lepton flavor violating $\tau \rightarrow 3\mu$ decay in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 853 \(2024\) 138633](#) | [arXiv:2312.02371](#) | [DOI:10.1016/j.physletb.2024.138633](#) | 2023-12-04

263. Higher-order moments of the elliptic flow distribution in PbPb collisions at $\sqrt{s_{\text{NN}}} = 5.02\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 2024 \(2024\) 106](#) | [arXiv:2311.11370](#) | [DOI:10.1007/JHEP02\(2024\)106](#) | 2023-11-19

264. Search for new Higgs bosons via same-sign top quark pair production in association with a jet in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 850 \(2024\) 138478](#) | [arXiv:2311.03261](#) | [DOI:10.1016/j.physletb.2024.138478](#) | 2023-11-06

265. Search for high-mass exclusive diphoton production with tagged protons in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.D 110 \(2024\) 012010](#) | [arXiv:2311.02725](#) | [DOI:10.1103/PhysRevD.110.012010](#) | 2023-11-05

266. Search for an exotic decay of the Higgs boson into a Z boson and a pseudoscalar particle in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 852 \(2024\) 138582](#) | [arXiv:2311.00130](#) | [DOI:10.1016/j.physletb.2024.138582](#) | 2023-10-31

267. Search for W' bosons decaying to a top and a bottom quark in leptonic final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 046](#) | [arXiv:2310.19893](#) | [DOI:10.1007/JHEP05\(2024\)046](#) | 2023-10-30

268. Search for Narrow Trijet Resonances in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 011801](#) | [arXiv:2310.14023](#) | [DOI:10.1103/PhysRevLett.133.011801](#) | 2023-10-21

269. Search for central exclusive production of top quark pairs in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$ with tagged protons

Tumasyan, Armen et al.

[JHEP 06 \(2024\) 187](#) | [arXiv:2310.11231](#) | [DOI:10.1007/JHEP06\(2024\)187](#) | 2023-10-17

270. Observation of $WW\gamma$ production and search for $H\gamma$ production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 121901](#) | [arXiv:2310.05164](#) | [DOI:10.1103/PhysRevLett.132.121901](#) | 2023-10-08

271. Muon identification using multivariate techniques in the CMS experiment in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P02031](#) | [arXiv:2310.03844](#) | [DOI:10.1088/1748-0221/19/02/P02031](#) | 2023-10-05

272. Study of azimuthal anisotropy of $\phi(1S)$ mesons in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 850 \(2024\) 138518](#) | [arXiv:2310.03233](#) | [DOI:10.1016/j.physletb.2024.138518](#) | 2023-10-04

273. Search for stealth supersymmetry in final states with two photons, jets, and low missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112009](#) | [arXiv:2310.03154](#) | [DOI:10.1103/PhysRevD.109.112009](#) | 2023-10-04

274. Search for a new resonance decaying into two spin-0 bosons in a final state with two photons and two bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2024\) 316](#) | [arXiv:2310.01643](#) | [DOI:10.1007/JHEP05\(2024\)316](#) | 2023-10-02

275. Search for supersymmetry in final states with disappearing tracks in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 072007](#) | [arXiv:2309.16823](#) | [DOI:10.1103/PhysRevD.109.072007](#) | 2023-09-28

276. Search for direct production of GeV-scale resonances decaying to a pair of muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 070](#) | [arXiv:2309.16003](#) | [DOI:10.1007/JHEP12\(2023\)070](#) | 2023-09-27

277. Inclusive and differential cross section measurements of $t\bar{t}b\bar{b}$ production in the lepton+jets channel at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 042](#) | [arXiv:2309.14442](#) | [DOI:10.1007/JHEP05\(2024\)042](#) | 2023-09-25

278. Measurement of the τ lepton polarization in Z boson decays in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 101](#) | [arXiv:2309.12408](#) | [DOI:10.1007/JHEP01\(2024\)101](#) | 2023-09-21

279. Development of the CMS detector for the CERN LHC Run 3

Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P05064](#) | [arXiv:2309.05466](#) | [DOI:10.1088/1748-0221/19/05/P05064](#) | 2023-09-11

280. Evidence for the Higgs Boson Decay to a Z Boson and a Photon at the LHC

Aad, Georges et al.

[Phys.Rev.Lett. 132 \(2024\) 021803](#) | [arXiv:2309.03501](#) | [DOI:10.1103/PhysRevLett.132.021803](#) | 2023-09-07

281. Luminosity determination using Z boson production at the CMS experiment

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 26](#) | [arXiv:2309.01008](#) | [DOI:10.1140/epjc/s10052-023-12268-2](#) | 2023-09-02

282. Search for a third-generation leptoquark coupled to a τ lepton and a b quark through single, pair, and nonresonant production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 311](#) | [arXiv:2308.07826](#) | [DOI:10.1007/JHEP05\(2024\)311](#) | 2023-08-15

283. Search for Scalar Leptoquarks Produced via τ -Lepton-Quark Scattering in pp Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 061801](#) | [arXiv:2308.06143](#) | [DOI:10.1103/PhysRevLett.132.061801](#) | 2023-08-11

284. Measurement of the production cross section for a W boson in association with a charm quark in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Tumasyan, Armen et al.

[Eur.Phys.J.C 84 \(2024\) 27](#) | [arXiv:2308.02285](#) | [DOI:10.1140/epjc/s10052-023-12258-4](#) | 2023-08-04

285. Measurement of the Higgs boson production via vector boson fusion and its decay into bottom quarks in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 173](#) | [arXiv:2308.01253](#) | [DOI:10.1007/JHEP01\(2024\)173](#) | 2023-08-02

286. Search for new physics in multijet events with at least one photon and large missing transverse momentum in proton-proton collisions at 13 TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 046](#) | [arXiv:2307.16216](#) | [DOI:10.1007/JHEP10\(2023\)046](#) | 2023-07-30

287. Search for physics beyond the standard model in top quark production with additional leptons in the context of effective field theory

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 068](#) | [arXiv:2307.15761](#) | [DOI:10.1007/JHEP12\(2023\)068](#) | 2023-07-28

288. Study of charm hadronization with prompt Λ_c^+ baryons in proton-proton and lead-lead collisions at $\sqrt{s_{NN}} = 5.02 \text{ TeV}$

Tumasyan, Armen et al.

[JHEP 01 \(2024\) 128](#) | [arXiv:2307.11186](#) | [DOI:10.1007/JHEP01\(2024\)128](#) | 2023-07-20

289. Multiplicity and transverse momentum dependence of charge-balance functions in pPb and PbPb collisions at LHC energies

Tumasyan, Armen et al.

[JHEP 08 \(2024\) 148](#) | [arXiv:2307.11185](#) | [DOI:10.1007/JHEP08\(2024\)148](#) | 2023-07-20

290. Search for a high-mass dimuon resonance produced in association with b quark jets at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 043](#) | [arXiv:2307.08708](#) | [DOI:10.1007/JHEP10\(2023\)043](#) | 2023-07-14

291. Search for Z' bosons decaying to pairs of heavy Majorana neutrinos in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Tumasyan, Armen et al.

[JHEP 11 \(2023\) 181](#) | [arXiv:2307.06959](#) | [DOI:10.1007/JHEP11\(2023\)181](#) | 2023-07-12

292. Two-particle Bose-Einstein correlations and their Lévy parameters in PbPb collisions at $\sqrt{s_{NN}}=5.02 \text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.C 109 \(2024\) 024914](#) | [arXiv:2306.11574](#) | [DOI:10.1103/PhysRevC.109.024914](#) | 2023-06-20

293. Performance of the local reconstruction algorithms for the CMS hadron calorimeter with Run 2 data

Tumasyan, Armen et al.

[JINST 18 \(2023\) P11017](#) | [arXiv:2306.10355](#) | [DOI:10.1088/1748-0221/18/11/P11017](#) | 2023-06-17

294. New Structures in the $J/\psi/\psi$ Mass Spectrum in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$
Hayrapetyan, Aram et al.
[Phys.Rev.Lett. 132 \(2024\) 111901](#) | [arXiv:2306.07164](#) | [DOI:10.1103/PhysRevLett.132.111901](#) | 2023-06-12
295. Search for the lepton-flavor violating decay of the Higgs boson and additional Higgs bosons in the $e\mu$ final state in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$
Hayrapetyan, Aram et al.
[Phys.Rev.D 108 \(2023\) 072004](#) | [arXiv:2305.18106](#) | [DOI:10.1103/PhysRevD.108.072004](#) | 2023-05-29
296. Measurements of the azimuthal anisotropy of prompt and nonprompt charmonia in PbPb collisions at $\sqrt{s_{\text{NN}}} = 5.02 \text{ TeV}$
Tumasyan, Armen et al.
[JHEP 10 \(2023\) 115](#) | [arXiv:2305.16928](#) | [DOI:10.1007/JHEP10\(2023\)115](#) | 2023-05-26
297. Observation of four top quark production in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$
Hayrapetyan, Aram et al.
[Phys.Lett.B 847 \(2023\) 138290](#) | [arXiv:2305.13439](#) | [DOI:10.1016/j.physletb.2023.138290](#) | 2023-05-22
298. Search for Inelastic Dark Matter in Events with Two Displaced Muons and Missing Transverse Momentum in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$
Hayrapetyan, Aram et al.
[Phys.Rev.Lett. 132 \(2024\) 041802](#) | [arXiv:2305.11649](#) | [DOI:10.1103/PhysRevLett.132.041802](#) | 2023-05-19
299. Search for resonances in events with photon and jet final states in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$
Tumasyan, Armen et al.
[JHEP 12 \(2023\) 189](#) | [arXiv:2305.07998](#) | [DOI:10.1007/JHEP12\(2023\)189](#) | 2023-05-13
300. Measurements of inclusive and differential cross sections for the Higgs boson production and decay to four-leptons in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$
Hayrapetyan, Aram et al.
[JHEP 08 \(2023\) 040](#) | [arXiv:2305.07532](#) | [DOI:10.1007/JHEP08\(2023\)040](#) | 2023-05-12
301. Observation of the rare decay of the η meson to four muons
Hayrapetyan, Aram et al.
[Phys.Rev.Lett. 131 \(2023\) 091903](#) | [arXiv:2305.04904](#) | [DOI:10.1103/PhysRevLett.131.091903](#) | 2023-05-08
302. Search for top squark pair production in a final state with at least one hadronically decaying tau lepton in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$
Tumasyan, Armen et al.
[JHEP 07 \(2023\) 110](#) | [arXiv:2304.07174](#) | [DOI:10.1007/JHEP07\(2023\)110](#) | 2023-04-14
303. Probing Small Bjorken- x Nuclear Gluonic Structure via Coherent J/ψ Photoproduction in Ultraperipheral Pb-Pb Collisions at $\sqrt{s_{\text{NN}}}=5.02\text{TeV}$
Tumasyan, Armen et al.
[Phys.Rev.Lett. 131 \(2023\) 262301](#) | [arXiv:2303.16984](#) | [DOI:10.1103/PhysRevLett.131.262301](#) | 2023-03-29
304. Observation of the $\chi(3S)$ Meson and Suppression of χ States in Pb-Pb Collisions at $\sqrt{s_{\text{NN}}}=5.02\text{TeV}$
Tumasyan, Armen et al.
[Phys.Rev.Lett. 133 \(2024\) 022302](#) | [arXiv:2303.17026](#) | [DOI:10.1103/PhysRevLett.133.022302](#) | 2023-03-29
305. First measurement of the top quark pair production cross section in proton-proton collisions at $\sqrt{s} = 13.6 \text{ TeV}$
Tumasyan, Armen et al.
[JHEP 08 \(2023\) 204](#) | [arXiv:2303.10680](#) | [DOI:10.1007/JHEP08\(2023\)204](#) | 2023-03-19

306. A search for new physics in central exclusive production using the missing mass technique with the CMS detector and the CMS-TOTEM precision proton spectrometer
Tumasyan, Armen et al.
[Eur.Phys.J.C 83 \(2023\) 827](#) | [arXiv:2303.04596](#) | [DOI:10.1140/epjc/s10052-023-11687-5](#) | 2023-03-08
307. Evidence for Four-Top Quark Production in Proton-Proton Collisions at $\sqrt{s}=13$ TeV
Tumasyan, Armen et al.
[Phys.Lett.B 844 \(2023\) 138076](#) | [arXiv:2303.03864](#) | [DOI:10.1016/j.physletb.2023.138076](#) | 2023-03-07
308. A search for decays of the Higgs boson to invisible particles in events with a top-antitop quark pair or a vector boson in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, Armen et al.
[Eur.Phys.J.C 83 \(2023\) 933](#) | [arXiv:2303.01214](#) | [DOI:10.1140/epjc/s10052-023-11952-7](#) | 2023-03-02
309. Search for dark matter particles in W^+W^- events with transverse momentum imbalance in proton-proton collisions at $\sqrt{s} = 13$ TeV
Hayrapetyan, Aram et al.
[JHEP 03 \(2024\) 134](#) | [arXiv:2310.12229](#) | [DOI:10.1007/JHEP03\(2024\)134](#) | 2023-03
310. Search for a vector-like quark $T' \rightarrow tH$ via the diphoton decay mode of the Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, Armen et al.
[JHEP 09 \(2023\) 057](#) | [arXiv:2302.12802](#) | [DOI:10.1007/JHEP09\(2023\)057](#) | 2023-02-24
311. Measurement of the top quark mass using a profile likelihood approach with the lepton + jets final states in proton-proton collisions at $\sqrt{s}=13$ TeV
Tumasyan, Armen et al.
[Eur.Phys.J.C 83 \(2023\) 963](#) | [arXiv:2302.01967](#) | [DOI:10.1140/epjc/s10052-023-12050-4](#) | 2023-02-03
312. Search for top squarks in the four-body decay mode with single lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, Armen et al.
[JHEP 06 \(2023\) 060](#) | [arXiv:2301.08096](#) | [DOI:10.1007/JHEP06\(2023\)060](#) | 2023-01-19
313. First Measurement of the Forward Rapidity Gap Distribution in pPb Collisions at $\sqrt{s_{NN}}=8.16$ TeV
Tumasyan, Armen et al.
[Phys.Rev.D 108 \(2023\) 092004](#) | [arXiv:2301.07630](#) | [DOI:10.1103/PhysRevD.108.092004](#) | 2023-01-18
314. K_S^0 and $\Lambda(\Lambda^0)$ two-particle femtoscopic correlations in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV
Tumasyan, Armen et al.
[Phys.Lett.B 857 \(2024\) 138936](#) | [arXiv:2301.05290](#) | [DOI:10.1016/j.physletb.2024.138936](#) | 2023-01-12
315. Search for new physics in the τ lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, A. et al.
[JHEP 09 \(2023\) 051](#) | [arXiv:2212.12604](#) | [DOI:10.1007/JHEP09\(2023\)051](#) | 2022-12-23
316. Measurement of the electroweak production of $W\gamma$ in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, Armen et al.
[Phys.Rev.D 108 \(2023\) 032017](#) | [arXiv:2212.12592](#) | [DOI:10.1103/PhysRevD.108.032017](#) | 2022-12-23
317. Measurement of the $B^0_S \rightarrow \mu^+ \mu^-$ decay properties and search for the $B^0 \rightarrow \mu^+ \mu^-$ decay in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137955](#) | [arXiv:2212.10311](#) | [DOI:10.1016/j.physletb.2023.137955](#) | 2022-12-20

318. Search for long-lived particles using out-of-time trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 210](#) | [arXiv:2212.06695](#) | [DOI:10.1007/JHEP07\(2023\)210](#) | 2022-12-13

319. Measurement of the dependence of the hadron production fraction ratio f_s / f_u and f_d / f_u on B meson kinematic variables in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 121901](#) | [arXiv:2212.02309](#) | [DOI:10.1103/PhysRevLett.131.121901](#) | 2022-12-05

320. Measurements of azimuthal anisotropy of nonprompt D0 mesons in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 850 \(2024\) 138389](#) | [arXiv:2212.01636](#) | [DOI:10.1016/j.physletb.2023.138389](#) | 2022-12-03

321. Search for high-mass exclusive $\gamma\gamma \rightarrow WW$ and $\gamma\gamma \rightarrow ZZ$ production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 229](#) | [arXiv:2211.16320](#) | [DOI:10.1007/JHEP07\(2023\)229](#) | 2022-11-29

322. Search for Higgs Boson and Observation of Z Boson through their Decay into a Charm Quark-Antiquark Pair in Boosted Topologies in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041801](#) | [arXiv:2211.14181](#) | [DOI:10.1103/PhysRevLett.131.041801](#) | 2022-11-25

323. Search for supersymmetry in final states with a single electron or muon using angular correlations and heavy-object identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 149](#) | [arXiv:2211.08476](#) | [DOI:10.1007/JHEP09\(2023\)149](#) | 2022-11-15

324. Measurement of the differential tt production cross section as a function of the jet mass and extraction of the top quark mass in hadronic decays of boosted top quarks

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 560](#) | [arXiv:2211.01456](#) | [DOI:10.1140/epjc/s10052-023-11587-8](#) | 2022-11-02

325. Azimuthal correlations in Z +jets events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 722](#) | [arXiv:2210.16139](#) | [DOI:10.1140/epjc/s10052-023-11833-z](#) | 2022-10-28

326. Measurements of jet multiplicity and jet transverse momentum in multijet events in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 742](#) | [arXiv:2210.13557](#) | [DOI:10.1140/epjc/s10052-023-11753-y](#) | 2022-10-24

327. Search for medium effects using jets from bottom quarks in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137849](#) | [arXiv:2210.08547](#) | [DOI:10.1016/j.physletb.2023.137849](#) | 2022-10-16

328. Azimuthal anisotropy of dijet events in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 139](#) | [arXiv:2210.08325](#) | [DOI:10.1007/JHEP07\(2023\)139](#) | 2022-10-15

329. Proton reconstruction with the CMS-TOTEM Precision Proton Spectrometer

Tumasyan, Armen et al.

[JINST 18 \(2023\) P09009](#) | [arXiv:2210.05854](#) | [DOI:10.1088/1748-0221/18/09/P09009](#) | 2022-10-11

330. Search for a heavy composite Majorana neutrino in events with dilepton signatures from proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 843 \(2023\) 137803](#) | [arXiv:2210.03082](#) | [DOI:10.1016/j.physletb.2023.137803](#) | 2022-10-06

331. Search for new heavy resonances decaying to WW, WZ, ZZ, WH, or ZH boson pairs in the all-jets final state in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137813](#) | [arXiv:2210.00043](#) | [DOI:10.1016/j.physletb.2023.137813](#) | 2022-09-30

332. Search for pair production of vector-like quarks in leptonic final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 020](#) | [arXiv:2209.07327](#) | [DOI:10.1007/JHEP07\(2023\)020](#) | 2022-09-15

333. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A} \rightarrow 4\gamma$ with events containing two merged diphotons in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 101801](#) | [arXiv:2209.06197](#) | [DOI:10.1103/PhysRevLett.131.101801](#) | 2022-09-13

334. Search for new physics using effective field theory in 13 TeV pp collision events that contain a top quark pair and a boosted Z or Higgs boson

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032008](#) | [arXiv:2208.12837](#) | [DOI:10.1103/PhysRevD.108.032008](#) | 2022-08-26

335. Measurement of the Higgs boson inclusive and differential fiducial production cross sections in the diphoton decay channel with pp collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 091](#) | [arXiv:2208.12279](#) | [DOI:10.1007/JHEP07\(2023\)091](#) | 2022-08-25

336. Search for pair-produced vector-like leptons in final states with third-generation leptons and at least three b quark jets in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137713](#) | [arXiv:2208.09700](#) | [DOI:10.1016/j.physletb.2023.137713](#) | 2022-08-20

337. Measurement of the cross section of top quark-antiquark pair production in association with a W boson in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 219](#) | [arXiv:2208.06485](#) | [DOI:10.1007/JHEP07\(2023\)219](#) | 2022-08-12

338. Search for CP violation in ttH and tH production in multilepton channels in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 092](#) | [arXiv:2208.02686](#) | [DOI:10.1007/JHEP07\(2023\)092](#) | 2022-08-04

339. Measurement of the $t\bar{t}$ charge asymmetry in events with highly Lorentz-boosted top quarks in pp collisions at $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137703](#) | [arXiv:2208.02751](#) | [DOI:10.1016/j.physletb.2023.137703](#) | 2022-08-04

340. Searches for additional Higgs bosons and for vector leptoquarks in $\tau\tau$ final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 073](#) | [arXiv:2208.02717](#) | [DOI:10.1007/JHEP07\(2023\)073](#) | 2022-08-04

341. Search for the exotic decay of the Higgs boson into two light pseudoscalars with four photons in the final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 148](#) | [arXiv:2208.01469](#) | [DOI:10.1007/JHEP07\(2023\)148](#) | 2022-08-02

342. Measurement of inclusive and differential cross sections for single top quark production in association with a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 046](#) | [arXiv:2208.00924](#) | [DOI:10.1007/JHEP07\(2023\)046](#) | 2022-08-01

343. Search for the Higgs boson decay to a pair of electrons in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137783](#) | [arXiv:2208.00265](#) | [DOI:10.1016/j.physletb.2023.137783](#) | 2022-07-30

344. Search for new physics in the lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 067](#) | [arXiv:2202.06075](#) | [DOI:10.1007/JHEP07\(2022\)067](#) | 2022-07-11

345. Measurement of the top quark pole mass using $t\bar{t}$ +jet events in the dilepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 077](#) | [arXiv:2207.02270](#) | [DOI:10.1007/JHEP07\(2023\)077](#) | 2022-07-05

346. Search for direct pair production of supersymmetric partners of τ leptons in the final state with two hadronically decaying τ leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 012011](#) | [arXiv:2207.02254](#) | [DOI:10.1103/PhysRevD.108.012011](#) | 2022-07-05

347. Search for a charged Higgs boson decaying into a heavy neutral Higgs boson and a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 032](#) | [arXiv:2207.01046](#) | [DOI:10.1007/JHEP09\(2023\)032](#) | 2022-07-03

348. Search for Higgs boson pairs decaying to WW^*WW^* , $WW^*\tau\tau$, and $\tau\tau\tau\tau$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 095](#) | [arXiv:2206.10268](#) | [DOI:10.1007/JHEP07\(2023\)095](#) | 2022-06-21

349. Search for nonresonant Higgs boson pair production in the four leptons plus two b jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 130](#) | [arXiv:2206.10657](#) | [DOI:10.1007/JHEP06\(2023\)130](#) | 2022-06-21

350. Search for nonresonant Higgs boson pair production in final state with two bottom quarks and two tau leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137531](#) | [arXiv:2206.09401](#) | [DOI:10.1016/j.physletb.2022.137531](#) | 2022-06-19

351. Measurements of the Higgs boson production cross section and couplings in the W boson pair decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 667](#) | [arXiv:2206.09466](#) | [DOI:10.1140/epjc/s10052-023-11632-6](#) | 2022-06-19

352. Probing Heavy Majorana Neutrinos and the Weinberg Operator through Vector Boson Fusion Processes in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 011803](#) | [arXiv:2206.08956](#) | [DOI:10.1103/PhysRevLett.131.011803](#) | 2022-06-17

353. Precision measurement of the Z boson invisible width in pp collisions at $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137563](#) | [arXiv:2206.07110](#) | [DOI:10.1016/j.physletb.2022.137563](#) | 2022-06-14

354. Observation of τ lepton pair production in ultraperipheral lead-lead collisions at $\sqrt{s_{\text{NN}}} = 5.02\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 151803](#) | [arXiv:2206.05192](#) | [DOI:10.1103/PhysRevLett.131.151803](#) | 2022-06-10

355. Search for Higgs boson decays into Z and J/ψ and for Higgs and Z boson decays into J/ψ or Y pairs in pp collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137534](#) | [arXiv:2206.03525](#) | [DOI:10.1016/j.physletb.2022.137534](#) | 2022-06-07

356. Observation of same-sign WW production from double parton scattering in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 091803](#) | [arXiv:2206.02681](#) | [DOI:10.1103/PhysRevLett.131.091803](#) | 2022-06-06

357. Combination of inclusive top-quark pair production cross-section measurements using ATLAS and CMS data at $\sqrt{s} = 7$ and 8 TeV

Aad, G. et al.

[JHEP 07 \(2023\) 213](#) | [arXiv:2205.13830](#) | [DOI:10.1007/JHEP07\(2023\)213](#) | 2022-05-27

358. Search for electroweak production of charginos and neutralinos at $\sqrt{s}=13\text{ TeV}$ in final states containing hadronic decays of WW, WZ, or WH and missing transverse momentum

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137460](#) | [arXiv:2205.09597](#) | [DOI:10.1016/j.physletb.2022.137460](#) | 2022-05-19

359. Search for long-lived particles decaying to a pair of muons in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 228](#) | [arXiv:2205.08582](#) | [DOI:10.1007/JHEP05\(2023\)228](#) | 2022-05-17

360. Search for CP violating top quark couplings in pp collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 023](#) | [arXiv:2205.07434](#) | [DOI:10.1007/JHEP07\(2023\)023](#) | 2022-05-15

361. Search for heavy resonances and quantum black holes in $e\mu$, $e\tau$, and $\mu\tau$ final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 227](#) | [arXiv:2205.06709](#) | [DOI:10.1007/JHEP05\(2023\)227](#) | 2022-05-13

362. Search for Nonresonant Pair Production of Highly Energetic Higgs Bosons Decaying to Bottom Quarks

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041803](#) | [arXiv:2205.06667](#) | [DOI:10.1103/PhysRevLett.131.041803](#) | 2022-05-13

363. Search for Higgs Boson Decay to a Charm Quark-Antiquark Pair in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 061801](#) | [arXiv:2205.05550](#) | [DOI:10.1103/PhysRevLett.131.061801](#) | 2022-05-11

364. Observation of electroweak $W+W^-$ pair production in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 841 \(2023\) 137495](#) | [arXiv:2205.05711](#) | [DOI:10.1016/j.physletb.2022.137495](#) | 2022-05-11

365. Measurement of the mass dependence of the transverse momentum of lepton pairs in Drell-Yan production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 628](#) | [arXiv:2205.04897](#) | [DOI:10.1140/epjc/s10052-023-11631-7](#) | 2022-05-10

366. Constraints on anomalous Higgs boson couplings to vector bosons and fermions from the production of Higgs bosons using the $\tau\tau$ final state

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032013](#) | [arXiv:2205.05120](#) | [DOI:10.1103/PhysRevD.108.032013](#) | 2022-05-10

367. CMS pythia 8 colour reconnection tunes based on underlying-event data

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 587](#) | [arXiv:2205.02905](#) | [DOI:10.1140/epjc/s10052-023-11630-8](#) | 2022-05-05

368. Measurement of differential cross sections for the production of a Z boson in association with jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 052004](#) | [arXiv:2205.02872](#) | [DOI:10.1103/PhysRevD.108.052004](#) | 2022-05-05

369. Search for CP violation using $t\bar{t}$ events in the lepton+jets channel in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 081](#) | [arXiv:2205.02314](#) | [DOI:10.1007/JHEP06\(2023\)081](#) | 2022-05-04

370. Search for narrow resonances in the b-tagged dijet mass spectrum in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 012009](#) | [arXiv:2205.01835](#) | [DOI:10.1103/PhysRevD.108.012009](#) | 2022-05-03

371. Strange hadron collectivity in pPb and PbPb collisions

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 007](#) | [arXiv:2205.00080](#) | [DOI:10.1007/JHEP05\(2023\)007](#) | 2022-04-29

372. Azimuthal Correlations within Exclusive Dijets with Large Momentum Transfer in Photon-Lead Collisions

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 051901](#) | [arXiv:2205.00045](#) | [DOI:10.1103/PhysRevLett.131.051901](#) | 2022-04-29

373. Two-particle azimuthal correlations in γp interactions using pPb collisions at $\sqrt{s_{NN}}=8.16$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137905](#) | [arXiv:2204.13486](#) | [DOI:10.1016/j.physletb.2023.137905](#) | 2022-04-28

374. Search for light Higgs bosons from supersymmetric cascade decays in pp collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 571](#) | [arXiv:2204.13532](#) | [DOI:10.1140/epjc/s10052-023-11581-0](#) | 2022-04-28

375. Search for Higgs boson decays to a Z boson and a photon in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 233](#) | [arXiv:2204.12945](#) | [DOI:10.1007/JHEP05\(2023\)233](#) | 2022-04-27

376. Measurements of Higgs boson production in the decay channel with a pair of τ leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 562](#) | [arXiv:2204.12957](#) | [DOI:10.1140/epjc/s10052-023-11452-8](#) | 2022-04-27

377. Search for a massive scalar resonance decaying to a light scalar and a Higgs boson in the four b quarks final state with boosted topology

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137392](#) | [arXiv:2204.12413](#) | [DOI:10.1016/j.physletb.2022.137392](#) | 2022-04-26

378. Reconstruction of decays to merged photons using end-to-end deep learning with domain continuation in the CMS detector

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 052002](#) | [arXiv:2204.12313](#) | [DOI:10.1103/PhysRevD.108.052002](#) | 2022-04-26

379. Search for new particles in an extended Higgs sector with four b quarks in the final state at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 835 \(2022\) 137566](#) | [arXiv:2203.00480](#) | [DOI:10.1016/j.physletb.2022.137566](#) | 2022-03-01

380. Search for a W' boson decaying to a vector-like quark and a top or bottom quark in the all-jets final state at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2022\) 088](#) | [arXiv:2202.12988](#) | [DOI:10.1007/JHEP09\(2022\)088](#) | 2022-02-25

381. Measurement of the Drell-Yan forward-backward asymmetry at high dilepton masses in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 2022 \(2022\) 063](#) | [arXiv:2202.12327](#) | [DOI:10.1007/JHEP08\(2022\)063](#) | 2022-02-24

382. Nuclear modification of Υ states in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 835 \(2022\) 137397](#) | [arXiv:2202.11807](#) | [DOI:10.1016/j.physletb.2022.137397](#) | 2022-02-23

383. Search for Higgs Boson Pair Production in the Four b Quark Final State in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 081802](#) | [arXiv:2202.09617](#) | [DOI:10.1103/PhysRevLett.129.081802](#) | 2022-02-19

384. Inclusive nonresonant multilepton probes of new phenomena at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 112007](#) | [arXiv:2202.08676](#) | [DOI:10.1103/PhysRevD.105.112007](#) | 2022-02-17

385. Measurement of the Higgs boson width and evidence of its off-shell contributions to ZZ production

Tumasyan, Armen et al.

[Nature Phys. 18 \(2022\) 1329](#) | [arXiv:2202.06923](#) | [DOI:10.1038/s41567-022-01682-0](#) | 2022-02-14

386. Search for invisible decays of the Higgs boson produced via vector boson fusion in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 092007](#) | [arXiv:2201.11585](#) | [DOI:10.1103/PhysRevD.105.092007](#) | 2022-01-27

387. Observation of $B^0 \rightarrow \psi(2S) K^0_S \pi^+ \pi^-$ and $B^0_S \rightarrow \psi(2S) K^0_S$ decays

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 499](#) | [arXiv:2201.09131](#) | [DOI:10.1140/epjc/s10052-022-10315-y](#) | 2022-01-22

388. Search for resonances decaying to three W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 021802](#) | [arXiv:2201.08476](#) | [DOI:10.1103/PhysRevLett.129.021802](#) | 2022-01-20

389. Identification of hadronic tau lepton decays using a deep neural network

Tumasyan, Armen et al.

[JINST 17 \(2022\) P07023](#) | [arXiv:2201.08458](#) | [DOI:10.1088/1748-0221/17/07/P07023](#) | 2022-01-20

390. Precision measurement of the W boson decay branching fractions in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 072008](#) | [arXiv:2201.07861](#) | [DOI:10.1103/PhysRevD.105.072008](#) | 2022-01-19

391. Search for charged-lepton flavor violation in top quark production and decay in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 082](#) | [arXiv:2201.07859](#) | [DOI:10.1007/JHEP06\(2022\)082](#) | 2022-01-19

392. Measurement of the inclusive and differential $t\bar{t}\gamma$ cross sections in the dilepton channel and effective field theory interpretation in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 091](#) | [arXiv:2201.07301](#) | [DOI:10.1007/JHEP05\(2022\)091](#) | 2022-01-18

393. Search for long-lived heavy neutral leptons with displaced vertices in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 081](#) | [arXiv:2201.05578](#) | [DOI:10.1007/JHEP07\(2022\)081](#) | 2022-01-14

394. Search for higgsinos decaying to two Higgs bosons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 014](#) | [arXiv:2201.04206](#) | [DOI:10.1007/JHEP05\(2022\)014](#) | 2022-01-11

395. Observation of the B_c^+ Meson in Pb-Pb and pp Collisions at $\sqrt{s_{NN}} = 5.02$ TeV and Measurement of its Nuclear Modification Factor

Tumasyan, Armen et al.

[Phys.Rev.Lett. 128 \(2022\) 252301](#) | [arXiv:2201.02659](#) | [DOI:10.1103/PhysRevLett.128.252301](#) | 2022-01-07

396. Search for high-mass resonances decaying to a jet and a Lorentz-boosted resonance in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 832 \(2022\) 137263](#) | [arXiv:2201.02140](#) | [DOI:10.1016/j.physletb.2022.137263](#) | 2022-01-06

397. Search for single production of a vector-like T quark decaying to a top quark and a Z boson in the final state with jets and missing transverse momentum at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 093](#) | [arXiv:2201.02227](#) | [DOI:10.1007/JHEP05\(2022\)093](#) | 2022-01-06

398. Search for long-lived particles decaying into muon pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV collected with a dedicated high-rate data stream

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 062](#) | [arXiv:2112.13769](#) | [DOI:10.1007/JHEP04\(2022\)062](#) | 2021-12-27

399. Search for resonances decaying to three W bosons in the hadronic final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 106 \(2022\) 012002](#) | [arXiv:2112.13090](#) | [DOI:10.1103/PhysRevD.106.012002](#) | 2021-12-24

400. Probing Charm Quark Dynamics via Multiparticle Correlations in Pb-Pb Collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 022001](#) | [arXiv:2112.12236](#) | [DOI:10.1103/PhysRevLett.129.022001](#) | 2021-12-22

401. Search for resonant production of strongly coupled dark matter in proton-proton collisions at 13 TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 156](#) | [arXiv:2112.11125](#) | [DOI:10.1007/JHEP06\(2022\)156](#) | 2021-12-21

402. Search for flavor-changing neutral current interactions of the top quark and the Higgs boson decaying to a bottom quark-antiquark pair at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 169](#) | [arXiv:2112.09734](#) | [DOI:10.1007/JHEP02\(2022\)169](#) | 2021-12-17

403. Measurement of the production cross section for Z+b jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 092014](#) | [arXiv:2112.09659](#) | [DOI:10.1103/PhysRevD.105.092014](#) | 2021-12-17

404. Measurement of the inclusive tt production cross section in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 144](#) | [arXiv:2112.09114](#) | [DOI:10.1007/JHEP04\(2022\)144](#) | 2021-12-16

405. Evidence for WW/WZ vector boson scattering in the decay channel $\ell\nu qq$ produced in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 834 \(2022\) 137438](#) | [arXiv:2112.05259](#) | [DOI:10.1016/j.physletb.2022.137438](#) | 2021-12-09

406. Search for a right-handed W boson and a heavy neutrino in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 047](#) | [arXiv:2112.03949](#) | [DOI:10.1007/JHEP04\(2022\)047](#) | 2021-12-07

407. Search for heavy resonances decaying to a pair of Lorentz-boosted Higgs bosons in final states with leptons and a bottom quark pair at $\sqrt{s}= 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 005](#) | [arXiv:2112.03161](#) | [DOI:10.1007/JHEP05\(2022\)005](#) | 2021-12-06

408. Measurements of the associated production of a W boson and a charm quark in proton-proton collisions at $\sqrt{s}=8$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 1094](#) | [arXiv:2112.00895](#) | [DOI:10.1140/epjc/s10052-022-10897-7](#) | 2021-12-01

409. Measurement of $W^{\pm}\gamma$ differential cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV and effective field theory constraints

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 052003](#) | [arXiv:2111.13948](#) | [DOI:10.1103/PhysRevD.105.052003](#) | 2021-11-27

410. Search for heavy resonances decaying to ZZ or ZW and axion-like particles mediating nonresonant ZZ or ZH production at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 087](#) | [arXiv:2111.13669](#) | [DOI:10.1007/JHEP04\(2022\)087](#) | 2021-11-26

411. Search for a heavy resonance decaying into a top quark and a W boson in the lepton+jets final state at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[JHEP 04 \(2022\) 048](#) | [arXiv:2111.10216](#) | [DOI:10.1007/JHEP04\(2022\)048](#) | 2021-11-19

412. Measurement and QCD analysis of double-differential inclusive jet cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 142](#) | [arXiv:2111.10431](#) | [DOI:10.1007/JHEP02\(2022\)142](#) | 2021-11-19

413. Strategies and performance of the CMS silicon tracker alignment during LHC Run 2

Tumasyan, Armen et al.

[Nucl.Instrum.Meth.A 1037 \(2022\) 166795](#) | [arXiv:2111.08757](#) | [DOI:10.1016/j.nima.2022.166795](#) | 2021-11-16

414. Search for supersymmetry in final states with two or three soft leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 091](#) | [arXiv:2111.06296](#) | [DOI:10.1007/JHEP04\(2022\)091](#) | 2021-11-11

415. Observation of triple J/ψ meson production in proton-proton collisions

Tumasyan, Armen et al.

[Nature Phys. 19 \(2023\) 338](#) | [arXiv:2111.05370](#) | [DOI:10.1038/s41567-022-01838-y](#) | 2021-11-09

416. Study of dijet events with large rapidity separation in proton-proton collisions at $\sqrt{s} = 2.76$ TeV

Tumasyan, Armen et al.

[JHEP 03 \(2022\) 189](#) | [arXiv:2111.04605](#) | [DOI:10.1007/JHEP03\(2022\)189](#) | 2021-11-08

417. A new calibration method for charm jet identification validated with proton-proton collision events at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JINST 17 \(2022\) P03014](#) | [arXiv:2111.03027](#) | [DOI:10.1088/1748-0221/17/03/P03014](#) | 2021-11-04

418. Inclusive and differential cross section measurements of single top quark production in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 107](#) | [arXiv:2111.02860](#) | [DOI:10.1007/JHEP02\(2022\)107](#) | 2021-11-04

419. Search for Flavor-Changing Neutral Current Interactions of the Top Quark and Higgs Boson in Final States with Two Photons in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[Phys.Rev.Lett. 129 \(2022\) 032001](#) | [arXiv:2111.02219](#) | [DOI:10.1103/PhysRevLett.129.032001](#) | 2021-11-03

420. Search for low-mass dilepton resonances in Higgs boson decays to four-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[Eur.Phys.J.C 82 \(2022\) 290](#) | [arXiv:2111.01299](#) | [DOI:10.1140/epjc/s10052-022-10127-0](#) | 2021-11-01

421. Search for long-lived particles produced in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 03 \(2022\) 160](#) | [arXiv:2110.13218](#) | [DOI:10.1007/JHEP03\(2022\)160](#) | 2021-10-25

422. Measurement of the inclusive and differential WZ production cross sections, polarization angles, and triple gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 032](#) | [arXiv:2110.11231](#) | [DOI:10.1007/JHEP07\(2022\)032](#) | 2021-10-21

423. First Search for Exclusive Diphoton Production at High Mass with Tagged Protons in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 011801](#) | [arXiv:2110.05916](#) | [DOI:10.1103/PhysRevLett.129.011801](#) | 2021-10-12

424. Search for long-lived particles decaying to leptons with large impact parameter in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 153](#) | [arXiv:2110.04809](#) | [DOI:10.1140/epjc/s10052-022-10027-3](#) | 2021-10-10

425. Analysis of the CP structure of the Yukawa coupling between the Higgs boson and τ leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 012](#) | [arXiv:2110.04836](#) | [DOI:10.1007/JHEP06\(2022\)012](#) | 2021-10-10

426. Measurement of double-parton scattering in inclusive production of four jets with low transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 177](#) | [arXiv:2109.13822](#) | [DOI:10.1007/JHEP01\(2022\)177](#) | 2021-09-28

427. Search for heavy resonances decaying to $Z(\nu\bar{\nu})V(q\bar{q}')$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 106 \(2022\) 012004](#) | [arXiv:2109.08268](#) | [DOI:10.1103/PhysRevD.106.012004](#) | 2021-09-17

428. Search for heavy resonances decaying to WW, WZ, or WH boson pairs in the lepton plus merged jet final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 032008](#) | [arXiv:2109.06055](#) | [DOI:10.1103/PhysRevD.105.032008](#) | 2021-09-13

429. Study of quark and gluon jet substructure in Z+jet and dijet events from pp collisions

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 188](#) | [arXiv:2109.03340](#) | [DOI:10.1007/JHEP01\(2022\)188](#) | 2021-09-07

430. Observation of B^0_s mesons and measurement of the B^0_s/B^+ yield ratio in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 829 \(2022\) 137062](#) | [arXiv:2109.01908](#) | [DOI:10.1016/j.physletb.2022.137062](#) | 2021-09-04

431. Observation of tW production in the single-lepton channel in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 111](#) | [arXiv:2109.01706](#) | [DOI:10.1007/JHEP11\(2021\)111](#) | 2021-09-03

432. Measurement of the top quark mass using events with a single reconstructed top quark in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 161](#) | [arXiv:2108.10407](#) | [DOI:10.1007/JHEP12\(2021\)161](#) | 2021-08-23

433. Measurement of differential $t\bar{t}$ production cross sections in the full kinematic range using lepton+jets events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 092013](#) | [arXiv:2108.02803](#) | [DOI:10.1103/PhysRevD.104.092013](#) | 2021-08-05

434. Probing effective field theory operators in the associated production of top quarks with a Z boson in multilepton final states at $\sqrt{s} = 13$ TeV

Lee, Kyeongpil et al.

[JHEP 12 \(2021\) 083](#) | [arXiv:2107.13896](#) | [DOI:10.1007/JHEP12\(2021\)083](#) | 2021-07-29

435. Search for new particles in events with energetic jets and large missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 153](#) | [arXiv:2107.13021](#) | [DOI:10.1007/JHEP11\(2021\)153](#) | 2021-07-27

436. Search for chargino-neutralino production in events with Higgs and W bosons using 137 fb^{-1} of proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 045](#) | [arXiv:2107.12553](#) | [DOI:10.1007/JHEP10\(2021\)045](#) | 2021-07-27

437. Measurement of the inclusive and differential Higgs boson production cross sections in the decay mode to a pair of τ leptons in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 128 \(2022\) 081805](#) | [arXiv:2107.11486](#) | [DOI:10.1103/PhysRevLett.128.081805](#) | 2021-07-23

438. Combined searches for the production of supersymmetric top quark partners in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 81 \(2021\) 970](#) | [arXiv:2107.10892](#) | [DOI:10.1140/epjc/s10052-021-09721-5](#) | 2021-07-22

439. Search for Long-Lived Particles Decaying in the CMS End Cap Muon Detectors in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 261804](#) | [arXiv:2107.04838](#) | [DOI:10.1103/PhysRevLett.127.261804](#) | 2021-07-10

440. Measurement of the inclusive and differential tt γ cross sections in the single-lepton channel and EFT interpretation at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 180](#) | [arXiv:2107.01508](#) | [DOI:10.1007/JHEP12\(2021\)180](#) | 2021-07-03

441. Measurement of prompt open-charm production cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 225](#) | [arXiv:2107.01476](#) | [DOI:10.1007/JHEP11\(2021\)225](#) | 2021-07-03

442. Measurements of the electroweak diboson production cross sections in proton-proton collisions at $\sqrt{s} = 5.02$ TeV using leptonic decays

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 191801](#) | [arXiv:2107.01137](#) | [DOI:10.1103/PhysRevLett.127.191801](#) | 2021-07-02

443. Search for electroweak production of charginos and neutralinos in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 147](#) | [arXiv:2106.14246](#) | [DOI:10.1007/JHEP04\(2022\)147](#) | 2021-06-27

444. Fragmentation of jets containing a prompt J/ ψ meson in PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 825 \(2022\) 136842](#) | [arXiv:2106.13235](#) | [DOI:10.1016/j.physletb.2021.136842](#) | 2021-06-23

445. Measurement of the electroweak production of $Z\gamma$ and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 072001](#) | [arXiv:2106.11082](#) | [DOI:10.1103/PhysRevD.104.072001](#) | 2021-06-21

446. Search for $W\gamma$ resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV using hadronic decays of Lorentz-boosted W bosons

Tumasyan, Armen et al.

[Phys.Lett.B 826 \(2022\) 136888](#) | [arXiv:2106.10509](#) | [DOI:10.1016/j.physletb.2022.136888](#) | 2021-06-19

447. Search for a heavy Higgs boson decaying into two lighter Higgs bosons in the $\tau\tau b\bar{b}$ final state at 13 TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 057](#) | [arXiv:2106.10361](#) | [DOI:10.1007/JHEP11\(2021\)057](#) | 2021-06-18

448. Study of Z boson plus jets events using variables sensitive to double-parton scattering in pp collisions at 13 TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 176](#) | [arXiv:2105.14511](#) | [DOI:10.1007/JHEP10\(2021\)176](#) | 2021-05-30

449. Measurements of the $pp \rightarrow W^\pm \gamma \gamma$ and $pp \rightarrow Z \gamma \gamma$ cross sections at $\sqrt{s} = 13$ TeV and limits on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 174](#) | [arXiv:2105.12780](#) | [DOI:10.1007/JHEP10\(2021\)174](#) | 2021-05-26

450. Search for strongly interacting massive particles generating trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 213](#) | [arXiv:2105.09178](#) | [DOI:10.1140/epjc/s10052-022-10095-5](#) | 2021-05-19

451. Search for lepton-flavor violating decays of the Higgs boson in the $\mu\tau$ and $e\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032013](#) | [arXiv:2105.03007](#) | [DOI:10.1103/PhysRevD.104.032013](#) | 2021-05-06

452. Search for long-lived particles decaying to jets with displaced vertices in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[Phys.Rev.D 104 \(2021\) 052011](#) | [arXiv:2104.13474](#) | [DOI:10.1103/PhysRevD.104.052011](#) | 2021-04-27

453. Search for a heavy resonance decaying to a top quark and a W boson at $\sqrt{s} = 13$ TeV in the fully hadronic final state

Sirunyan, Albert M et al.

[JHEP 12 \(2021\) 106](#) | [arXiv:2104.12853](#) | [DOI:10.1007/JHEP12\(2021\)106](#) | 2021-04-26

454. Constraints on anomalous Higgs boson couplings to vector bosons and fermions in its production and decay using the four-lepton final state

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 052004](#) | [arXiv:2104.12152](#) | [DOI:10.1103/PhysRevD.104.052004](#) | 2021-04-25

455. Search for W' bosons decaying to a top and a bottom quark at $\sqrt{s}=13$ TeV in the hadronic final state

Sirunyan, Albert M et al.

[Phys.Lett.B 820 \(2021\) 136535](#) | [arXiv:2104.04831](#) | [DOI:10.1016/j.physletb.2021.136535](#) | 2021-04-10

456. Search for charged Higgs bosons produced in vector boson fusion processes and decaying into vector boson pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 723](#) | [arXiv:2104.04762](#) | [DOI:10.1140/epjc/s10052-021-09472-3](#) | 2021-04-10

457. Precision luminosity measurement in proton-proton collisions at $\sqrt{s} = 13$ TeV in 2015 and 2016 at CMS

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 800](#) | [arXiv:2104.01927](#) | [DOI:10.1140/epjc/s10052-021-09538-2](#) | 2021-04-05

458. Constraints on the Initial State of Pb-Pb Collisions via Measurements of Z-Boson Yields and Azimuthal Anisotropy at $\sqrt{s_{\text{NN}}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 127 \(2021\) 102002](#) | [arXiv:2103.14089](#) | [DOI:10.1103/PhysRevLett.127.102002](#) | 2021-03-25

459. Measurements of Higgs boson production cross sections and couplings in the diphoton decay channel at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 027](#) | [arXiv:2103.06956](#) | [DOI:10.1007/JHEP07\(2021\)027](#) | 2021-03-11

460. Measurements of production cross sections of the Higgs boson in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 488](#) | [arXiv:2103.04956](#) | [DOI:10.1140/epjc/s10052-021-09200-x](#) | 2021-03-08

461. Using Z Boson Events to Study Parton-Medium Interactions in Pb-Pb Collisions

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 128 \(2022\) 122301](#) | [arXiv:2103.04377](#) | [DOI:10.1103/PhysRevLett.128.122301](#) | 2021-03-07

462. Search for resonant and nonresonant new phenomena in high-mass dilepton final states at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 208](#) | [arXiv:2103.02708](#) | [DOI:10.1007/JHEP07\(2021\)208](#) | 2021-03-03

463. Search for top squark production in fully-hadronic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 052001](#) | [arXiv:2103.01290](#) | [DOI:10.1103/PhysRevD.104.052001](#) | 2021-03-01

464. Study of Drell-Yan dimuon production in proton-lead collisions at $\sqrt{s_{\text{NN}}} = 8.16$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 182](#) | [arXiv:2102.13648](#) | [DOI:10.1007/JHEP05\(2021\)182](#) | 2021-02-26

465. First measurement of large area jet transverse momentum spectra in heavy-ion collisions

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 284](#) | [arXiv:2102.13080](#) | [DOI:10.1007/JHEP05\(2021\)284](#) | 2021-02-25

466. Evidence for X(3872) in Pb-Pb Collisions and Studies of its Prompt Production at $\sqrt{s_{\text{NN}}}=5.02$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 128 \(2022\) 032001](#) | [arXiv:2102.13048](#) | [DOI:10.1103/PhysRevLett.128.032001](#) | 2021-02-25

467. Measurements of angular distance and momentum ratio distributions in three-jet and Z + two-jet final states in pp collisions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 852](#) | [arXiv:2102.08816](#) | [DOI:10.1140/epjc/s10052-021-09570-2](#) | 2021-02-17

468. Search for a heavy vector resonance decaying to a Z^0 boson and a Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 688](#) | [arXiv:2102.08198](#) | [DOI:10.1140/epjc/s10052-021-09348-6](#) | 2021-02-16

469. Search for top squarks in final states with two top quarks and several light-flavor jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032006](#) | [arXiv:2102.06976](#) | [DOI:10.1103/PhysRevD.104.032006](#) | 2021-02-13

470. Hard color-singlet exchange in dijet events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032009](#) | [arXiv:2102.06945](#) | [DOI:10.1103/PhysRevD.104.032009](#) | 2021-02-13

471. Performance of the CMS muon trigger system in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JINST 16 \(2021\) P07001](#) | [arXiv:2102.04790](#) | [DOI:10.1088/1748-0221/16/07/P07001](#) | 2021-02-09

472. Observation of a New Excited Beauty Strange Baryon Decaying to $\Xi^- \pi^+ \pi^-$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 126 \(2021\) 252003](#) | [arXiv:2102.04524](#) | [DOI:10.1103/PhysRevLett.126.252003](#) | 2021-02-08

473. Measurement of the $W\gamma$ Production Cross Section in Proton-Proton Collisions at $\sqrt{s}=13$ TeV and Constraints on Effective Field Theory Coefficients

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 126 \(2021\) 252002](#) | [arXiv:2102.02283](#) | [DOI:10.1103/PhysRevLett.126.252002](#) | 2021-02-03

474. Measurements of the differential cross sections of the production of $Z + \text{jets}$ and $\gamma + \text{jets}$ and of Z boson emission collinear with a jet in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 285](#) | [arXiv:2102.02238](#) | [DOI:10.1007/JHEP05\(2021\)285](#) | 2021-02-03

475. In-medium modification of dijets in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 116](#) | [arXiv:2101.04720](#) | [DOI:10.1007/JHEP05\(2021\)116](#) | 2021-01-12

476. Measurement of the Z boson differential production cross section using its invisible decay mode ($Z\nu\bar{\nu}$) in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 205](#) | [arXiv:2012.09254](#) | [DOI:10.1007/JHEP05\(2021\)205](#) | 2020-12-16

477. First measurement of the cross section for top quark pair production with additional charm jets using dileptonic final states in pp collisions at $s=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 820 \(2021\) 136565](#) | [arXiv:2012.09225](#) | [DOI:10.1016/j.physletb.2021.136565](#) | 2020-12-16

478. Search for supersymmetry in final states with two oppositely charged same-flavor leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2021\) 123](#) | [arXiv:2012.08600](#) | [DOI:10.1007/JHEP04\(2021\)123](#) | 2020-12-15

479. Electron and photon reconstruction and identification with the CMS experiment at the CERN LHC

Sirunyan, Albert M et al.

[JINST 16 \(2021\) P05014](#) | [arXiv:2012.06888](#) | [DOI:10.1088/1748-0221/16/05/P05014](#) | 2020-12-12

480. Measurement of differential cross sections for Z bosons produced in association with charm jets in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2021\) 109](#) | [arXiv:2012.04119](#) | [DOI:10.1007/JHEP04\(2021\)109](#) | 2020-12-07

481. Search for new physics in top quark production with additional leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV using effective field theory

Sirunyan, Albert M et al.

[JHEP 03 \(2021\) 095](#) | [arXiv:2012.04120](#) | [DOI:10.1007/JHEP03\(2021\)095](#) | 2020-12-07

482. Search for singly and pair-produced leptoquarks coupling to third-generation fermions in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 819 \(2021\) 136446](#) | [arXiv:2012.04178](#) | [DOI:10.1016/j.physletb.2021.136446](#) | 2020-12-07

483. Search for long-lived particles using displaced jets in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 012015](#) | [arXiv:2012.01581](#) | [DOI:10.1103/PhysRevD.104.012015](#) | 2020-12-02

484. Search for nonresonant Higgs boson pair production in final states with two bottom quarks and two photons in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 03 \(2021\) 257](#) | [arXiv:2011.12373](#) | [DOI:10.1007/JHEP03\(2021\)257](#) | 2020-11-24

485. Search for the rare decay of the W boson into a pion and a photon in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 819 \(2021\) 136409](#) | [arXiv:2011.06028](#) | [DOI:10.1016/j.physletb.2021.136409](#) | 2020-11-11

486. Observation of Forward Neutron Multiplicity Dependence of Dimuon Acoplanarity in Ultraperipheral Pb-Pb Collisions at $\sqrt{s_{\text{NN}}}=5.02\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 127 \(2021\) 122001](#) | [arXiv:2011.05239](#) | [DOI:10.1103/PhysRevLett.127.122001](#) | 2020-11-10

487. Development and validation of HERWIG 7 tunes from CMS underlying-event measurements

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 312](#) | [arXiv:2011.03422](#) | [DOI:10.1140/epjc/s10052-021-08949-5](#) | 2020-11-06

488. Measurement of the Higgs boson production rate in association with top quarks in final states with electrons, muons, and hadronically decaying tau leptons at $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 378](#) | [arXiv:2011.03652](#) | [DOI:10.1140/epjc/s10052-021-09014-x](#) | 2020-11-06

489. The very forward CASTOR calorimeter of the CMS experiment

Khachatryan, Vardan et al.

[JINST 16 \(2021\) P02010](#) | [arXiv:2011.01185](#) | [DOI:10.1088/1748-0221/16/02/P02010](#) | 2020-11-02

490. Angular analysis of the decay $B^+ \rightarrow K^{*(892)^+} \mu^+ \mu^-$ in proton-proton collisions at $\sqrt{s} = 8\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 04 \(2021\) 124](#) | [arXiv:2010.13968](#) | [DOI:10.1007/JHEP04\(2021\)124](#) | 2020-10-26

491. MUSiC: a model-unspecific search for new physics in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 629](#) | [arXiv:2010.02984](#) | [DOI:10.1140/epjc/s10052-021-09236-z](#) | 2020-10-06

492. Search for dark photons in Higgs boson production via vector boson fusion in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 03 \(2021\) 011](#) | [arXiv:2009.14009](#) | [DOI:10.1007/JHEP03\(2021\)011](#) | 2020-09-29

493. Measurement of prompt D^0 and $\bar{D}^0$ meson azimuthal anisotropy and search for strong electric fields in PbPb collisions at $\sqrt{s_{\text{NN}}} = 5.02\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 816 \(2021\) 136253](#) | [arXiv:2009.12628](#) | [DOI:10.1016/j.physletb.2021.136253](#) | 2020-09-26

494. Measurements of production cross sections of polarized same-sign W boson pairs in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 812 \(2021\) 136018](#) | [arXiv:2009.09429](#) | [DOI:10.1016/j.physletb.2020.136018](#) | 2020-09-20

495. Measurement of the top quark Yukawa coupling from $t\bar{t}$ kinematic distributions in the dilepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 092013](#) | [arXiv:2009.07123](#) | [DOI:10.1103/PhysRevD.102.092013](#) | 2020-09-15

496. Studies of charm and beauty hadron long-range correlations in pp and pPb collisions at LHC energies

Sirunyan, Albert M et al.

[Phys.Lett.B 813 \(2021\) 136036](#) | [arXiv:2009.07065](#) | [DOI:10.1016/j.physletb.2020.136036](#) | 2020-09-15

497. Evidence for Higgs boson decay to a pair of muons

Sirunyan, Albert M et al.

[JHEP 01 \(2021\) 148](#) | [arXiv:2009.04363](#) | [DOI:10.1007/JHEP01\(2021\)148](#) | 2020-09-09

498. Measurements of $pp \rightarrow ZZ$ production cross sections and constraints on anomalous triple gauge couplings at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 200](#) | [arXiv:2009.01186](#) | [DOI:10.1140/epjc/s10052-020-08817-8](#) | 2020-09-02

499. W^+W^- boson pair production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 092001](#) | [arXiv:2009.00119](#) | [DOI:10.1103/PhysRevD.102.092001](#) | 2020-08-31

500. Observation of electroweak production of $W\gamma$ with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 811 \(2020\) 135988](#) | [arXiv:2008.10521](#) | [DOI:10.1016/j.physletb.2020.135988](#) | 2020-08-24

501. A search for bottom-type, vector-like quark pair production in a fully hadronic final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 112004](#) | [arXiv:2008.09835](#) | [DOI:10.1103/PhysRevD.102.112004](#) | 2020-08-22

502. Measurement of $B_c(2S)^+$ and $B_c^{*+}(2S)^+$ cross section ratios in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 092007](#) | [arXiv:2008.08629](#) | [DOI:10.1103/PhysRevD.102.092007](#) | 2020-08-19

503. Measurement of differential $t\bar{t}$ production cross sections using top quarks at large transverse momenta in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 103 \(2021\) 052008](#) | [arXiv:2008.07860](#) | [DOI:10.1103/PhysRevD.103.052008](#) | 2020-08-18

504. Evidence for electroweak production of four charged leptons and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 812 \(2021\) 135992](#) | [arXiv:2008.07013](#) | [DOI:10.1016/j.physletb.2020.135992](#) | 2020-08-16

505. Search for top squark pair production using dilepton final states in pp collision data collected at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 3](#) | [arXiv:2008.05936](#) | [DOI:10.1140/epjc/s10052-020-08701-5](#) | 2020-08-13

506. Search for dark matter produced in association with a leptonically decaying Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 13](#) | [arXiv:2008.04735](#) | [DOI:10.1140/epjc/s10052-020-08739-5](#) | 2020-08-11

507. Measurements of the W boson rapidity, helicity, double-differential cross sections, and charge asymmetry in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 092012](#) | [arXiv:2008.04174](#) | [DOI:10.1103/PhysRevD.102.092012](#) | 2020-08-10

508. Search for supersymmetry in proton-proton collisions at $\sqrt{s} = 13$ TeV in events with high-momentum Z bosons and missing transverse momentum

Sirunyan, Albert M et al.

[JHEP 09 \(2020\) 149](#) | [arXiv:2008.04422](#) | [DOI:10.1007/JHEP09\(2020\)149](#) | 2020-08-10

509. Search for decays of the 125 GeV Higgs boson into a Z boson and a ρ or ϕ meson

Sirunyan, Albert M et al.

[JHEP 11 \(2020\) 039](#) | [arXiv:2007.05122](#) | [DOI:10.1007/JHEP11\(2020\)039](#) | 2020-07-10

510. Search for the lepton flavor violating decay $\tau \rightarrow 3\mu$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2021\) 163](#) | [arXiv:2007.05658](#) | [DOI:10.1007/JHEP01\(2021\)163](#) | 2020-07-10

511. Investigation into the event-activity dependence of $Y(nS)$ relative production in proton-proton collisions at $\sqrt{s} = 7$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2020\) 001](#) | [arXiv:2007.04277](#) | [DOI:10.1007/JHEP11\(2020\)001](#) | 2020-07-09

512. Measurement of the CP-violating phase ϕ_s in the $B_s^0 \rightarrow J/\psi \phi(1020) \rightarrow \mu^+ \mu^- K^+ K^-$ channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 816 \(2021\) 136188](#) | [arXiv:2007.02434](#) | [DOI:10.1016/j.physletb.2021.136188](#) | 2020-07-05

513. Measurement of the inclusive and differential Higgs boson production cross sections in the leptonic WW decay mode at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2021\) 003](#) | [arXiv:2007.01984](#) | [DOI:10.1007/JHEP03\(2021\)003](#) | 2020-07-03

514. Search for resonant and nonresonant production of pairs of dijet resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 161](#) | [arXiv:2206.09997](#) | [DOI:10.1007/JHEP07\(2023\)161](#) | 2020

515. Reconstruction of signal amplitudes in the CMS electromagnetic calorimeter in the presence of overlapping proton-proton interactions

Sirunyan, Albert M et al.

[JINST 15 \(2020\) P10002](#) | [arXiv:2006.14359](#) | [DOI:10.1088/1748-0221/15/10/P10002](#) | 2020-06-25

516. Inclusive search for highly boosted Higgs bosons decaying to bottom quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2020\) 085](#) | [arXiv:2006.13251](#) | [DOI:10.1007/JHEP12\(2020\)085](#) | 2020-06-23

517. Evidence for Top Quark Production in Nucleus-Nucleus Collisions

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 125 \(2020\) 222001](#) | [arXiv:2006.11110](#) | [DOI:10.1103/PhysRevLett.125.222001](#) | 2020-06-19

518. Performance of the CMS Level-1 trigger in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JINST 15 \(2020\) P10017](#) | [arXiv:2006.10165](#) | [DOI:10.1088/1748-0221/15/10/P10017](#) | 2020-06-17

519. Measurement of the azimuthal anisotropy of $\Upsilon(1S)$ and $\Upsilon(2S)$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 819 \(2021\) 136385](#) | [arXiv:2006.07707](#) | [DOI:10.1016/j.physletb.2021.136385](#) | 2020-06-13

520. Search for resonant pair production of Higgs bosons in the $bbZZ$ channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 032003](#) | [arXiv:2006.06391](#) | [DOI:10.1103/PhysRevD.102.032003](#) | 2020-06-11

521. Measurement of b jet shapes in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 054](#) | [arXiv:2005.14219](#) | [DOI:10.1007/JHEP05\(2021\)054](#) | 2020-05-28

522. Search for a light pseudoscalar Higgs boson in the boosted $\mu\mu\tau\tau$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2020\) 139](#) | [arXiv:2005.08694](#) | [DOI:10.1007/JHEP08\(2020\)139](#) | 2020-05-18

523. Search for a light charged Higgs boson in the $H^\pm \rightarrow cs$ channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 072001](#) | [arXiv:2005.08900](#) | [DOI:10.1103/PhysRevD.102.072001](#) | 2020-05-18

524. Dependence of inclusive jet production on the anti- k_T distance parameter in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2020\) 082](#) | [arXiv:2005.05159](#) | [DOI:10.1007/JHEP12\(2020\)082](#) | 2020-05-11

525. Observation of the $B^0_s \rightarrow X(3872)\phi$ decay

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 125 \(2020\) 152001](#) | [arXiv:2005.04764](#) | [DOI:10.1103/PhysRevLett.125.152001](#) | 2020-05-10

526. Combination of the W boson polarization measurements in top quark decays using ATLAS and CMS data at $\sqrt{s} = 8$ TeV

Aad, Georges et al.

[JHEP 08 \(2020\) 051](#) | [arXiv:2005.03799](#) | [DOI:10.1007/JHEP08\(2020\)051](#) | 2020-05-07

527. Measurements of production cross sections of WZ and same-sign WW boson pairs in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 809 \(2020\) 135710](#) | [arXiv:2005.01173](#) | [DOI:10.1016/j.physletb.2020.135710](#) | 2020-05-03

528. Measurement of CKM matrix elements in single top quark t -channel production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 808 \(2020\) 135609](#) | [arXiv:2004.12181](#) | [DOI:10.1016/j.physletb.2020.135609](#) | 2020-04-25

529. Identification of heavy, energetic, hadronically decaying particles using machine-learning techniques

Sirunyan, Albert M et al.

[JINST 15 \(2020\) P06005](#) | [arXiv:2004.08262](#) | [DOI:10.1088/1748-0221/15/06/P06005](#) | 2020-04-17

530. Search for disappearing tracks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 806 \(2020\) 135502](#) | [arXiv:2004.05153](#) | [DOI:10.1016/j.physletb.2020.135502](#) | 2020-04-10

531. Measurement of quark- and gluon-like jet fractions using jet charge in PbPb and pp collisions at 5.02 TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2020\) 115](#) | [arXiv:2004.00602](#) | [DOI:10.1007/JHEP07\(2020\)115](#) | 2020-04-01

532. The production of isolated photons in PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2020\) 116](#) | [arXiv:2003.12797](#) | [DOI:10.1007/JHEP07\(2020\)116](#) | 2020-03-28

533. Search for supersymmetry in pp collisions at $\sqrt{s} = 13$ TeV with 137 fb^{-1} in final states with a single lepton using the sum of masses of large-radius jets

Sirunyan, Albert M et al.

[Phys.Rev.D 101 \(2020\) 052010](#) | [arXiv:1911.07558](#) | [DOI:10.1103/PhysRevD.101.052010](#) | 2020-03-25

534. Measurements of $t\bar{t}H$ Production and the CP Structure of the Yukawa Interaction between the Higgs Boson and Top Quark in the Diphoton Decay Channel

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 125 \(2020\) 061801](#) | [arXiv:2003.10866](#) | [DOI:10.1103/PhysRevLett.125.061801](#) | 2020-03-24

535. Measurement of the cross section for $t\bar{t}$ production with additional jets and b jets in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2020\) 125](#) | [arXiv:2003.06467](#) | [DOI:10.1007/JHEP07\(2020\)125](#) | 2020-03-13

536. Study of central exclusive $\pi^+ \pi^-$ production in proton-proton collisions at $\sqrt{s} = 5.02$ and 13 TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 718](#) | [arXiv:2003.02811](#) | [DOI:10.1140/epjc/s10052-020-8166-5](#) | 2020-03-05

537. Pileup mitigation at CMS in 13 TeV data

Sirunyan, Albert M et al.

[JINST 15 \(2020\) P09018](#) | [arXiv:2003.00503](#) | [DOI:10.1088/1748-0221/15/09/P09018](#) | 2020-03-01

538. Measurement of single-diffractive dijet production in proton-proton collisions at $\sqrt{s} = 8$ TeV with the CMS and TOTEM experiments

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 1164](#) | [arXiv:2002.12146](#) | [DOI:10.1140/epjc/s10052-020-08562-y](#) | 2020-02-27

539. Measurement of the cross section for electroweak production of a Z boson, a photon and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on anomalous quartic couplings

Sirunyan, Albert M et al.

[JHEP 06 \(2020\) 076](#) | [arXiv:2002.09902](#) | [DOI:10.1007/JHEP06\(2020\)076](#) | 2020-02-23

540. Measurement of the $\Upsilon(1S)$ pair production cross section and search for resonances decaying to $\Upsilon(1S)\mu^+ \mu^-$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 808 \(2020\) 135578](#) | [arXiv:2002.06393](#) | [DOI:10.1016/j.physletb.2020.135578](#) | 2020-02-15

541. A measurement of the Higgs boson mass in the diphoton decay channel

Sirunyan, Albert M et al.

[Phys.Lett.B 805 \(2020\) 135425](#) | [arXiv:2002.06398](#) | [DOI:10.1016/j.physletb.2020.135425](#) | 2020-02-15

542. Search for physics beyond the standard model in events with jets and two same-sign or at least three charged leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 752](#) | [arXiv:2001.10086](#) | [DOI:10.1140/epjc/s10052-020-8168-3](#) | 2020-01-27

543. Search for charged Higgs bosons decaying into a top and a bottom quark in the all-jet final state of pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2020\) 126](#) | [arXiv:2001.07763](#) | [DOI:10.1007/JHEP07\(2020\)126](#) | 2020-01-21

544. Measurement of the associated production of a Z boson with charm or bottom quark jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 032007](#) | [arXiv:2001.06899](#) | [DOI:10.1103/PhysRevD.102.032007](#) | 2020-01-19

545. Measurements with silicon photomultipliers of dose-rate effects in the radiation damage of plastic scintillator tiles in the CMS hadron endcap calorimeter

Sirunyan, Albert M et al.

[JINST 15 \(2020\) P06009](#) | [arXiv:2001.06553](#) | [DOI:10.1088/1748-0221/15/06/P06009](#) | 2020-01-17

546. Study of excited Λ_b^0 states decaying to $\Lambda_b^0\pi^+\pi^-$ in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 803 \(2020\) 135345](#) | [arXiv:2001.06533](#) | [DOI:10.1016/j.physletb.2020.135345](#) | 2020-01-17

547. Search for an excited lepton that decays via a contact interaction to a lepton and two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2020\) 052](#) | [arXiv:2001.04521](#) | [DOI:10.1007/JHEP05\(2020\)052](#) | 2020-01-13

548. A deep neural network to search for new long-lived particles decaying to jets

Sirunyan, Albert M et al.

[Mach.Learn.Sci.Tech. 1 \(2020\) 035012](#) | [arXiv:1912.12238](#) | [DOI:10.1088/2632-2153/ab9023](#) | 2019-12-27

549. Measurement of the top quark forward-backward production asymmetry and the anomalous chromoelectric and chromomagnetic moments in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2020\) 146](#) | [arXiv:1912.09540](#) | [DOI:10.1007/JHEP06\(2020\)146](#) | 2019-12-19

550. Search for direct top squark pair production in events with one lepton, jets, and missing transverse momentum at 13 TeV with the CMS experiment

Sirunyan, Albert M et al.

[JHEP 05 \(2020\) 032](#) | [arXiv:1912.08887](#) | [DOI:10.1007/JHEP05\(2020\)032](#) | 2019-12-18

551. Constraints on the χ_{c1} versus χ_{c2} Polarizations in Proton-Proton Collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 124 \(2020\) 162002](#) | [arXiv:1912.07706](#) | [DOI:10.1103/PhysRevLett.124.162002](#) | 2019-12-16

552. A Deep Neural Network for Simultaneous Estimation of b Jet Energy and Resolution

Sirunyan, Albert M et al.

[Comput.Softw.Big Sci. 4 \(2020\) 10](#) | [arXiv:1912.06046](#) | [DOI:10.1007/s41781-020-00041-z](#) | 2019-12-12

553. Search for a Narrow Resonance Lighter than 200 GeV Decaying to a Pair of Muons in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 124 \(2020\) 131802](#) | [arXiv:1912.04776](#) | [DOI:10.1103/PhysRevLett.124.131802](#) | 2019-12-10

554. Determination of the strong coupling constant $\alpha_S(m_Z)$ from measurements of inclusive $W^\pm$ and Z boson production cross sections in proton-proton collisions at $\sqrt{s} = 7$ and 8 TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2020\) 018](#) | [arXiv:1912.04387](#) | [DOI:10.1007/JHEP06\(2020\)018](#) | 2019-12-09

555. Performance of the reconstruction and identification of high-momentum muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JINST 15 \(2020\) P02027](#) | [arXiv:1912.03516](#) | [DOI:10.1088/1748-0221/15/02/P02027](#) | 2019-12-07

556. Search for a heavy Higgs boson decaying to a pair of W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 034](#) | [arXiv:1912.01594](#) | [DOI:10.1007/JHEP03\(2020\)034](#) | 2019-12-03

557. A search for the standard model Higgs boson decaying to charm quarks

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 131](#) | [arXiv:1912.01662](#) | [DOI:10.1007/JHEP03\(2020\)131](#) | 2019-12-03

558. Measurement of the top quark pair production cross section in dilepton final states containing one τ lepton in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 02 \(2020\) 191](#) | [arXiv:1911.13204](#) | [DOI:10.1007/JHEP02\(2020\)191](#) | 2019-11-29

559. Search for lepton flavour violating decays of a neutral heavy Higgs boson to $\mu\tau$ and $e\tau$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 103](#) | [arXiv:1911.10267](#) | [DOI:10.1007/JHEP03\(2020\)103](#) | 2019-11-22

560. Search for physics beyond the standard model in multilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 051](#) | [arXiv:1911.04968](#) | [DOI:10.1007/JHEP03\(2020\)051](#) | 2019-11-12

561. Search for high mass dijet resonances with a new background prediction method in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2020\) 033](#) | [arXiv:1911.03947](#) | [DOI:10.1007/JHEP05\(2020\)033](#) | 2019-11-10

562. Measurement of the Jet Mass Distribution and Top Quark Mass in Hadronic Decays of Boosted Top Quarks in pp Collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 124 \(2020\) 202001](#) | [arXiv:1911.03800](#) | [DOI:10.1103/PhysRevLett.124.202001](#) | 2019-11-09

563. Search for dijet resonances using events with three jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 805 \(2020\) 135448](#) | [arXiv:1911.03761](#) | [DOI:10.1016/j.physletb.2020.135448](#) | 2019-11-09

564. Observation of the $\Lambda_b^0 \rightarrow J/\psi \Lambda \phi$ decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 802 \(2020\) 135203](#) | [arXiv:1911.03789](#) | [DOI:10.1016/j.physletb.2020.135203](#) | 2019-11-09

565. Search for new neutral Higgs bosons through the $H \rightarrow ZA \rightarrow \ell^+ \ell^- b\bar{b}$ process in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 055](#) | [arXiv:1911.03781](#) | [DOI:10.1007/JHEP03\(2020\)055](#) | 2019-11-09

566. Studies of charm quark diffusion inside jets using PbPb and pp collisions at $\sqrt{s_{\text{NN}}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 125 \(2020\) 102001](#) | [arXiv:1911.01461](#) | [DOI:10.1103/PhysRevLett.125.102001](#) | 2019-11-04

567. Search for top squark pair production in a final state with two tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 02 \(2020\) 015](#) | [arXiv:1910.12932](#) | [DOI:10.1007/JHEP02\(2020\)015](#) | 2019-10-28

568. Measurement of properties of $B^0_s \rightarrow \mu^+ \mu^-$ decays and search for $B^0 \rightarrow \mu^+ \mu^-$ with the CMS experiment

Sirunyan, Albert M et al.

[JHEP 04 \(2020\) 188](#) | [arXiv:1910.12127](#) | [DOI:10.1007/JHEP04\(2020\)188](#) | 2019-10-26

569. Search for a heavy pseudoscalar Higgs boson decaying into a 125 GeV Higgs boson and a Z boson in final states with two tau and two light leptons at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 065](#) | [arXiv:1910.11634](#) | [DOI:10.1007/JHEP03\(2020\)065](#) | 2019-10-25

570. Bose-Einstein correlations of charged hadrons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 014](#) | [arXiv:1910.08815](#) | [DOI:10.1007/JHEP03\(2020\)014](#) | 2019-10-19

571. Mixed higher-order anisotropic flow and nonlinear response coefficients of charged particles in PbPb collisions at $\sqrt{\langle b \rangle s_{\text{NN}}} = 2.76$ and 5.02 TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 534](#) | [arXiv:1910.08789](#) | [DOI:10.1140/epjc/s10052-020-7834-9](#) | 2019-10-19

572. Strange hadron production in pp and pPb collisions at $\sqrt{s_{\text{NN}}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.C 101 \(2020\) 064906](#) | [arXiv:1910.04812](#) | [DOI:10.1103/PhysRevC.101.064906](#) | 2019-10-10

573. Measurement of the top quark polarization and $t\bar{t}$ spin correlations using dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 072002](#) | [arXiv:1907.03729](#) | [DOI:10.1103/PhysRevD.100.072002](#) | 2019-10-09

574. Study of J/ψ meson production inside jets in pp collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 804 \(2020\) 135409](#) | [arXiv:1910.01686](#) | [DOI:10.1016/j.physletb.2020.135409](#) | 2019-10-03

575. Search for Supersymmetry with a Compressed Mass Spectrum in Events with a Soft τ Lepton, a Highly Energetic Jet, and Large Missing Transverse Momentum in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 124 \(2020\) 041803](#) | [arXiv:1910.01185](#) | [DOI:10.1103/PhysRevLett.124.041803](#) | 2019-10-02

576. Calibration of the CMS hadron calorimeters using proton-proton collision data at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JINST 15 \(2020\) P05002](#) | [arXiv:1910.00079](#) | [DOI:10.1088/1748-0221/15/05/P05002](#) | 2019-09-30

577. Running of the top quark mass from proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 803 \(2020\) 135263](#) | [arXiv:1909.09193](#) | [DOI:10.1016/j.physletb.2020.135263](#) | 2019-09-19

578. Search for long-lived particles using delayed photons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 112003](#) | [arXiv:1909.06166](#) | [DOI:10.1103/PhysRevD.100.112003](#) | 2019-09-13

579. Evidence for WW production from double-parton interactions in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 41](#) | [arXiv:1909.06265](#) | [DOI:10.1140/epjc/s10052-019-7541-6](#) | 2019-09-13

580. Measurement of the $t\bar{t}b\bar{b}$ production cross section in the all-jet final state in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 803 \(2020\) 135285](#) | [arXiv:1909.05306](#) | [DOI:10.1016/j.physletb.2020.135285](#) | 2019-09-11

581. Search for electroweak production of a vector-like T quark using fully hadronic final states

Sirunyan, Albert M et al.

[JHEP 01 \(2020\) 036](#) | [arXiv:1909.04721](#) | [DOI:10.1007/JHEP01\(2020\)036](#) | 2019-09-10

582. Search for low mass vector resonances decaying into quark-antiquark pairs in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 112007](#) | [arXiv:1909.04114](#) | [DOI:10.1103/PhysRevD.100.112007](#) | 2019-09-09

583. Measurements of differential Z boson production cross sections in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2019\) 061](#) | [arXiv:1909.04133](#) | [DOI:10.1007/JHEP12\(2019\)061](#) | 2019-09-09

584. Searches for physics beyond the standard model with the M_{T2} variable in hadronic final states with and without disappearing tracks in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 3](#) | [arXiv:1909.03460](#) | [DOI:10.1140/epjc/s10052-019-7493-x](#) | 2019-09-08

585. Search for a charged Higgs boson decaying into top and bottom quarks in events with electrons or muons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2020\) 096](#) | [arXiv:1908.09206](#) | [DOI:10.1007/JHEP01\(2020\)096](#) | 2019-08-24

586. Search for supersymmetry using Higgs boson to diphoton decays at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2019\) 109](#) | [arXiv:1908.08500](#) | [DOI:10.1007/JHEP11\(2019\)109](#) | 2019-08-22

587. Search for production of four top quarks in final states with same-sign or multiple leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 75](#) | [arXiv:1908.06463](#) | [DOI:10.1140/epjc/s10052-019-7593-7](#) | 2019-08-18

588. Search for supersymmetry in proton-proton collisions at 13 TeV in final states with jets and missing transverse momentum

Collaboration, The Cms et al.

[JHEP 10 \(2019\) 244](#) | [arXiv:1908.04722](#) | [DOI:10.1007/JHEP10\(2019\)244](#) | 2019-08-13

589. Search for dark photons in decays of Higgs bosons produced in association with Z bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2019\) 139](#) | [arXiv:1908.02699](#) | [DOI:10.1007/JHEP10\(2019\)139](#) | 2019-08-07

590. Measurement of the average very forward energy as a function of the track multiplicity at central pseudorapidities in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 893](#) | [arXiv:1908.01750](#) | [DOI:10.1140/epjc/s10052-019-7402-3](#) | 2019-08-05

591. Search for dark matter particles produced in association with a Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 025](#) | [arXiv:1908.01713](#) | [DOI:10.1007/JHEP03\(2020\)025](#) | 2019-08-05

592. Search for direct pair production of supersymmetric partners to the τ lepton in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 189](#) | [arXiv:1907.13179](#) | [DOI:10.1140/epjc/s10052-020-7739-7](#) | 2019-07-30

593. Measurement of top quark pair production in association with a Z boson in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 056](#) | [arXiv:1907.11270](#) | [DOI:10.1007/JHEP03\(2020\)056](#) | 2019-07-25

594. Measurements of triple-differential cross sections for inclusive isolated-photon+jet events in pp collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 969](#) | [arXiv:1907.08155](#) | [DOI:10.1140/epjc/s10052-019-7451-7](#) | 2019-07-18

595. Search for anomalous triple gauge couplings in WW and WZ production in lepton + jet events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2019\) 062](#) | [arXiv:1907.08354](#) | [DOI:10.1007/JHEP12\(2019\)062](#) | 2019-07-18

596. Measurement of differential cross sections and charge ratios for t-channel single top quark production in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 370](#) | [arXiv:1907.08330](#) | [DOI:10.1140/epjc/s10052-020-7858-1](#) | 2019-07-18

597. Search for light pseudoscalar boson pairs produced from decays of the 125 GeV Higgs boson in final states with two muons and two nearby tracks in pp collisions at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 800 \(2020\) 135087](#) | [arXiv:1907.07235](#) | [DOI:10.1016/j.physletb.2019.135087](#) | 2019-07-16

598. Search for Physics beyond the Standard Model in Events with Overlapping Photons and Jets

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 123 \(2019\) 241801](#) | [arXiv:1907.06275](#) | [DOI:10.1103/PhysRevLett.123.241801](#) | 2019-07-14

599. Study of the $B^+ \rightarrow J/\psi \Lambda p$ decay in proton-proton collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2019\) 100](#) | [arXiv:1907.05461](#) | [DOI:10.1007/JHEP12\(2019\)100](#) | 2019-07-11

600. Search for MSSM Higgs bosons decaying to $\mu + \mu -$ in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 798 \(2019\) 134992](#) | [arXiv:1907.03152](#) | [DOI:10.1016/j.physletb.2019.134992](#) | 2019-07-06

601. Measurement of the top quark Yukawa coupling from $t\bar{t}$ kinematic distributions in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 072007](#) | [arXiv:1907.01590](#) | [DOI:10.1103/PhysRevD.100.072007](#) | 2019-07-02

602. Combined search for supersymmetry with photons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 801 \(2020\) 135183](#) | [arXiv:1907.00857](#) | [DOI:10.1016/j.physletb.2019.135183](#) | 2019-07-01

603. Search for pair production of vectorlike quarks in the fully hadronic final state

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 072001](#) | [arXiv:1906.11903](#) | [DOI:10.1103/PhysRevD.100.072001](#) | 2019-06-27

604. Search for long-lived particles using nonprompt jets and missing transverse momentum with proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 797 \(2019\) 134876](#) | [arXiv:1906.06441](#) | [DOI:10.1016/j.physletb.2019.134876](#) | 2019-06-14

605. A multi-dimensional search for new heavy resonances decaying to boosted WW, WZ, or ZZ boson pairs in the dijet final state at 13 TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 237](#) | [arXiv:1906.05977](#) | [DOI:10.1140/epjc/s10052-020-7773-5](#) | 2019-06-13

606. Production of Λ_c^+ baryons in proton-proton and lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 803 \(2020\) 135328](#) | [arXiv:1906.03322](#) | [DOI:10.1016/j.physletb.2020.135328](#) | 2019-06-07

607. Search for the production of four top quarks in the single-lepton and opposite-sign dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2019\) 082](#) | [arXiv:1906.02805](#) | [DOI:10.1007/JHEP11\(2019\)082](#) | 2019-06-06

608. Combination of CMS searches for heavy resonances decaying to pairs of bosons or leptons

Sirunyan, Albert M et al.

[Phys.Lett.B 798 \(2019\) 134952](#) | [arXiv:1906.00057](#) | [DOI:10.1016/j.physletb.2019.134952](#) | 2019-05-31

609. Search for supersymmetry with a compressed mass spectrum in the vector boson fusion topology with 1-lepton and 0-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2019\) 150](#) | [arXiv:1905.13059](#) | [DOI:10.1007/JHEP08\(2019\)150](#) | 2019-05-30

610. Search for vector-like leptons in multilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 052003](#) | [arXiv:1905.10853](#) | [DOI:10.1103/PhysRevD.100.052003](#) | 2019-05-26

611. Search for Higgs and Z boson decays to J/ψ or Y pairs in the four-muon final state in proton-proton collisions at $s = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 797 \(2019\) 134811](#) | [arXiv:1905.10408](#) | [DOI:10.1016/j.physletb.2019.134811](#) | 2019-05-24

612. Search for Low-Mass Quark-Antiquark Resonances Produced in Association with a Photon at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 123 \(2019\) 231803](#) | [arXiv:1905.10331](#) | [DOI:10.1103/PhysRevLett.123.231803](#) | 2019-05-24

613. Correlations of azimuthal anisotropy Fourier harmonics with subevent cumulants in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.C 103 \(2021\) 014902](#) | [arXiv:1905.09935](#) | [DOI:10.1103/PhysRevC.103.014902](#) | 2019-05-23

614. Search for a light charged Higgs boson decaying to a W boson and a CP-odd Higgs boson in final states with $e\mu\mu$ or $\mu\mu\mu$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 123 \(2019\) 131802](#) | [arXiv:1905.07453](#) | [DOI:10.1103/PhysRevLett.123.131802](#) | 2019-05-17

615. Search for anomalous electroweak production of vector boson pairs in association with two jets in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 798 \(2019\) 134985](#) | [arXiv:1905.07445](#) | [DOI:10.1016/j.physletb.2019.134985](#) | 2019-05-17

616. Observation of nuclear modifications in $W^{\pm}$ boson production in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 800 \(2020\) 135048](#) | [arXiv:1905.01486](#) | [DOI:10.1016/j.physletb.2019.135048](#) | 2019-05-04

617. Multiparticle correlation studies in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.C 101 \(2020\) 014912](#) | [arXiv:1904.11519](#) | [DOI:10.1103/PhysRevC.101.014912](#) | 2019-04-25

618. Measurement of $t\bar{t}$ normalised multi-differential cross sections in pp collisions at $\sqrt{s} = 13$ TeV, and simultaneous determination of the strong coupling strength, top quark pole mass, and parton distribution functions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 658](#) | [arXiv:1904.05237](#) | [DOI:10.1140/epjc/s10052-020-7917-7](#) | 2019-04-10

619. Search for resonances decaying to a pair of Higgs bosons in the $b\bar{b}q\bar{q}'\ell\nu$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2019\) 125](#) | [arXiv:1904.04193](#) | [DOI:10.1007/JHEP10\(2019\)125](#) | 2019-04-08

620. Extraction and validation of a new set of CMS PYTHIA8 tunes from underlying-event measurements

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 4](#) | [arXiv:1903.12179](#) | [DOI:10.1140/epjc/s10052-019-7499-4](#) | 2019-03-28

621. Search for new physics in top quark production in dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 886](#) | [arXiv:1903.11144](#) | [DOI:10.1140/epjc/s10052-019-7387-y](#) | 2019-03-26

622. Search for a low-mass $\tau^+\tau^-$ resonance in association with a bottom quark in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2019\) 210](#) | [arXiv:1903.10228](#) | [DOI:10.1007/JHEP05\(2019\)210](#) | 2019-03-25

623. Search for supersymmetry in final states with photons and missing transverse momentum in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2019\) 143](#) | [arXiv:1903.07070](#) | [DOI:10.1007/JHEP06\(2019\)143](#) | 2019-03-17

624. Constraints on anomalous HVV couplings from the production of Higgs bosons decaying to τ lepton pairs

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 112002](#) | [arXiv:1903.06973](#) | [DOI:10.1103/PhysRevD.100.112002](#) | 2019-03-16

625. Performance of missing transverse momentum reconstruction in proton-proton collisions at $\sqrt{s} = 13$ TeV using the CMS detector

Sirunyan, Albert M et al.

[JINST 14 \(2019\) P07004](#) | [arXiv:1903.06078](#) | [DOI:10.1088/1748-0221/14/07/P07004](#) | 2019-03-14

626. Search for charged Higgs bosons in the $H^{\pm} \rightarrow \tau^{\pm} \nu_{\tau}$ decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2019\) 142](#) | [arXiv:1903.04560](#) | [DOI:10.1007/JHEP07\(2019\)142](#) | 2019-03-11

627. Measurement of electroweak production of a W boson in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 43](#) | [arXiv:1903.04040](#) | [DOI:10.1140/epjc/s10052-019-7585-7](#) | 2019-03-10

628. An embedding technique to determine $\tau\tau$ backgrounds in proton-proton collision data

Sirunyan, Albert M et al.

[JINST 14 \(2019\) P06032](#) | [arXiv:1903.01216](#) | [DOI:10.1088/1748-0221/14/06/P06032](#) | 2019-03-04

629. Search for a heavy pseudoscalar boson decaying to a Z and a Higgs boson at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 564](#) | [arXiv:1903.00941](#) | [DOI:10.1140/epjc/s10052-019-7058-z](#) | 2019-03-03

630. Combinations of single-top-quark production cross-section measurements and $|f_{LVV_{tb}}|$ determinations at $\sqrt{s} = 7$ and 8 TeV with the ATLAS and CMS experiments

Aaboud, Morad et al.

[JHEP 05 \(2019\) 088](#) | [arXiv:1902.07158](#) | [DOI:10.1007/JHEP05\(2019\)088](#) | 2019-02-19

631. Azimuthal separation in nearly back-to-back jet topologies in inclusive 2- and 3-jet events in pp collisions at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 773](#) | [arXiv:1902.04374](#) | [DOI:10.1140/epjc/s10052-019-7276-4](#) | 2019-02-12

632. Pseudorapidity distributions of charged hadrons in xenon-xenon collisions at $\sqrt{s_{NN}} = 5.44$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 799 \(2019\) 135049](#) | [arXiv:1902.03603](#) | [DOI:10.1016/j.physletb.2019.135049](#) | 2019-02-10

633. Measurement of exclusive $\rho(770)^0$ photoproduction in ultraperipheral pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 702](#) | [arXiv:1902.01339](#) | [DOI:10.1140/epjc/s10052-019-7202-9](#) | 2019-02-04

634. Observation of Two Excited B^+_c States and Measurement of the $B^+_c(2S)$ Mass in pp Collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 132001](#) | [arXiv:1902.00571](#) | [DOI:10.1103/PhysRevLett.122.132001](#) | 2019-02-01

635. Search for W boson decays to three charged pions

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 151802](#) | [arXiv:1901.11201](#) | [DOI:10.1103/PhysRevLett.122.151802](#) | 2019-01-30

636. Charged-particle angular correlations in XeXe collisions at $\sqrt{s_{NN}} = 5.44$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.C 100 \(2019\) 044902](#) | [arXiv:1901.07997](#) | [DOI:10.1103/PhysRevC.100.044902](#) | 2019-01-23

637. Search for supersymmetry in events with a photon, jets, b-jets, and missing transverse momentum in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 444](#) | [arXiv:1901.06726](#) | [DOI:10.1140/epjc/s10052-019-6926-x](#) | 2019-01-20

638. Measurement of electroweak WZ boson production and search for new physics in WZ + two jets events in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 795 \(2019\) 281](#) | [arXiv:1901.04060](#) | [DOI:10.1016/j.physletb.2019.05.042](#) | 2019-01-13

639. Measurements of the $pp \rightarrow WZ$ inclusive and differential production cross section and constraints on charged anomalous triple gauge couplings at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2019\) 122](#) | [arXiv:1901.03428](#) | [DOI:10.1007/JHEP04\(2019\)122](#) | 2019-01-10

640. Search for dark matter produced in association with a single top quark or a top quark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 141](#) | [arXiv:1901.01553](#) | [DOI:10.1007/JHEP03\(2019\)141](#) | 2019-01-06

641. Search for the pair production of light top squarks in the $e^{\pm}\mu^{\mp}$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 101](#) | [arXiv:1901.01288](#) | [DOI:10.1007/JHEP03\(2019\)101](#) | 2019-01-04

642. Measurements of the Higgs boson width and anomalous HVV couplings from on-shell and off-shell production in the four-lepton final state

Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 112003](#) | [arXiv:1901.00174](#) | [DOI:10.1103/PhysRevD.99.112003](#) | 2019-01-01

643. Measurement of the tt production cross section, the top quark mass, and the strong coupling constant using dilepton events in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 368](#) | [arXiv:1812.10505](#) | [DOI:10.1140/epjc/s10052-019-6863-8](#) | 2018-12-26

644. Measurement of the differential Drell-Yan cross section in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2019\) 059](#) | [arXiv:1812.10529](#) | [DOI:10.1007/JHEP12\(2019\)059](#) | 2018-12-26

645. Search for contact interactions and large extra dimensions in the dilepton mass spectra from proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2019\) 114](#) | [arXiv:1812.10443](#) | [DOI:10.1007/JHEP04\(2019\)114](#) | 2018-12-26

646. Measurement of the top quark mass in the all-jets final state at $\sqrt{s} = 13$ TeV and combination with the lepton+jets channel

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 313](#) | [arXiv:1812.10534](#) | [DOI:10.1140/epjc/s10052-019-6788-2](#) | 2018-12-26

647. Measurement of the single top quark and antiquark production cross sections in the t channel and their ratio in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 800 \(2020\) 135042](#) | [arXiv:1812.10514](#) | [DOI:10.1016/j.physletb.2019.135042](#) | 2018-12-26

648. Search for vector-like quarks in events with two oppositely charged leptons and jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 364](#) | [arXiv:1812.09768](#) | [DOI:10.1140/epjc/s10052-019-6855-8](#) | 2018-12-23

649. Measurement and interpretation of differential cross sections for Higgs boson production at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 792 \(2019\) 369](#) | [arXiv:1812.06504](#) | [DOI:10.1016/j.physletb.2019.03.059](#) | 2018-12-16

650. Search for a heavy resonance decaying to a top quark and a vector-like top quark in the lepton+jets final state in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 208](#) | [arXiv:1812.06489](#) | [DOI:10.1140/epjc/s10052-019-6688-5](#) | 2018-12-16

651. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state with two muons and two b quarks in pp collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 795 \(2019\) 398](#) | [arXiv:1812.06359](#) | [DOI:10.1016/j.physletb.2019.06.021](#) | 2018-12-15

652. Inclusive search for supersymmetry in pp collisions at $\sqrt{s}=13$ TeV using razor variables and boosted object identification in zero and one lepton final states

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 031](#) | [arXiv:1812.06302](#) | [DOI:10.1007/JHEP03\(2019\)031](#) | 2018-12-15

653. Observation of Single Top Quark Production in Association with a Z Boson in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 132003](#) | [arXiv:1812.05900](#) | [DOI:10.1103/PhysRevLett.122.132003](#) | 2018-12-14

654. Search for supersymmetry in events with a photon, a lepton, and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2019\) 154](#) | [arXiv:1812.04066](#) | [DOI:10.1007/JHEP01\(2019\)154](#) | 2018-12-10

655. Measurement of the energy density as a function of pseudorapidity in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 391](#) | [arXiv:1812.04095](#) | [DOI:10.1140/epjc/s10052-019-6861-x](#) | 2018-12-10

656. Measurement of inclusive very forward jet cross sections in proton-lead collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2019\) 043](#) | [arXiv:1812.01691](#) | [DOI:10.1007/JHEP05\(2019\)043](#) | 2018-12-04

657. A search for pair production of new light bosons decaying into muons in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 796 \(2019\) 131](#) | [arXiv:1812.00380](#) | [DOI:10.1016/j.physletb.2019.07.013](#) | 2018-12-02

658. Measurement of associated production of a W boson and a charm quark in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 269](#) | [arXiv:1811.10021](#) | [DOI:10.1140/epjc/s10052-019-6752-1](#) | 2018-11-25

659. Search for dark matter in events with a leptoquark and missing transverse momentum in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 795 \(2019\) 76](#) | [arXiv:1811.10151](#) | [DOI:10.1016/j.physletb.2019.05.046](#) | 2018-11-25

660. Combination of searches for Higgs boson pair production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 121803](#) | [arXiv:1811.09689](#) | [DOI:10.1103/PhysRevLett.122.121803](#) | 2018-11-23

661. Search for associated production of a Higgs boson and a single top quark in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 092005](#) | [arXiv:1811.09696](#) | [DOI:10.1103/PhysRevD.99.092005](#) | 2018-11-23

662. Search for resonant production of second-generation sleptons with same-sign dimuon events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 305](#) | [arXiv:1811.09760](#) | [DOI:10.1140/epjc/s10052-019-6800-x](#) | 2018-11-23

663. Search for a standard model-like Higgs boson in the mass range between 70 and 110 GeV in the diphoton final state in proton-proton collisions at $\sqrt{s} = 8$ and 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 793 \(2019\) 320](#) | [arXiv:1811.08459](#) | [DOI:10.1016/j.physletb.2019.03.064](#) | 2018-11-20

664. Search for long-lived particles decaying into displaced jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 032011](#) | [arXiv:1811.07991](#) | [DOI:10.1103/PhysRevD.99.032011](#) | 2018-11-19

665. Search for a W' boson decaying to a vector-like quark and a top or bottom quark in the all-jets final state

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 127](#) | [arXiv:1811.07010](#) | [DOI:10.1007/JHEP03\(2019\)127](#) | 2018-11-16

666. Measurements of $t\bar{t}$ differential cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV using events containing two leptons

Sirunyan, Albert M et al.

[JHEP 02 \(2019\) 149](#) | [arXiv:1811.06625](#) | [DOI:10.1007/JHEP02\(2019\)149](#) | 2018-11-15

667. Search for excited leptons in $\ell\ell\gamma$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2019\) 015](#) | [arXiv:1811.03052](#) | [DOI:10.1007/JHEP04\(2019\)015](#) | 2018-11-07

668. Search for pair production of first-generation scalar leptoquarks at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 052002](#) | [arXiv:1811.01197](#) | [DOI:10.1103/PhysRevD.99.052002](#) | 2018-11-03

669. Search for heavy neutrinos and third-generation leptoquarks in hadronic states of two τ leptons and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 170](#) | [arXiv:1811.00806](#) | [DOI:10.1007/JHEP03\(2019\)170](#) | 2018-11-02

670. Event shape variables measured using multijet final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2018\) 117](#) | [arXiv:1811.00588](#) | [DOI:10.1007/JHEP12\(2018\)117](#) | 2018-11-01

671. Search for low-mass resonances decaying into bottom quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 012005](#) | [arXiv:1810.11822](#) | [DOI:10.1103/PhysRevD.99.012005](#) | 2018-10-28

672. Search for nonresonant Higgs boson pair production in the bbbb final state at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2019\) 112](#) | [arXiv:1810.11854](#) | [DOI:10.1007/JHEP04\(2019\)112](#) | 2018-10-28

673. Studies of Beauty Suppression via Nonprompt D^0 Mesons in Pb-Pb Collisions at $Q^2 = 4 \text{ GeV}^2$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 123 \(2019\) 022001](#) | [arXiv:1810.11102](#) | [DOI:10.1103/PhysRevLett.123.022001](#) | 2018-10-25

674. Search for new particles decaying to a jet and an emerging jet

Sirunyan, Albert M et al.

[JHEP 02 \(2019\) 179](#) | [arXiv:1810.10069](#) | [DOI:10.1007/JHEP02\(2019\)179](#) | 2018-10-23

675. Search for rare decays of Z and Higgs bosons to J/ψ and a photon in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 94](#) | [arXiv:1810.10056](#) | [DOI:10.1140/epjc/s10052-019-6562-5](#) | 2018-10-23

676. Search for pair-produced three-jet resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 012010](#) | [arXiv:1810.10092](#) | [DOI:10.1103/PhysRevD.99.012010](#) | 2018-10-23

677. Search for resonant $t\bar{t}$ production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2019\) 031](#) | [arXiv:1810.05905](#) | [DOI:10.1007/JHEP04\(2019\)031](#) | 2018-10-13

678. Centrality and pseudorapidity dependence of the transverse energy density in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.C 100 \(2019\) 024902](#) | [arXiv:1810.05745](#) | [DOI:10.1103/PhysRevC.100.024902](#) | 2018-10-12

679. Evidence for light-by-light scattering and searches for axion-like particles in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 797 \(2019\) 134826](#) | [arXiv:1810.04602](#) | [DOI:10.1016/j.physletb.2019.134826](#) | 2018-10-10

680. Search for top quark partners with charge $5/3$ in the same-sign dilepton and single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 082](#) | [arXiv:1810.03188](#) | [DOI:10.1007/JHEP03\(2019\)082](#) | 2018-10-07

681. Measurement of B^0_s meson production in pp and PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 796 \(2019\) 168](#) | [arXiv:1810.03022](#) | [DOI:10.1016/j.physletb.2019.07.014](#) | 2018-10-06

682. Observation of prompt J/ψ meson elliptic flow in high-multiplicity pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 791 \(2019\) 172](#) | [arXiv:1810.01473](#) | [DOI:10.1016/j.physletb.2019.02.018](#) | 2018-10-02

683. Search for new physics in final states with a single photon and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 02 \(2019\) 074](#) | [arXiv:1810.00196](#) | [DOI:10.1007/JHEP02\(2019\)074](#) | 2018-09-29

684. Measurement of exclusive Υ photoproduction from protons in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M. et al.

[Eur.Phys.J.C 79 \(2019\) 277](#) | [arXiv:1809.11080](#) | [DOI:10.1140/epjc/s10052-019-6774-8](#) | 2018-09-28

685. Combined measurements of Higgs boson couplings in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 421](#) | [arXiv:1809.10733](#) | [DOI:10.1140/epjc/s10052-019-6909-y](#) | 2018-09-27

686. Search for single production of vector-like quarks decaying to a top quark and a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 90](#) | [arXiv:1809.08597](#) | [DOI:10.1140/epjc/s10052-019-6556-3](#) | 2018-09-23

687. Jet Shapes of Isolated Photon-Tagged Jets in Pb-Pb and pp Collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 152001](#) | [arXiv:1809.08602](#) | [DOI:10.1103/PhysRevLett.122.152001](#) | 2018-09-23

688. Search for invisible decays of a Higgs boson produced through vector boson fusion in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 793 \(2019\) 520](#) | [arXiv:1809.05937](#) | [DOI:10.1016/j.physletb.2019.04.025](#) | 2018-09-16

689. Search for leptoquarks coupled to third-generation quarks in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 121 \(2018\) 241802](#) | [arXiv:1809.05558](#) | [DOI:10.1103/PhysRevLett.121.241802](#) | 2018-09-14

690. Studies of $B^*_{s2}(5840)^0$ and $B_{s1}(5830)^0$ mesons including the observation of the $B^*_{s2}(5840)^0 \rightarrow B^0 K^0_S$ decay in proton-proton collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 939](#) | [arXiv:1809.03578](#) | [DOI:10.1140/epjc/s10052-018-6390-z](#) | 2018-09-10

691. Search for the associated production of the Higgs boson and a vector boson in proton-proton collisions at $\sqrt{s}=13$ TeV via Higgs boson decays to τ leptons

Sirunyan, Albert M et al.

[JHEP 06 \(2019\) 093](#) | [arXiv:1809.03590](#) | [DOI:10.1007/JHEP06\(2019\)093](#) | 2018-09-10

692. Performance of reconstruction and identification of τ leptons decaying to hadrons and ν_τ in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JINST 13 \(2018\) P10005](#) | [arXiv:1809.02816](#) | [DOI:10.1088/1748-0221/13/10/P10005](#) | 2018-09-08

693. Search for physics beyond the standard model in high-mass diphoton events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A. M. et al.

[Phys.Rev.D 98 \(2018\) 092001](#) | [arXiv:1809.00327](#) | [DOI:10.1103/PhysRevD.98.092001](#) | 2018-09-02

694. Charged-particle nuclear modification factors in XeXe collisions at $\sqrt{s_{\text{NN}}} = 5.44$ TeV
Sirunyan, Albert M et al.
[JHEP 10 \(2018\) 138](#) | [arXiv:1809.00201](#) | [DOI:10.1007/JHEP10\(2018\)138](#) | 2018-09-01
695. Observation of Higgs boson decay to bottom quarks
Sirunyan, A. M. et al.
[Phys.Rev.Lett. 121 \(2018\) 121801](#) | [arXiv:1808.08242](#) | [DOI:10.1103/PhysRevLett.121.121801](#) | 2018-08-24
696. Measurement of jet substructure observables in tt events from proton-proton collisions at $\sqrt{s} = 13$ TeV
Sirunyan, A. M. et al.
[Phys.Rev.D 98 \(2018\) 092014](#) | [arXiv:1808.07340](#) | [DOI:10.1103/PhysRevD.98.092014](#) | 2018-08-22
697. Search for a charged Higgs boson decaying to charm and bottom quarks in proton-proton collisions at $\sqrt{s} = 8$ TeV
Sirunyan, Albert M et al.
[JHEP 11 \(2018\) 115](#) | [arXiv:1808.06575](#) | [DOI:10.1007/JHEP11\(2018\)115](#) | 2018-08-20
698. Search for pair production of second-generation leptoquarks at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[Phys.Rev.D 99 \(2019\) 032014](#) | [arXiv:1808.05082](#) | [DOI:10.1103/PhysRevD.99.032014](#) | 2018-08-15
699. Search for an L_μ - L_τ gauge boson using $Z \rightarrow 4\mu$ events in proton-proton collisions at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[Phys.Lett.B 792 \(2019\) 345](#) | [arXiv:1808.03684](#) | [DOI:10.1016/j.physletb.2019.01.072](#) | 2018-08-10
700. Search for pair-produced resonances decaying to quark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[Phys.Rev.D 98 \(2018\) 112014](#) | [arXiv:1808.03124](#) | [DOI:10.1103/PhysRevD.98.112014](#) | 2018-08-09
701. Search for long-lived particles with displaced vertices in multijet events in proton-proton collisions at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[Phys.Rev.D 98 \(2018\) 092011](#) | [arXiv:1808.03078](#) | [DOI:10.1103/PhysRevD.98.092011](#) | 2018-08-09
702. Evidence for the associated production of a single top quark and a photon in proton-proton collisions at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[Phys.Rev.Lett. 121 \(2018\) 221802](#) | [arXiv:1808.02913](#) | [DOI:10.1103/PhysRevLett.121.221802](#) | 2018-08-08
703. Search for resonances in the mass spectrum of muon pairs produced in association with b quark jets in proton-proton collisions at $\sqrt{s} = 8$ and 13 TeV
Sirunyan, A.M. et al.
[JHEP 11 \(2018\) 161](#) | [arXiv:1808.01890](#) | [DOI:10.1007/JHEP11\(2018\)161](#) | 2018-08-06
704. Search for production of Higgs boson pairs in the four b quark final state using large-area jets in proton-proton collisions at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[JHEP 01 \(2019\) 040](#) | [arXiv:1808.01473](#) | [DOI:10.1007/JHEP01\(2019\)040](#) | 2018-08-04
705. Search for narrow $H\gamma$ resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[Phys.Rev.Lett. 122 \(2019\) 081804](#) | [arXiv:1808.01257](#) | [DOI:10.1103/PhysRevLett.122.081804](#) | 2018-08-03
706. Search for heavy resonances decaying into two Higgs bosons or into a Higgs boson and a W or Z boson in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2019\) 051](#) | [arXiv:1808.01365](#) | [DOI:10.1007/JHEP01\(2019\)051](#) | 2018-08-03

707. Search for a W' boson decaying to a τ lepton and a neutrino in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 792 \(2019\) 107](#) | [arXiv:1807.11421](#) | [DOI:10.1016/j.physletb.2019.01.069](#) | 2018-07-30

708. Searches for pair production of charginos and top squarks in final states with two oppositely charged leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 079](#) | [arXiv:1807.07799](#) | [DOI:10.1007/JHEP11\(2018\)079](#) | 2018-07-20

709. Search for the Higgs boson decaying to two muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 122 \(2019\) 021801](#) | [arXiv:1807.06325](#) | [DOI:10.1103/PhysRevLett.122.021801](#) | 2018-07-17

710. Search for dark matter particles produced in association with a top quark pair at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 122 \(2019\) 011803](#) | [arXiv:1807.06522](#) | [DOI:10.1103/PhysRevLett.122.011803](#) | 2018-07-17

711. Measurements of the differential jet cross section as a function of the jet mass in dijet events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 113](#) | [arXiv:1807.05974](#) | [DOI:10.1007/JHEP11\(2018\)113](#) | 2018-07-16

712. Measurement of inclusive and differential Higgs boson production cross sections in the diphoton decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 01 \(2019\) 183](#) | [arXiv:1807.03825](#) | [DOI:10.1007/JHEP01\(2019\)183](#) | 2018-07-10

713. Precision measurement of the structure of the CMS inner tracking system using nuclear interactions

Sirunyan, Albert M et al.

[JINST 13 \(2018\) P10034](#) | [arXiv:1807.03289](#) | [DOI:10.1088/1748-0221/13/10/P10034](#) | 2018-07-09

714. Search for heavy resonances decaying into a vector boson and a Higgs boson in final states with charged leptons, neutrinos and b quarks at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 172](#) | [arXiv:1807.02826](#) | [DOI:10.1007/JHEP11\(2018\)172](#) | 2018-07-08

715. Study of the underlying event in top quark pair production in pp collisions at 13 TeV

Sirunyan, Albert M. et al.

[Eur.Phys.J.C 79 \(2019\) 123](#) | [arXiv:1807.02810](#) | [DOI:10.1140/epjc/s10052-019-6620-z](#) | 2018-07-08

716. Search for supersymmetry in events with a τ lepton pair and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 151](#) | [arXiv:1807.02048](#) | [DOI:10.1007/JHEP11\(2018\)151](#) | 2018-07-05

717. Measurement of differential cross sections for inclusive isolated-photon and photon+jets production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 20](#) | [arXiv:1807.00782](#) | [DOI:10.1140/epjc/s10052-018-6482-9](#) | 2018-07-02

718. Measurement of charged particle spectra in minimum-bias events from proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M. et al.

[Eur.Phys.J.C 78 \(2018\) 697](#) | [arXiv:1806.11245](#) | [DOI:10.1140/epjc/s10052-018-6144-y](#) | 2018-06-29

719. Measurement of differential cross sections for Z boson pair production in association with jets at $\sqrt{s} = 8$ and 13 TeV

Sirunyan, Albert M. et al.

[Phys.Lett.B 789 \(2019\) 19](#) | [arXiv:1806.11073](#) | [DOI:10.1016/j.physletb.2018.11.007](#) | 2018-06-28

720. Search for heavy Majorana neutrinos in same-sign dilepton channels in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2019\) 122](#) | [arXiv:1806.10905](#) | [DOI:10.1007/JHEP01\(2019\)122](#) | 2018-06-28

721. Search for the decay of a Higgs boson in the $\ell\ell\gamma$ channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 152](#) | [arXiv:1806.05996](#) | [DOI:10.1007/JHEP11\(2018\)152](#) | 2018-06-15

722. Search for supersymmetric partners of electrons and muons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M. et al.

[Phys.Lett.B 790 \(2019\) 140](#) | [arXiv:1806.05264](#) | [DOI:10.1016/j.physletb.2019.01.005](#) | 2018-06-13

723. Measurements of properties of the Higgs boson decaying to a W boson pair in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M. et al.

[Phys.Lett.B 791 \(2019\) 96](#) | [arXiv:1806.05246](#) | [DOI:10.1016/j.physletb.2018.12.073](#) | 2018-06-13

724. Search for dark matter produced in association with a Higgs boson decaying to $\gamma\gamma$ or $\tau^+\tau^-$ at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 09 \(2018\) 046](#) | [arXiv:1806.04771](#) | [DOI:10.1007/JHEP09\(2018\)046](#) | 2018-06-12

725. Observation of the $Z\rightarrow\psi\ell^+\ell^-$ decay in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 121 \(2018\) 141801](#) | [arXiv:1806.04213](#) | [DOI:10.1103/PhysRevLett.121.141801](#) | 2018-06-11

726. Search for resonant pair production of Higgs bosons decaying to bottom quark-antiquark pairs in proton-proton collisions at 13 TeV

Sirunyan, Albert M. et al.

[JHEP 08 \(2018\) 152](#) | [arXiv:1806.03548](#) | [DOI:10.1007/JHEP08\(2018\)152](#) | 2018-06-09

727. Search for a singly produced third-generation scalar leptoquark decaying to a τ lepton and a bottom quark in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 07 \(2018\) 115](#) | [arXiv:1806.03472](#) | [DOI:10.1007/JHEP07\(2018\)115](#) | 2018-06-09

728. Search for pair-produced resonances each decaying into at least four quarks in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 121 \(2018\) 141802](#) | [arXiv:1806.01058](#) | [DOI:10.1103/PhysRevLett.121.141802](#) | 2018-06-04

729. Measurement of the weak mixing angle using the forward-backward asymmetry of Drell-Yan events in pp collisions at 8 TeV

Sirunyan, Albert M. et al.

[Eur.Phys.J.C 78 \(2018\) 701](#) | [arXiv:1806.00863](#) | [DOI:10.1140/epjc/s10052-018-6148-7](#) | 2018-06-03

730. Search for narrow and broad dijet resonances in proton-proton collisions at $\sqrt{s}=13$ TeV and constraints on dark matter mediators and other new particles

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 130](#) | [arXiv:1806.00843](#) | [DOI:10.1007/JHEP08\(2018\)130](#) | 2018-06-03

731. Angular analysis of the decay $B^+ \rightarrow K^+ \mu^+ \mu^-$ in proton-proton collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 98 \(2018\) 112011](#) | [arXiv:1806.00636](#) | [DOI:10.1103/PhysRevD.98.112011](#) | 2018-06-02

732. Search for Higgs boson pair production in the $\gamma\gamma b\bar{b}$ final state in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 788 \(2019\) 7](#) | [arXiv:1806.00408](#) | [DOI:10.1016/j.physletb.2018.10.056](#) | 2018-06-01

733. Search for beyond the standard model Higgs bosons decaying into a $b\bar{b}$ pair in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 113](#) | [arXiv:1805.12191](#) | [DOI:10.1007/JHEP08\(2018\)113](#) | 2018-05-30

734. Observation of the $\chi_{b1}(3P)$ and $\chi_{b2}(3P)$ and measurement of their masses

Sirunyan, A. M. et al.

[Phys.Rev.Lett. 121 \(2018\) 092002](#) | [arXiv:1805.11192](#) | [DOI:10.1103/PhysRevLett.121.092002](#) | 2018-05-28

735. Constraints on models of scalar and vector leptoquarks decaying to a quark and a neutrino at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 98 \(2018\) 032005](#) | [arXiv:1805.10228](#) | [DOI:10.1103/PhysRevD.98.032005](#) | 2018-05-25

736. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state with two b quarks and two τ leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 785 \(2018\) 462](#) | [arXiv:1805.10191](#) | [DOI:10.1016/j.physletb.2018.08.057](#) | 2018-05-25

737. Measurement of nuclear modification factors of $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 790 \(2019\) 270](#) | [arXiv:1805.09215](#) | [DOI:10.1016/j.physletb.2019.01.006](#) | 2018-05-23

738. Measurement of the production cross section for single top quarks in association with W bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2018\) 117](#) | [arXiv:1805.07399](#) | [DOI:10.1007/JHEP10\(2018\)117](#) | 2018-05-18

739. Search for black holes and sphalerons in high-multiplicity final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2018\) 042](#) | [arXiv:1805.06013](#) | [DOI:10.1007/JHEP11\(2018\)042](#) | 2018-05-15

740. Search for top squarks decaying via four-body or chargino-mediated modes in single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[JHEP 09 \(2018\) 065](#) | [arXiv:1805.05784](#) | [DOI:10.1007/JHEP09\(2018\)065](#) | 2018-05-15

741. Measurement of the groomed jet mass in PbPb and pp collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2018\) 161](#) | [arXiv:1805.05145](#) | [DOI:10.1007/JHEP10\(2018\)161](#) | 2018-05-14

742. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state of two muons and two τ leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2018\) 018](#) | [arXiv:1805.04865](#) | [DOI:10.1007/JHEP11\(2018\)018](#) | 2018-05-13

743. Constraining gluon distributions in nuclei using dijets in proton-proton and proton-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 121 \(2018\) 062002](#) | [arXiv:1805.04736](#) | [DOI:10.1103/PhysRevLett.121.062002](#) | 2018-05-12

744. Search for vector-like T and B quark pairs in final states with leptons at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 177](#) | [arXiv:1805.04758](#) | [DOI:10.1007/JHEP08\(2018\)177](#) | 2018-05-12

745. Measurement of prompt $\psi(2S)$ production cross sections in proton-lead and proton-proton collisions at $\sqrt{s_{NN}}= 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 790 \(2019\) 509](#) | [arXiv:1805.02248](#) | [DOI:10.1016/j.physletb.2019.01.058](#) | 2018-05-06

746. Measurement of the top quark mass with lepton+jets final states using p p collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 891](#) | [arXiv:1805.01428](#) | [DOI:10.1140/epjc/s10052-018-6332-9](#) | 2018-05-03

747. Elliptic flow of charm and strange hadrons in high-multiplicity pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, A. M. et al.

[Phys.Rev.Lett. 121 \(2018\) 082301](#) | [arXiv:1804.09767](#) | [DOI:10.1103/PhysRevLett.121.082301](#) | 2018-04-25

748. Search for disappearing tracks as a signature of new long-lived particles in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 016](#) | [arXiv:1804.07321](#) | [DOI:10.1007/JHEP08\(2018\)016](#) | 2018-04-19

749. Measurement of differential cross sections for Z boson production in association with jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 965](#) | [arXiv:1804.05252](#) | [DOI:10.1140/epjc/s10052-018-6373-0](#) | 2018-04-14

750. Performance of the CMS muon detector and muon reconstruction with proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, A.M. et al.

[JINST 13 \(2018\) P06015](#) | [arXiv:1804.04528](#) | [DOI:10.1088/1748-0221/13/06/P06015](#) | 2018-04-12

751. Search for ttH production in the $H \rightarrow b\bar{b}$ decay channel with leptonic tt decays in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 026](#) | [arXiv:1804.03682](#) | [DOI:10.1007/JHEP03\(2019\)026](#) | 2018-04-10

752. Measurements of Higgs boson properties in the diphoton decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[JHEP 11 \(2018\) 185](#) | [arXiv:1804.02716](#) | [DOI:10.1007/JHEP11\(2018\)185](#) | 2018-04-08

753. Observation of ttH production

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 231801](#) | [arXiv:1804.02610](#) | [DOI:10.1103/PhysRevLett.120.231801](#) | 2018-04-07

754. Search for a new scalar resonance decaying to a pair of Z bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 127](#) | [arXiv:1804.01939](#) | [DOI:10.1007/JHEP06\(2018\)127](#) | 2018-04-05

755. Search for a heavy right-handed W boson and a heavy neutrino in events with two same-flavor leptons and two jets at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 148](#) | [arXiv:1803.11116](#) | [DOI:10.1007/JHEP05\(2018\)148](#) | 2018-03-29

756. Search for high-mass resonances in final states with a lepton and missing transverse momentum at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 128](#) | [arXiv:1803.11133](#) | [DOI:10.1007/JHEP06\(2018\)128](#) | 2018-03-29

757. Search for a heavy resonance decaying into a Z boson and a Z or W boson in $2\ell 2q$ final states at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 09 \(2018\) 101](#) | [arXiv:1803.10093](#) | [DOI:10.1007/JHEP09\(2018\)101](#) | 2018-03-27

758. Measurement of differential cross sections for the production of top quark pairs and of additional jets in lepton+jets events from pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 112003](#) | [arXiv:1803.08856](#) | [DOI:10.1103/PhysRevD.97.112003](#) | 2018-03-23

759. Search for new physics in dijet angular distributions using proton-proton collisions at $\sqrt{s}=13$ TeV and constraints on dark matter and other models

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 789](#) | [arXiv:1803.08030](#) | [DOI:10.1140/epjc/s10052-018-6242-x](#) | 2018-03-21

760. Search for ttH production in the all-jet final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 101](#) | [arXiv:1803.06986](#) | [DOI:10.1007/JHEP06\(2018\)101](#) | 2018-03-19

761. Search for additional neutral MSSM Higgs bosons in the $\tau\tau$ final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 09 \(2018\) 007](#) | [arXiv:1803.06553](#) | [DOI:10.1007/JHEP09\(2018\)007](#) | 2018-03-17

762. Search for high-mass resonances in dilepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 120](#) | [arXiv:1803.06292](#) | [DOI:10.1007/JHEP06\(2018\)120](#) | 2018-03-16

763. Evidence for associated production of a Higgs boson with a top quark pair in final states with electrons, muons, and hadronically decaying τ leptons at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 066](#) | [arXiv:1803.05485](#) | [DOI:10.1007/JHEP08\(2018\)066](#) | 2018-03-14

764. Observation of proton-tagged, central (semi)exclusive production of high-mass lepton pairs in pp collisions at 13 TeV with the CMS-TOTEM precision proton spectrometer

Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 153](#) | [arXiv:1803.04496](#) | [DOI:10.1007/JHEP07\(2018\)153](#) | 2018-03-12

765. Measurements of differential cross sections of top quark pair production as a function of kinematic event variables in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JHEP 06 \(2018\) 002](#) | [arXiv:1803.03991](#) | [DOI:10.1007/JHEP06\(2018\)002](#) | 2018-03-11

766. Search for a heavy resonance decaying into a Z boson and a vector boson in the $\nu\nu q\bar{q}$ final state

Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 075](#) | [arXiv:1803.03838](#) | [DOI:10.1007/JHEP07\(2018\)075](#) | 2018-03-10

767. Search for third-generation scalar leptoquarks decaying to a top quark and a τ lepton at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 707](#) | [arXiv:1803.02864](#) | [DOI:10.1140/epjc/s10052-018-6143-z](#) | 2018-03-07

768. Jet properties in PbPb and pp collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 006](#) | [arXiv:1803.00042](#) | [DOI:10.1007/JHEP05\(2018\)006](#) | 2018-02-28

769. Search for a heavy resonance decaying to a pair of vector bosons in the lepton plus merged jet final state at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 088](#) | [arXiv:1802.09407](#) | [DOI:10.1007/JHEP05\(2018\)088](#) | 2018-02-26

770. Search for narrow resonances in the b-tagged dijet mass spectrum in proton-proton collisions at $\sqrt{s}=8$ TeV

Sirunyan, A. M. et al.

[Phys.Rev.Lett. 120 \(2018\) 201801](#) | [arXiv:1802.06149](#) | [DOI:10.1103/PhysRevLett.120.201801](#) | 2018-02-16

771. Measurement of the Λ_b polarization and angular parameters in $\Lambda_b \rightarrow J/\psi \Lambda$ decays from pp collisions at $\sqrt{s}=7$ and 8 TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 072010](#) | [arXiv:1802.04867](#) | [DOI:10.1103/PhysRevD.97.072010](#) | 2018-02-13

772. Search for heavy neutral leptons in events with three charged leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 221801](#) | [arXiv:1802.02965](#) | [DOI:10.1103/PhysRevLett.120.221801](#) | 2018-02-08

773. Measurement of the inelastic proton-proton cross section at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 161](#) | [arXiv:1802.02613](#) | [DOI:10.1007/JHEP07\(2018\)161](#) | 2018-02-07

774. Search for natural and split supersymmetry in proton-proton collisions at $\sqrt{s}=13$ TeV in final states with jets and missing transverse momentum

Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 025](#) | [arXiv:1802.02110](#) | [DOI:10.1007/JHEP05\(2018\)025](#) | 2018-02-06

775. Search for single production of vector-like quarks decaying to a b quark and a Higgs boson

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 031](#) | [arXiv:1802.01486](#) | [DOI:10.1007/JHEP06\(2018\)031](#) | 2018-02-05

776. Search for lepton-flavor violating decays of heavy resonances and quantum black holes to $e\mu$ final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2018\) 073](#) | [arXiv:1802.01122](#) | [DOI:10.1007/JHEP04\(2018\)073](#) | 2018-02-04

777. Comparing transverse momentum balance of b jet pairs in pp and PbPb collisions at $\sqrt{s_{\text{NN}}}=5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2018\) 181](#) | [arXiv:1802.00707](#) | [DOI:10.1007/JHEP03\(2018\)181](#) | 2018-02-02

778. Observation of Medium-Induced Modifications of Jet Fragmentation in Pb-Pb Collisions at $\sqrt{s_{\text{NN}}}=5.02$ TeV Using Isolated Photon-Tagged Jets

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 121 \(2018\) 242301](#) | [arXiv:1801.04895](#) | [DOI:10.1103/PhysRevLett.121.242301](#) | 2018-01-15

779. Measurement of the $Z\gamma^*\rightarrow\tau\tau$ cross section in pp collisions at $\sqrt{s}=13$ TeV and validation of τ lepton analysis techniques

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 708](#) | [arXiv:1801.03535](#) | [DOI:10.1140/epjc/s10052-018-6146-9](#) | 2018-01-10

780. Search for new physics in events with two soft oppositely charged leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 782 \(2018\) 440](#) | [arXiv:1801.01846](#) | [DOI:10.1016/j.physletb.2018.05.062](#) | 2018-01-05

781. Search for decays of stopped exotic long-lived particles produced in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JHEP 05 \(2018\) 127](#) | [arXiv:1801.00359](#) | [DOI:10.1007/JHEP05\(2018\)127](#) | 2017-12-31

782. Electroweak production of two jets in association with a Z boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 589](#) | [arXiv:1712.09814](#) | [DOI:10.1140/epjc/s10052-018-6049-9](#) | 2017-12-28

783. Search for R-parity violating supersymmetry in pp collisions at $\sqrt{s}=13$ TeV using b jets in a final state with a single lepton, many jets, and high sum of large-radius jet masses

Sirunyan, Albert M et al.

[Phys.Lett.B 783 \(2018\) 114](#) | [arXiv:1712.08920](#) | [DOI:10.1016/j.physletb.2018.06.028](#) | 2017-12-24

784. Measurement of prompt and nonprompt charmonium suppression in PbPb collisions at 5.02 TeV

Sirunyan, Albert M. et al.

[Eur.Phys.J.C 78 \(2018\) 509](#) | [arXiv:1712.08959](#) | [DOI:10.1140/epjc/s10052-018-5950-6](#) | 2017-12-24

785. Search for Physics Beyond the Standard Model in Events with High-Momentum Higgs Bosons and Missing Transverse Momentum in Proton-Proton Collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 241801](#) | [arXiv:1712.08501](#) | [DOI:10.1103/PhysRevLett.120.241801](#) | 2017-12-22

786. Search for lepton flavour violating decays of the Higgs boson to $\mu\tau$ and $e\tau$ in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 001](#) | [arXiv:1712.07173](#) | [DOI:10.1007/JHEP06\(2018\)001](#) | 2017-12-19

787. Bose-Einstein correlations in pp, pPb, and PbPb collisions at $\sqrt{s_{\text{NN}}}=0.9-7$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.C 97 \(2018\) 064912](#) | [arXiv:1712.07198](#) | [DOI:10.1103/PhysRevC.97.064912](#) | 2017-12-19

788. Identification of heavy-flavour jets with the CMS detector in pp collisions at 13 TeV

Sirunyan, A.M. et al.

[JINST 13 \(2018\) P05011](#) | [arXiv:1712.07158](#) | [DOI:10.1088/1748-0221/13/05/P05011](#) | 2017-12-19

789. Search for the X(5568) state decaying into $B^0_{\text{str}} \pi^\pm$ in proton-proton collisions at $\sqrt{s} = 8 \text{ TeV}$

Sirunyan, A. M. et al.

[Phys.Rev.Lett. 120 \(2018\) 202005](#) | [arXiv:1712.06144](#) | [DOI:10.1103/PhysRevLett.120.202005](#) | 2017-12-17

790. Azimuthal correlations for inclusive 2-jet, 3-jet, and 4-jet events in pp collisions at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 566](#) | [arXiv:1712.05471](#) | [DOI:10.1140/epjc/s10052-018-6033-4](#) | 2017-12-14

791. Search for $Z\gamma$ resonances using leptonic and hadronic final states in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 09 \(2018\) 148](#) | [arXiv:1712.03143](#) | [DOI:10.1007/JHEP09\(2018\)148](#) | 2017-12-08

792. Measurement of the associated production of a single top quark and a Z boson in pp collisions at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 779 \(2018\) 358](#) | [arXiv:1712.02825](#) | [DOI:10.1016/j.physletb.2018.02.025](#) | 2017-12-07

793. Search for the flavor-changing neutral current interactions of the top quark and the Higgs boson which decays into a pair of b quarks at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 102](#) | [arXiv:1712.02399](#) | [DOI:10.1007/JHEP06\(2018\)102](#) | 2017-12-06

794. Constraints on the double-parton scattering cross section from same-sign W boson pair production in proton-proton collisions at $\sqrt{s} = 8 \text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 02 \(2018\) 032](#) | [arXiv:1712.02280](#) | [DOI:10.1007/JHEP02\(2018\)032](#) | 2017-12-06

795. Search for new physics in final states with an energetic jet or a hadronically decaying W or Z boson and transverse momentum imbalance at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, A. M. et al.

[Phys.Rev.D 97 \(2018\) 092005](#) | [arXiv:1712.02345](#) | [DOI:10.1103/PhysRevD.97.092005](#) | 2017-12-06

796. Search for pair production of excited top quarks in the lepton + jets final state

Sirunyan, Albert M et al.

[Phys.Lett.B 778 \(2018\) 349](#) | [arXiv:1711.10949](#) | [DOI:10.1016/j.physletb.2018.01.049](#) | 2017-11-29

797. Study of jet quenching with isolated-photon+jet correlations in PbPb and pp collisions at $\sqrt{s_{\text{NN}}} = 5.02 \text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 785 \(2018\) 14](#) | [arXiv:1711.09738](#) | [DOI:10.1016/j.physletb.2018.07.061](#) | 2017-11-27

798. Search for new long-lived particles at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 780 \(2018\) 432](#) | [arXiv:1711.09120](#) | [DOI:10.1016/j.physletb.2018.03.019](#) | 2017-11-24

799. Search for gauge-mediated supersymmetry in events with at least one photon and missing transverse momentum in pp collisions at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 780 \(2018\) 118](#) | [arXiv:1711.08008](#) | [DOI:10.1016/j.physletb.2018.02.045](#) | 2017-11-21

800. Non-Gaussian elliptic-flow fluctuations in PbPb collisions at $\sqrt{s_{\text{NN}}} = 5.02 \text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 789 \(2019\) 643](#) | [arXiv:1711.05594](#) | [DOI:10.1016/j.physletb.2018.11.063](#) | 2017-11-15

801. Search for excited quarks of light and heavy flavor in γ + jet final states in proton-proton collisions at $\sqrt{s} = 13\text{TeV}$
Sirunyan, Albert M et al.
[Phys.Lett.B 781 \(2018\) 390](#) | [arXiv:1711.04652](#) | [DOI:10.1016/j.physletb.2018.04.007](#) | 2017-11-13
802. Search for ZZ resonances in the $2\ell 2\nu$ final state in proton-proton collisions at 13 TeV
Sirunyan, Albert M et al.
[JHEP 03 \(2018\) 003](#) | [arXiv:1711.04370](#) | [DOI:10.1007/JHEP03\(2018\)003](#) | 2017-11-12
803. Measurement of the underlying event activity in inclusive Z boson production in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$
Sirunyan, A.M. et al.
[JHEP 07 \(2018\) 032](#) | [arXiv:1711.04299](#) | [DOI:10.1007/JHEP07\(2018\)032](#) | 2017-11-12
804. Measurement of the inclusive tt cross section in pp collisions at $\sqrt{s}=5.02\text{ TeV}$ using final states with at least one charged lepton
Sirunyan, A.M. et al.
[JHEP 03 \(2018\) 115](#) | [arXiv:1711.03143](#) | [DOI:10.1007/JHEP03\(2018\)115](#) | 2017-11-08
805. Measurement of the cross section for top quark pair production in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$
Sirunyan, Albert M et al.
[JHEP 08 \(2018\) 011](#) | [arXiv:1711.02547](#) | [DOI:10.1007/JHEP08\(2018\)011](#) | 2017-11-07
806. Measurement of associated Z + charm production in proton-proton collisions at $\sqrt{s} = 8\text{ TeV}$
Sirunyan, A.M. et al.
[Eur.Phys.J.C 78 \(2018\) 287](#) | [arXiv:1711.02143](#) | [DOI:10.1140/epjc/s10052-018-5752-x](#) | 2017-11-06
807. Search for top squarks and dark matter particles in opposite-charge dilepton final states at $\sqrt{s}= 13\text{ TeV}$
Sirunyan, Albert M et al.
[Phys.Rev.D 97 \(2018\) 032009](#) | [arXiv:1711.00752](#) | [DOI:10.1103/PhysRevD.97.032009](#) | 2017-11-02
808. Search for new physics in events with a leptonically decaying Z boson and a large transverse momentum imbalance in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$
Sirunyan, A.M. et al.
[Eur.Phys.J.C 78 \(2018\) 291](#) | [arXiv:1711.00431](#) | [DOI:10.1140/epjc/s10052-018-5740-1](#) | 2017-11-01
809. Measurement of quarkonium production cross sections in pp collisions at $\sqrt{s}= 13\text{ TeV}$
Sirunyan, A.M. et al.
[Phys.Lett.B 780 \(2018\) 251](#) | [arXiv:1710.11002](#) | [DOI:10.1016/j.physletb.2018.02.033](#) | 2017-10-30
810. Search for supersymmetry in proton-proton collisions at 13 TeV using identified top quarks
Sirunyan, Albert M et al.
[Phys.Rev.D 97 \(2018\) 012007](#) | [arXiv:1710.11188](#) | [DOI:10.1103/PhysRevD.97.012007](#) | 2017-10-30
811. Search for supersymmetry in events with at least three electrons or muons, jets, and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$
Sirunyan, Albert M et al.
[JHEP 02 \(2018\) 067](#) | [arXiv:1710.09154](#) | [DOI:10.1007/JHEP02\(2018\)067](#) | 2017-10-25
812. Pseudorapidity distributions of charged hadrons in proton-lead collisions at $\sqrt{s_{NN}} = 5.02$ and 8.16 TeV
Sirunyan, Albert M et al.
[JHEP 01 \(2018\) 045](#) | [arXiv:1710.09355](#) | [DOI:10.1007/JHEP01\(2018\)045](#) | 2017-10-25

813. Measurement of b hadron lifetimes in pp collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 457](#) | [arXiv:1710.08949](#) | [DOI:10.1140/epjc/s10052-018-5929-3](#) | 2017-10-24

814. Measurement of differential cross sections in the kinematic angular variable ϕ^* for inclusive Z boson production in pp collisions at $\sqrt{s} = 8$ TeV

Sirunyan, A.M. et al.

[JHEP 03 \(2018\) 172](#) | [arXiv:1710.07955](#) | [DOI:10.1007/JHEP03\(2018\)172](#) | 2017-10-22

815. Pseudorapidity and transverse momentum dependence of flow harmonics in pPb and PbPb collisions

Sirunyan, Albert M et al.

[Phys.Rev.C 98 \(2018\) 044902](#) | [arXiv:1710.07864](#) | [DOI:10.1103/PhysRevC.98.044902](#) | 2017-10-21

816. Search for a massive resonance decaying to a pair of Higgs bosons in the four b quark final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[Phys.Lett.B 781 \(2018\) 244](#) | [arXiv:1710.04960](#) | [DOI:10.1016/j.physletb.2018.03.084](#) | 2017-10-13

817. Measurement of angular parameters from the decay $B^0 \rightarrow K^{*0} \mu^+ \mu^-$ in proton-proton collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 781 \(2018\) 517](#) | [arXiv:1710.02846](#) | [DOI:10.1016/j.physletb.2018.04.030](#) | 2017-10-08

818. Study of dijet events with a large rapidity gap between the two leading jets in pp collisions at $\sqrt{s} = 7$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 242](#) | [arXiv:1710.02586](#) | [DOI:10.1140/epjc/s10052-018-5691-6](#) | 2017-10-06

819. Search for pair production of vector-like quarks in the bWbW channel from proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 779 \(2018\) 82](#) | [arXiv:1710.01539](#) | [DOI:10.1016/j.physletb.2018.01.077](#) | 2017-10-04

820. Search for low mass vector resonances decaying into quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2018\) 097](#) | [arXiv:1710.00159](#) | [DOI:10.1007/JHEP01\(2018\)097](#) | 2017-09-30

821. Search for supersymmetry in events with one lepton and multiple jets exploiting the angular correlation between the lepton and the missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[Phys.Lett.B 780 \(2018\) 384](#) | [arXiv:1709.09814](#) | [DOI:10.1016/j.physletb.2018.03.028](#) | 2017-09-28

822. Observation of Correlated Azimuthal Anisotropy Fourier Harmonics in pp and p+Pb Collisions at the LHC

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 092301](#) | [arXiv:1709.09189](#) | [DOI:10.1103/PhysRevLett.120.092301](#) | 2017-09-26

823. Measurements of the $pp \rightarrow ZZ$ production cross section and the $Z \rightarrow 4\ell$ branching fraction, and constraints on anomalous triple gauge couplings at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 165](#) | [arXiv:1709.08601](#) | [DOI:10.1140/epjc/s10052-018-5567-9](#) | 2017-09-25

824. Observation of top quark production in proton-nucleus collisions

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 242001](#) | [arXiv:1709.07411](#) | [DOI:10.1103/PhysRevLett.119.242001](#) | 2017-09-21

825. Evidence for the Higgs boson decay to a bottom quark-antiquark pair

Sirunyan, Albert M et al.

[Phys.Lett.B 780 \(2018\) 501](#) | [arXiv:1709.07497](#) | [DOI:10.1016/j.physletb.2018.02.050](#) | 2017-09-21

826. Observation of electroweak production of same-sign W boson pairs in the two jet and two same-sign lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 081801](#) | [arXiv:1709.05822](#) | [DOI:10.1103/PhysRevLett.120.081801](#) | 2017-09-18

827. Inclusive search for a highly boosted Higgs boson decaying to a bottom quark-antiquark pair

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 071802](#) | [arXiv:1709.05543](#) | [DOI:10.1103/PhysRevLett.120.071802](#) | 2017-09-16

828. Combination of inclusive and differential tt charge asymmetry measurements using ATLAS and CMS data at $\sqrt{s}=7$ and 8 TeV

Aaboud, Morad et al.

[JHEP 04 \(2018\) 033](#) | [arXiv:1709.05327](#) | [DOI:10.1007/JHEP04\(2018\)033](#) | 2017-09-15

829. Search for electroweak production of charginos and neutralinos in multilepton final states in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, A.M. et al.

[JHEP 03 \(2018\) 166](#) | [arXiv:1709.05406](#) | [DOI:10.1007/JHEP03\(2018\)166](#) | 2017-09-15

830. Search for Higgsino pair production in pp collisions at $\sqrt{s} = 13$ TeV in final states with large missing transverse momentum and two Higgs bosons decaying via $H \rightarrow b \bar{b}$

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 032007](#) | [arXiv:1709.04896](#) | [DOI:10.1103/PhysRevD.97.032007](#) | 2017-09-14

831. Search for supersymmetry with Higgs boson to diphoton decays using the razor variables at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 779 \(2018\) 166](#) | [arXiv:1709.00384](#) | [DOI:10.1016/j.physletb.2017.12.069](#) | 2017-09-01

832. Measurement of the Splitting Function in pp and Pb-Pb Collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 142302](#) | [arXiv:1708.09429](#) | [DOI:10.1103/PhysRevLett.120.142302](#) | 2017-08-30

833. Probing the chiral magnetic wave in pPb and PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV using charge-dependent azimuthal anisotropies

Sirunyan, Albert M et al.

[Phys.Rev.C 100 \(2019\) 064908](#) | [arXiv:1708.08901](#) | [DOI:10.1103/PhysRevC.100.064908](#) | 2017-08-29

834. Search for heavy resonances decaying to a top quark and a bottom quark in the lepton+jets final state in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 777 \(2018\) 39](#) | [arXiv:1708.08539](#) | [DOI:10.1016/j.physletb.2017.12.006](#) | 2017-08-28

835. Search for Evidence of the Type-III Seesaw Mechanism in Multilepton Final States in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 221802](#) | [arXiv:1708.07962](#) | [DOI:10.1103/PhysRevLett.119.221802](#) | 2017-08-26

836. Measurement of normalized differential tt cross sections in the dilepton channel from pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JHEP 04 \(2018\) 060](#) | [arXiv:1708.07638](#) | [DOI:10.1007/JHEP04\(2018\)060](#) | 2017-08-25

837. Principal-component analysis of two-particle azimuthal correlations in PbPb and pPb collisions at CMS

Sirunyan, A.M. et al.

[Phys.Rev.C 96 \(2017\) 064902](#) | [arXiv:1708.07113](#) | [DOI:10.1103/PhysRevC.96.064902](#) | 2017-08-23

838. Search for massive resonances decaying into WW, WZ, ZZ, qW, and qZ with dijet final states at $\sqrt{s}=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 072006](#) | [arXiv:1708.05379](#) | [DOI:10.1103/PhysRevD.97.072006](#) | 2017-08-17

839. Nuclear modification factor of D^0 mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 782 \(2018\) 474](#) | [arXiv:1708.04962](#) | [DOI:10.1016/j.physletb.2018.05.074](#) | 2017-08-16

840. Search for resonant and nonresonant Higgs boson pair production in the $b\bar{b}\nu\ell\nu$ final state in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 01 \(2018\) 054](#) | [arXiv:1708.04188](#) | [DOI:10.1007/JHEP01\(2018\)054](#) | 2017-08-14

841. Measurement of prompt D^0 meson azimuthal anisotropy in Pb-Pb collisions at $\sqrt{s_{NN}} = 5.02\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 202301](#) | [arXiv:1708.03497](#) | [DOI:10.1103/PhysRevLett.120.202301](#) | 2017-08-11

842. Measurement of vector boson scattering and constraints on anomalous quartic couplings from events with four leptons and two jets in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 774 \(2017\) 682](#) | [arXiv:1708.02812](#) | [DOI:10.1016/j.physletb.2017.10.020](#) | 2017-08-09

843. Search for vectorlike light-flavor quark partners in proton-proton collisions at $\sqrt{s}=8\text{ TeV}$

Sirunyan, A. M. et al.

[Phys.Rev.D 97 \(2018\) 072008](#) | [arXiv:1708.02510](#) | [DOI:10.1103/PhysRevD.97.072008](#) | 2017-08-08

844. Constraints on the chiral magnetic effect using charge-dependent azimuthal correlations in pPb and PbPb collisions at the CERN Large Hadron Collider

Sirunyan, Albert M et al.

[Phys.Rev.C 97 \(2018\) 044912](#) | [arXiv:1708.01602](#) | [DOI:10.1103/PhysRevC.97.044912](#) | 2017-08-04

845. Search for single production of a vector-like T quark decaying to a Z boson and a top quark in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Sirunyan, A.M. et al.

[Phys.Lett.B 781 \(2018\) 574](#) | [arXiv:1708.01062](#) | [DOI:10.1016/j.physletb.2018.04.036](#) | 2017-08-03

846. Observation of the Higgs boson decay to a pair of τ leptons with the CMS detector

Sirunyan, Albert M et al.

[Phys.Lett.B 779 \(2018\) 283](#) | [arXiv:1708.00373](#) | [DOI:10.1016/j.physletb.2018.02.004](#) | 2017-08-01

847. Search for a light pseudoscalar Higgs boson produced in association with bottom quarks in pp collisions at $\sqrt{s}=8\text{ TeV}$

Sirunyan, A.M. et al.

[JHEP 11 \(2017\) 010](#) | [arXiv:1707.07283](#) | [DOI:10.1007/JHEP11\(2017\)010](#) | 2017-07-23

848. Search for the pair production of third-generation squarks with two-body decays to a bottom or charm quark and a neutralino in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 778 \(2018\) 263](#) | [arXiv:1707.07274](#) | [DOI:10.1016/j.physletb.2018.01.012](#) | 2017-07-23

849. Measurement of the differential cross sections for the associated production of a W boson and jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 96 \(2017\) 072005](#) | [arXiv:1707.05979](#) | [DOI:10.1103/PhysRevD.96.072005](#) | 2017-07-19

850. Search for supersymmetry in events with at least one photon, missing transverse momentum, and large transverse event activity in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2017\) 142](#) | [arXiv:1707.06193](#) | [DOI:10.1007/JHEP12\(2017\)142](#) | 2017-07-19

851. Search for direct production of supersymmetric partners of the top quark in the all-jets final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 005](#) | [arXiv:1707.03316](#) | [DOI:10.1007/JHEP10\(2017\)005](#) | 2017-07-11

852. Search for natural supersymmetry in events with top quark pairs and photons in pp collisions at $\sqrt{s} = 8$ TeV

Sirunyan, A.M. et al.

[JHEP 03 \(2018\) 167](#) | [arXiv:1707.03325](#) | [DOI:10.1007/JHEP03\(2018\)167](#) | 2017-07-11

853. Search for Higgs boson pair production in events with two bottom quarks and two tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 778 \(2018\) 101](#) | [arXiv:1707.02909](#) | [DOI:10.1016/j.physletb.2018.01.001](#) | 2017-07-10

854. Search for heavy resonances that decay into a vector boson and a Higgs boson in hadronic final states at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 636](#) | [arXiv:1707.01303](#) | [DOI:10.1140/epjc/s10052-017-5192-z](#) | 2017-07-05

855. Constraints on anomalous Higgs boson couplings using production and decay information in the four-lepton final state

Sirunyan, Albert M et al.

[Phys.Lett.B 775 \(2017\) 1](#) | [arXiv:1707.00541](#) | [DOI:10.1016/j.physletb.2017.10.021](#) | 2017-07-03

856. Search for Higgs boson pair production in the $b\bar{b}\tau\tau$ final state in proton-proton collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 96 \(2017\) 072004](#) | [arXiv:1707.00350](#) | [DOI:10.1103/PhysRevD.96.072004](#) | 2017-07-02

857. Measurement of charged pion, kaon, and proton production in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 96 \(2017\) 112003](#) | [arXiv:1706.10194](#) | [DOI:10.1103/PhysRevD.96.112003](#) | 2017-06-30

858. The Phase-2 Upgrade of the CMS Tracker

Klein, K. et al.

[DOI:10.17181/CERN.QZ28.FLHW](#) | 2017-06-30

859. Measurements of properties of the Higgs boson decaying into the four-lepton final state in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2017\) 047](#) | [arXiv:1706.09936](#) | [DOI:10.1007/JHEP11\(2017\)047](#) | 2017-06-29

860. Exclusive and semi-exclusive $\pi^+\pi^-$ production in proton-proton collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[arXiv:1706.08310](#) | 2017-06-26

861. Search for a heavy composite Majorana neutrino in the final state with two leptons and two quarks at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[Phys.Lett.B 775 \(2017\) 315](#) | [arXiv:1706.08578](#) | [DOI:10.1016/j.physletb.2017.11.001](#) | 2017-06-26

862. Measurement of the semileptonic $t\bar{t} + \gamma$ production cross section in pp collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 006](#) | [arXiv:1706.08128](#) | [DOI:10.1007/JHEP10\(2017\)006](#) | 2017-06-25

863. Measurements of jet charge with dijet events in pp collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 131](#) | [arXiv:1706.05868](#) | [DOI:10.1007/JHEP10\(2017\)131](#) | 2017-06-19

864. Suppression of Excited Υ States Relative to the Ground State in Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 142301](#) | [arXiv:1706.05984](#) | [DOI:10.1103/PhysRevLett.120.142301](#) | 2017-06-19

865. Particle-flow reconstruction and global event description with the CMS detector

Sirunyan, A.M. et al.

[JINST 12 \(2017\) P10003](#) | [arXiv:1706.04965](#) | [DOI:10.1088/1748-0221/12/10/P10003](#) | 2017-06-15

866. Searches for W' bosons decaying to a top quark and a bottom quark in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2017\) 029](#) | [arXiv:1706.04260](#) | [DOI:10.1007/JHEP08\(2017\)029](#) | 2017-06-13

867. Search for new physics in the monophoton final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 073](#) | [arXiv:1706.03794](#) | [DOI:10.1007/JHEP10\(2017\)073](#) | 2017-06-12

868. Search for pair production of vector-like T and B quarks in single-lepton final states using boosted jet substructure in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JHEP 11 \(2017\) 085](#) | [arXiv:1706.03408](#) | [DOI:10.1007/JHEP11\(2017\)085](#) | 2017-06-11

869. Search for dark matter produced in association with heavy-flavor quark pairs in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[Eur.Phys.J.C 77 \(2017\) 845](#) | [arXiv:1706.02581](#) | [DOI:10.1140/epjc/s10052-017-5317-4](#) | 2017-06-08

870. Search for top quark partners with charge $5/3$ in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2017\) 073](#) | [arXiv:1705.10967](#) | [DOI:10.1007/JHEP08\(2017\)073](#) | 2017-05-31

871. Search for Low Mass Vector Resonances Decaying to Quark-Antiquark Pairs in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 111802](#) | [arXiv:1705.10532](#) | [DOI:10.1103/PhysRevLett.119.111802](#) | 2017-05-30

872. Measurements of $t\bar{t}$ cross sections in association with b jets and inclusive jets and their ratio using dilepton final states in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 776 \(2018\) 355](#) | [arXiv:1705.10141](#) | [DOI:10.1016/j.physletb.2017.11.043](#) | 2017-05-29

873. Combination of searches for heavy resonances decaying to WW , WZ , ZZ , WH , and ZH boson pairs in proton–proton collisions at $\sqrt{s}=8$ and 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 774 \(2017\) 533](#) | [arXiv:1705.09171](#) | [DOI:10.1016/j.physletb.2017.09.083](#) | 2017-05-25

874. Search for Supersymmetry in pp Collisions at $\sqrt{s}=13$ TeV in the Single-Lepton Final State Using the Sum of Masses of Large-Radius Jets

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 151802](#) | [arXiv:1705.04673](#) | [DOI:10.1103/PhysRevLett.119.151802](#) | 2017-05-12

875. Measurement of the $B^{\pm}$ Meson Nuclear Modification Factor in Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 152301](#) | [arXiv:1705.04727](#) | [DOI:10.1103/PhysRevLett.119.152301](#) | 2017-05-12

876. Search for Charged Higgs Bosons Produced via Vector Boson Fusion and Decaying into a Pair of W and Z Bosons Using pp Collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 141802](#) | [arXiv:1705.02942](#) | [DOI:10.1103/PhysRevLett.119.141802](#) | 2017-05-08

877. Measurement of the triple-differential dijet cross section in proton-proton collisions at $\sqrt{s}=8$ TeV and constraints on parton distribution functions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 746](#) | [arXiv:1705.02628](#) | [DOI:10.1140/epjc/s10052-017-5286-7](#) | 2017-05-07

878. Search for black holes in high-multiplicity final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 774 \(2017\) 279](#) | [arXiv:1705.01403](#) | [DOI:10.1016/j.physletb.2017.09.053](#) | 2017-05-03

879. Search for supersymmetry in multijet events with missing transverse momentum in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 96 \(2017\) 032003](#) | [arXiv:1704.07781](#) | [DOI:10.1103/PhysRevD.96.032003](#) | 2017-04-25

880. Measurement of the top quark mass in the dileptonic $t\bar{t}$ decay channel using the mass observables $M_{b\ell}$, M_{T2} , and $M_{b\ell\nu}$ in pp collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 96 \(2017\) 032002](#) | [arXiv:1704.06142](#) | [DOI:10.1103/PhysRevD.96.032002](#) | 2017-04-20

881. Search for $t\bar{t}$ resonances in highly boosted lepton+jets and fully hadronic final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2017\) 001](#) | [arXiv:1704.03366](#) | [DOI:10.1007/JHEP07\(2017\)001](#) | 2017-04-11

882. Measurements of the $pp \rightarrow W\gamma\gamma$ and $pp \rightarrow Z\gamma\gamma$ cross sections and limits on anomalous quartic gauge couplings at $\sqrt{s}=8$ TeV

Sirunyan, A.M. et al.

[JHEP 10 \(2017\) 072](#) | [arXiv:1704.00366](#) | [DOI:10.1007/JHEP10\(2017\)072](#) | 2017-04-02

883. Search for new physics with dijet angular distributions in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2017\) 013](#) | [arXiv:1703.09986](#) | [DOI:10.1007/JHEP07\(2017\)013](#) | 2017-03-29

884. Search for a heavy resonance decaying to a top quark and a vector-like top quark at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 09 \(2017\) 053](#) | [arXiv:1703.06352](#) | [DOI:10.1007/JHEP09\(2017\)053](#) | 2017-03-18

885. Measurement of the jet mass in highly boosted tt events from pp collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 467](#) | [arXiv:1703.06330](#) | [DOI:10.1140/epjc/s10052-017-5030-3](#) | 2017-03-18

886. Search for anomalous couplings in boosted WW/WZ $\rightarrow \ell \nu$ q $\bar{q}$ production in proton-proton collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 772 \(2017\) 21](#) | [arXiv:1703.06095](#) | [DOI:10.1016/j.physletb.2017.06.009](#) | 2017-03-17

887. Search for associated production of dark matter with a Higgs boson decaying to bb or $\gamma\gamma$ at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 180](#) | [arXiv:1703.05236](#) | [DOI:10.1007/JHEP10\(2017\)180](#) | 2017-03-15

888. Search for third-generation scalar leptoquarks and heavy right-handed neutrinos in final states with two tau leptons and two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2017\) 121](#) | [arXiv:1703.03995](#) | [DOI:10.1007/JHEP07\(2017\)121](#) | 2017-03-11

889. Measurement of the top quark mass using single top quark events in proton-proton collisions at $\sqrt{s}= 8$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 354](#) | [arXiv:1703.02530](#) | [DOI:10.1140/epjc/s10052-017-4912-8](#) | 2017-03-07

890. Measurement of double-differential cross sections for top quark pair production in pp collisions at $\sqrt{s} = 8$ TeV and impact on parton distribution functions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 459](#) | [arXiv:1703.01630](#) | [DOI:10.1140/epjc/s10052-017-4984-5](#) | 2017-03-05

891. Search for dark matter produced with an energetic jet or a hadronically decaying W or Z boson at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2017\) 014](#) | [arXiv:1703.01651](#) | [DOI:10.1007/JHEP07\(2017\)014](#) | 2017-03-05

892. Search for standard model production of four top quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[Phys.Lett.B 772 \(2017\) 336](#) | [arXiv:1702.06164](#) | [DOI:10.1016/j.physletb.2017.06.064](#) | 2017-02-20

893. Measurement of the cross section for electroweak production of $Z\gamma$ in association with two jets and constraints on anomalous quartic gauge couplings in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 770 \(2017\) 380](#) | [arXiv:1702.03025](#) | [DOI:10.1016/j.physletb.2017.04.071](#) | 2017-02-09

894. Measurement of prompt and nonprompt J/ψ production in pp and pPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 269](#) | [arXiv:1702.01462](#) | [DOI:10.1140/epjc/s10052-017-4828-3](#) | 2017-02-05

895. Search for associated production of a Z boson with a single top quark and for tZ flavour-changing interactions in pp collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2017\) 003](#) | [arXiv:1702.01404](#) | [DOI:10.1007/JHEP07\(2017\)003](#) | 2017-02-05

896. Study of Jet Quenching with Z+jet Correlations in Pb-Pb and pp Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 082301](#) | [arXiv:1702.01060](#) | [DOI:10.1103/PhysRevLett.119.082301](#) | 2017-02-03

897. Azimuthal anisotropy of charged particles with transverse momentum up to 100 GeV/ c in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, A.M. et al.

[Phys.Lett.B 776 \(2018\) 195](#) | [arXiv:1702.00630](#) | [DOI:10.1016/j.physletb.2017.11.041](#) | 2017-02-02

898. Measurement of the inclusive energy spectrum in the very forward direction in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2017\) 046](#) | [arXiv:1701.08695](#) | [DOI:10.1007/JHEP08\(2017\)046](#) | 2017-01-30

899. Search for single production of vector-like quarks decaying into a b quark and a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 772 \(2017\) 634](#) | [arXiv:1701.08328](#) | [DOI:10.1016/j.physletb.2017.07.022](#) | 2017-01-28

900. Search for single production of vector-like quarks decaying to a Z boson and a top or a bottom quark in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2017\) 029](#) | [arXiv:1701.07409](#) | [DOI:10.1007/JHEP05\(2017\)029](#) | 2017-01-25

901. Search for new phenomena with multiple charged leptons in proton-proton collisions at $\sqrt{s}= 13$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 77 \(2017\) 635](#) | [arXiv:1701.06940](#) | [DOI:10.1140/epjc/s10052-017-5182-1](#) | 2017-01-24

902. Measurement of the $t\bar{t}$ production cross section using events with one lepton and at least one jet in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 09 \(2017\) 051](#) | [arXiv:1701.06228](#) | [DOI:10.1007/JHEP09\(2017\)051](#) | 2017-01-22

903. Mechanical stability of the CMS strip tracker measured with a laser alignment system

Sirunyan, Albert M et al.

[JINST 12 \(2017\) P04023](#) | [arXiv:1701.02022](#) | [DOI:10.1088/1748-0221/12/04/P04023](#) | 2017-01-08

904. Search for light bosons in decays of the 125 GeV Higgs boson in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, V. et al.

[JHEP 10 \(2017\) 076](#) | [arXiv:1701.02032](#) | [DOI:10.1007/JHEP10\(2017\)076](#) | 2017-01-08

905. Search for supersymmetry in the all-hadronic final state using top quark tagging in pp collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 96 \(2017\) 012004](#) | [arXiv:1701.01954](#) | [DOI:10.1103/PhysRevD.96.012004](#) | 2017-01-08

906. Search for dark matter and unparticles in events with a Z boson and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2017\) 061](#) | [arXiv:1701.02042](#) | [DOI:10.1007/JHEP03\(2017\)061](#) | 2017-01-08

907. Search for leptophobic Z' bosons decaying into four-lepton final states in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, V. et al.

[Phys.Lett.B 773 \(2017\) 563](#) | [arXiv:1701.01345](#) | [DOI:10.1016/j.physletb.2017.08.069](#) | 2017-01-05

908. Search for high-mass Z γ resonances in proton-proton collisions at $\sqrt{s}=8$ and 13 TeV using jet substructure techniques

Sirunyan, Albert M et al.

[Phys.Lett.B 772 \(2017\) 363](#) | [arXiv:1612.09516](#) | [DOI:10.1016/j.physletb.2017.06.062](#) | 2016-12-30

909. Measurement of electroweak-induced production of $W\gamma$ with two jets in pp collisions at $\sqrt{s}=8$ TeV and constraints on anomalous quartic gauge couplings

Khachatryan, Vardan et al.

[JHEP 06 \(2017\) 106](#) | [arXiv:1612.09256](#) | [DOI:10.1007/JHEP06\(2017\)106](#) | 2016-12-29

910. Search for heavy gauge W' boson in events with an energetic lepton and large missing transverse momentum at $\sqrt{s}=13$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 770 \(2017\) 278](#) | [arXiv:1612.09274](#) | [DOI:10.1016/j.physletb.2017.04.043](#) | 2016-12-29

911. Search for massive resonances decaying into WW, WZ or ZZ bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2017\) 162](#) | [arXiv:1612.09159](#) | [DOI:10.1007/JHEP03\(2017\)162](#) | 2016-12-29

912. Measurements of the charm jet cross section and nuclear modification factor in pPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 772 \(2017\) 306](#) | [arXiv:1612.08972](#) | [DOI:10.1016/j.physletb.2017.06.053](#) | 2016-12-28

913. Search for electroweak production of a vector-like quark decaying to a top quark and a Higgs boson using boosted topologies in fully hadronic final states

Sirunyan, Albert M et al.

[JHEP 04 \(2017\) 136](#) | [arXiv:1612.05336](#) | [DOI:10.1007/JHEP04\(2017\)136](#) | 2016-12-15

914. Search for heavy neutrinos or third-generation leptoquarks in final states with two hadronically decaying τ leptons and two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2017\) 077](#) | [arXiv:1612.01190](#) | [DOI:10.1007/JHEP03\(2017\)077](#) | 2016-12-04

915. Search for single production of a heavy vector-like T quark decaying to a Higgs boson and a top quark with a lepton and jets in the final state

Khachatryan, Vardan et al.

[Phys.Lett.B 771 \(2017\) 80](#) | [arXiv:1612.00999](#) | [DOI:10.1016/j.physletb.2017.05.019](#) | 2016-12-03

916. Search for CP violation in $t\bar{t}$ production and decay in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2017\) 101](#) | [arXiv:1611.08931](#) | [DOI:10.1007/JHEP03\(2017\)101](#) | 2016-11-27

917. Measurements of the associated production of a Z boson and b jets in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, V. et al.

[Eur.Phys.J.C 77 \(2017\) 751](#) | [arXiv:1611.06507](#) | [DOI:10.1140/epjc/s10052-017-5140-y](#) | 2016-11-20

918. Search for supersymmetry in events with photons and missing transverse energy in pp collisions at 13 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 769 \(2017\) 391](#) | [arXiv:1611.06604](#) | [DOI:10.1016/j.physletb.2017.04.005](#) | 2016-11-20

919. Search for heavy resonances decaying to tau lepton pairs in proton-proton collisions at $\sqrt{s}=13$ TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2017\) 048](#) | [arXiv:1611.06594](#) | [DOI:10.1007/JHEP02\(2017\)048](#) | 2016-11-20

920. Measurement of the $t\bar{t}$ production cross section using events in the $e\mu$ final state in pp collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 77 \(2017\) 172](#) | [arXiv:1611.04040](#) | [DOI:10.1140/epjc/s10052-017-4718-8](#) | 2016-11-12

921. Measurements of differential production cross sections for a Z boson in association with jets in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2017\) 022](#) | [arXiv:1611.03844](#) | [DOI:10.1007/JHEP04\(2017\)022](#) | 2016-11-11

922. Search for dijet resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on dark matter and other models

Sirunyan, Albert M et al.

[Phys.Lett.B 769 \(2017\) 520](#) | [arXiv:1611.03568](#) | [DOI:10.1016/j.physletb.2017.02.012](#) | 2016-11-10

923. Charged-particle nuclear modification factors in PbPb and pPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2017\) 039](#) | [arXiv:1611.01664](#) | [DOI:10.1007/JHEP04\(2017\)039](#) | 2016-11-05

924. Suppression of $\Upsilon(1S)$, $\Upsilon(2S)$ and $\Upsilon(3S)$ production in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 770 \(2017\) 357](#) | [arXiv:1611.01510](#) | [DOI:10.1016/j.physletb.2017.04.031](#) | 2016-11-04

925. Relative Modification of Prompt $\psi(2S)$ and J/ψ Yields from pp to PbPb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 118 \(2017\) 162301](#) | [arXiv:1611.01438](#) | [DOI:10.1103/PhysRevLett.118.162301](#) | 2016-11-04

926. A search for new phenomena in pp collisions at $\sqrt{s} = 13$ TeV in final states with missing transverse momentum and at least one jet using the α_T variable

Khachatryan, Vardan et al.

[Eur.Phys.J.C 77 \(2017\) 294](#) | [arXiv:1611.00338](#) | [DOI:10.1140/epjc/s10052-017-4787-8](#) | 2016-11-01

927. Measurement of the mass difference between top quark and antiquark in pp collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 770 \(2017\) 50](#) | [arXiv:1610.09551](#) | [DOI:10.1016/j.physletb.2017.04.028](#) | 2016-10-29

928. Searches for invisible decays of the Higgs boson in pp collisions at $\sqrt{s} = 7, 8,$ and 13 TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2017\) 135](#) | [arXiv:1610.09218](#) | [DOI:10.1007/JHEP02\(2017\)135](#) | 2016-10-28

929. Search for heavy resonances decaying into a vector boson and a Higgs boson in final states with charged leptons, neutrinos, and b quarks

Khachatryan, Vardan et al.

[Phys.Lett.B 768 \(2017\) 137](#) | [arXiv:1610.08066](#) | [DOI:10.1016/j.physletb.2017.02.040](#) | 2016-10-25

930. Observation of $\Upsilon(1S)$ pair production in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 05 \(2017\) 013](#) | [arXiv:1610.07095](#) | [DOI:10.1007/JHEP05\(2017\)013](#) | 2016-10-22

931. Search for R-parity violating supersymmetry with displaced vertices in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 95 \(2017\) 012009](#) | [arXiv:1610.05133](#) | [DOI:10.1103/PhysRevD.95.012009](#) | 2016-10-17

932. Search for electroweak production of charginos in final states with two τ leptons in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2017\) 018](#) | [arXiv:1610.04870](#) | [DOI:10.1007/JHEP04\(2017\)018](#) | 2016-10-16

933. Search for top quark decays via Higgs-boson-mediated flavor-changing neutral currents in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2017\) 079](#) | [arXiv:1610.04857](#) | [DOI:10.1007/JHEP02\(2017\)079](#) | 2016-10-16

934. Measurement of differential cross sections for top quark pair production using the lepton+jets final state in proton-proton collisions at 13 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 95 \(2017\) 092001](#) | [arXiv:1610.04191](#) | [DOI:10.1103/PhysRevD.95.092001](#) | 2016-10-13

935. Measurements of differential cross sections for associated production of a W boson and jets in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 95 \(2017\) 052002](#) | [arXiv:1610.04222](#) | [DOI:10.1103/PhysRevD.95.052002](#) | 2016-10-13

936. Search for anomalous Wtb couplings and flavour-changing neutral currents in t-channel single top quark production in pp collisions at $\sqrt{s} = 7$ and 8 TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2017\) 028](#) | [arXiv:1610.03545](#) | [DOI:10.1007/JHEP02\(2017\)028](#) | 2016-10-11

937. Search for high-mass $Z\gamma$ resonances in $e^+e^-\gamma$ and $\mu^+\mu^-\gamma$ final states in proton-proton collisions at $\sqrt{s} = 8$ and 13 TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2017\) 076](#) | [arXiv:1610.02960](#) | [DOI:10.1007/JHEP01\(2017\)076](#) | 2016-10-10

938. Cross section measurement of t-channel single top quark production in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 772 \(2017\) 752](#) | [arXiv:1610.00678](#) | [DOI:10.1016/j.physletb.2017.07.047](#) | 2016-10-03

939. Suppression and azimuthal anisotropy of prompt and nonprompt J/ψ production in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 77 \(2017\) 252](#) | [arXiv:1610.00613](#) | [DOI:10.1140/epjc/s10052-017-4781-1](#) | 2016-10-03

940. Observation of charge-dependent azimuthal correlations in p-Pb collisions and its implication for the search for the chiral magnetic effect

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 118 \(2017\) 122301](#) | [arXiv:1610.00263](#) | [DOI:10.1103/PhysRevLett.118.122301](#) | 2016-10-02

941. Search for supersymmetry in events with one lepton and multiple jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 95 \(2017\) 012011](#) | [arXiv:1609.09386](#) | [DOI:10.1103/PhysRevD.95.012011](#) | 2016-09-29

942. Search for long-lived charged particles in proton-proton collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 94 \(2016\) 112004](#) | [arXiv:1609.08382](#) | [DOI:10.1103/PhysRevD.94.112004](#) | 2016-09-27

943. Inclusive search for supersymmetry using razor variables in pp collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 95 \(2017\) 012003](#) | [arXiv:1609.07658](#) | [DOI:10.1103/PhysRevD.95.012003](#) | 2016-09-24

944. Measurement of the WZ production cross section in pp collisions at $\sqrt{s} = 7$ and 8 TeV and search for anomalous triple gauge couplings at $\sqrt{s} = 8$ TeV

Khachatryan, V. et al.

[Eur.Phys.J.C 77 \(2017\) 236](#) | [arXiv:1609.05721](#) | [DOI:10.1140/epjc/s10052-017-4730-z](#) | 2016-09-19

945. Search for narrow resonances in dilepton mass spectra in proton-proton collisions at $\sqrt{s} = 13$ TeV and combination with 8 TeV data

Khachatryan, Vardan et al.

[Phys.Lett.B 768 \(2017\) 57](#) | [arXiv:1609.05391](#) | [DOI:10.1016/j.physletb.2017.02.010](#) | 2016-09-17

946. Measurement and QCD analysis of double-differential inclusive jet cross sections in pp collisions at $\sqrt{s} = 8$ TeV and cross section ratios to 2.76 and 7 TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2017\) 156](#) | [arXiv:1609.05331](#) | [DOI:10.1007/JHEP03\(2017\)156](#) | 2016-09-17

947. Measurement of inclusive jet cross sections in pp and PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.C 96 \(2017\) 015202](#) | [arXiv:1609.05383](#) | [DOI:10.1103/PhysRevC.96.015202](#) | 2016-09-17

948. Studies of inclusive four-jet production with two b-tagged jets in proton-proton collisions at 7 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 94 \(2016\) 112005](#) | [arXiv:1609.03489](#) | [DOI:10.1103/PhysRevD.94.112005](#) | 2016-09-12

949. Decomposing transverse momentum balance contributions for quenched jets in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Khachatryan, Vardan et al.

[JHEP 11 \(2016\) 055](#) | [arXiv:1609.02466](#) | [DOI:10.1007/JHEP11\(2016\)055](#) | 2016-09-08

950. Search for high-mass diphoton resonances in proton-proton collisions at 13 TeV and combination with 8 TeV search

Khachatryan, Vardan et al.

[Phys.Lett.B 767 \(2017\) 147](#) | [arXiv:1609.02507](#) | [DOI:10.1016/j.physletb.2017.01.027](#) | 2016-09-08

951. The CMS trigger system

Khachatryan, Vardan et al.

[JINST 12 \(2017\) P01020](#) | [arXiv:1609.02366](#) | [DOI:10.1088/1748-0221/12/01/P01020](#) | 2016-09-08

952. Measurement of the total and differential inclusive B^+ hadron cross sections in pp collisions at $\sqrt{s} = 13$ TeV
 Khachatryan, Vardan et al.
[Phys.Lett.B 771 \(2017\) 435](#) | [arXiv:1609.00873](#) | [DOI:10.1016/j.physletb.2017.05.074](#) | 2016-09-03
953. Measurement of the production cross section of a W boson in association with two b jets in pp collisions at $\sqrt{s} = 8$ TeV
 Khachatryan, Vardan et al.
[Eur.Phys.J.C 77 \(2017\) 92](#) | [arXiv:1608.07561](#) | [DOI:10.1140/epjc/s10052-016-4573-z](#) | 2016-08-26
954. Measurement of the mass of the top quark in decays with a J/ψ meson in pp collisions at 8 TeV
 Khachatryan, Vardan et al.
[JHEP 12 \(2016\) 123](#) | [arXiv:1608.03560](#) | [DOI:10.1007/JHEP12\(2016\)123](#) | 2016-08-11
955. Search for new phenomena in events with high jet multiplicity and low missing transverse momentum in proton-proton collisions at $\sqrt{s}=8$ TeV
 Khachatryan, Vardan et al.
[Phys.Lett.B 770 \(2017\) 257](#) | [arXiv:1608.01224](#) | [DOI:10.1016/j.physletb.2017.01.073](#) | 2016-08-03
956. Measurement of the ZZ production cross section and $Z \rightarrow \ell^+ \ell^- \ell'^+ \ell'^-$ branching fraction in pp collisions at $\sqrt{s}=13$ TeV
 Khachatryan, Vardan et al.
[Phys.Lett.B 763 \(2016\) 280](#) | [arXiv:1607.08834](#) | [DOI:10.1016/j.physletb.2016.10.054](#) | 2016-07-29
957. Measurement of electroweak production of a W boson and two forward jets in proton-proton collisions at $\sqrt{s}=8$ TeV
 Khachatryan, Vardan et al.
[JHEP 11 \(2016\) 147](#) | [arXiv:1607.06975](#) | [DOI:10.1007/JHEP11\(2016\)147](#) | 2016-07-23
958. Measurement of the WZ production cross section in pp collisions at $\sqrt{s} = 13$ TeV
 Khachatryan, Vardan et al.
[Phys.Lett.B 766 \(2017\) 268](#) | [arXiv:1607.06943](#) | [DOI:10.1016/j.physletb.2017.01.011](#) | 2016-07-23
959. Search for dark matter in proton-proton collisions at 8 TeV with missing transverse momentum and vector boson tagged jets
 Khachatryan, Vardan et al.
[JHEP 12 \(2016\) 083](#) | [arXiv:1607.05764](#) | [DOI:10.1007/JHEP12\(2016\)083](#) | 2016-07-19
960. Jet energy scale and resolution in the CMS experiment in pp collisions at 8 TeV
 Khachatryan, Vardan et al.
[JINST 12 \(2017\) P02014](#) | [arXiv:1607.03663](#) | [DOI:10.1088/1748-0221/12/02/P02014](#) | 2016-07-13
961. Search for lepton flavour violating decays of the Higgs boson to $e \tau$ and $e \mu$ in proton-proton collisions at $\sqrt{s}=8$ TeV
 Khachatryan, Vardan et al.
[Phys.Lett.B 763 \(2016\) 472](#) | [arXiv:1607.03561](#) | [DOI:10.1016/j.physletb.2016.09.062](#) | 2016-07-12
962. Observation of the decay $B^+ \rightarrow \psi(2S) \phi(1020) K^+$ in pp collisions at $\sqrt{s} = 8$ TeV
 Khachatryan, Vardan et al.
[Phys.Lett.B 764 \(2017\) 66](#) | [arXiv:1607.02638](#) | [DOI:10.1016/j.physletb.2016.11.001](#) | 2016-07-09
963. Measurement of the differential cross sections for top quark pair production as a function of kinematic event variables in pp collisions at $\sqrt{s}=7$ and 8 TeV
 Khachatryan, Vardan et al.
[Phys.Rev.D 94 \(2016\) 052006](#) | [arXiv:1607.00837](#) | [DOI:10.1103/PhysRevD.94.052006](#) | 2016-07-04

964. Searches for R-parity-violating supersymmetry in pp collisions at $\sqrt{s} = 8$ TeV in final states with 0-4 leptons
Khachatryan, Vardan et al.
[Phys.Rev.D 94 \(2016\) 112009](#) | [arXiv:1606.08076](#) | [DOI:10.1103/PhysRevD.94.112009](#) | 2016-06-26
965. Evidence for collectivity in pp collisions at the LHC
Khachatryan, Vardan et al.
[Phys.Lett.B 765 \(2017\) 193](#) | [arXiv:1606.06198](#) | [DOI:10.1016/j.physletb.2016.12.009](#) | 2016-06-20
966. Measurement of the transverse momentum spectra of weak vector bosons produced in proton-proton collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[JHEP 02 \(2017\) 096](#) | [arXiv:1606.05864](#) | [DOI:10.1007/JHEP02\(2017\)096](#) | 2016-06-19
967. Search for Resonant Production of High-Mass Photon Pairs in Proton-Proton Collisions at $\sqrt{s}=8$ and 13 TeV
Khachatryan, Vardan et al.
[Phys.Rev.Lett. 117 \(2016\) 051802](#) | [arXiv:1606.04093](#) | [DOI:10.1103/PhysRevLett.117.051802](#) | 2016-06-13
968. Phenomenological MSSM interpretation of CMS searches in pp collisions at $\sqrt{s} = 7$ and 8 TeV
Khachatryan, Vardan et al.
[JHEP 10 \(2016\) 129](#) | [arXiv:1606.03577](#) | [DOI:10.1007/JHEP10\(2016\)129](#) | 2016-06-11
969. Measurements of the Higgs boson production and decay rates and constraints on its couplings from a combined ATLAS and CMS analysis of the LHC pp collision data at $\sqrt{s}=7$ and 8 TeV
Aad, Georges et al.
[JHEP 08 \(2016\) 045](#) | [arXiv:1606.02266](#) | [DOI:10.1007/JHEP08\(2016\)045](#) | 2016-06-07
970. Measurement of the transverse momentum spectrum of the Higgs boson produced in pp collisions at $\sqrt{s}=8$ TeV using $H \rightarrow WW$ decays
Khachatryan, Vardan et al.
[JHEP 03 \(2017\) 032](#) | [arXiv:1606.01522](#) | [DOI:10.1007/JHEP03\(2017\)032](#) | 2016-06-05
971. Search for Dark Matter and Supersymmetry with a Compressed Mass Spectrum in the Vector Boson Fusion Topology in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV
Khachatryan, Vardan et al.
[Phys.Rev.Lett. 118 \(2017\) 021802](#) | [arXiv:1605.09305](#) | [DOI:10.1103/PhysRevLett.118.021802](#) | 2016-05-30
972. Measurement of the W boson helicity fractions in the decays of top quark pairs to lepton + jets final states produced in pp collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[Phys.Lett.B 762 \(2016\) 512](#) | [arXiv:1605.09047](#) | [DOI:10.1016/j.physletb.2016.10.007](#) | 2016-05-29
973. Search for top squark pair production in compressed-mass-spectrum scenarios in proton-proton collisions at $\sqrt{s} = 8$ TeV using the α_T variable
Khachatryan, Vardan et al.
[Phys.Lett.B 767 \(2017\) 403](#) | [arXiv:1605.08993](#) | [DOI:10.1016/j.physletb.2017.02.007](#) | 2016-05-29
974. Coherent J/ψ photoproduction in ultra-peripheral PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV with the CMS experiment
Khachatryan, Vardan et al.
[Phys.Lett.B 772 \(2017\) 489](#) | [arXiv:1605.06966](#) | [DOI:10.1016/j.physletb.2017.07.001](#) | 2016-05-23
975. Multiplicity and rapidity dependence of strange hadron production in pp, pPb, and PbPb collisions at the LHC

Khachatryan, Vardan et al.

[Phys.Lett.B 768 \(2017\) 103](#) | [arXiv:1605.06699](#) | [DOI:10.1016/j.physletb.2017.01.075](#) | 2016-05-21

976. Search for supersymmetry in pp collisions at $\sqrt{s}=13$ TeV in the single-lepton final state using the sum of masses of large-radius jets

Khachatryan, Vardan et al.

[JHEP 08 \(2016\) 122](#) | [arXiv:1605.04608](#) | [DOI:10.1007/JHEP08\(2016\)122](#) | 2016-05-15

977. Measurement of the double-differential inclusive jet cross section in proton-proton collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 451](#) | [arXiv:1605.04436](#) | [DOI:10.1140/epjc/s10052-016-4286-3](#) | 2016-05-14

978. Search for Higgs boson off-shell production in proton-proton collisions at 7 and 8 TeV and derivation of constraints on its total decay width

Khachatryan, Vardan et al.

[JHEP 09 \(2016\) 051](#) | [arXiv:1605.02329](#) | [DOI:10.1007/JHEP09\(2016\)051](#) | 2016-05-08

979. Measurement of the integrated and differential $t\bar{t}$ production cross sections for high- p_t top quarks in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 94 \(2016\) 072002](#) | [arXiv:1605.00116](#) | [DOI:10.1103/PhysRevD.94.072002](#) | 2016-04-30

980. Search for narrow resonances in dijet final states at $\sqrt{s}=8$ TeV with the novel CMS technique of data scouting

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 117 \(2016\) 031802](#) | [arXiv:1604.08907](#) | [DOI:10.1103/PhysRevLett.117.031802](#) | 2016-04-29

981. Search for lepton flavour violating decays of heavy resonances and quantum black holes to an $e\mu$ pair in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 317](#) | [arXiv:1604.05239](#) | [DOI:10.1140/epjc/s10052-016-4149-y](#) | 2016-04-18

982. Pseudorapidity dependence of long-range two-particle correlations in pPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.C 96 \(2017\) 014915](#) | [arXiv:1604.05347](#) | [DOI:10.1103/PhysRevC.96.014915](#) | 2016-04-18

983. Evidence for exclusive $\gamma\gamma \rightarrow W^+ W^-$ production and constraints on anomalous quartic gauge couplings in pp collisions at $\sqrt{s}=7$ and 8 TeV

Khachatryan, Vardan et al.

[JHEP 08 \(2016\) 119](#) | [arXiv:1604.04464](#) | [DOI:10.1007/JHEP08\(2016\)119](#) | 2016-04-15

984. Search for dark matter particles in proton-proton collisions at $\sqrt{s}=8$ TeV using the razor variables

Khachatryan, Vardan et al.

[JHEP 12 \(2016\) 088](#) | [arXiv:1603.08914](#) | [DOI:10.1007/JHEP12\(2016\)088](#) | 2016-03-29

985. Search for two Higgs bosons in final states containing two photons and two bottom quarks in proton-proton collisions at 8 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 94 \(2016\) 052012](#) | [arXiv:1603.06896](#) | [DOI:10.1103/PhysRevD.94.052012](#) | 2016-03-22

986. Measurement of the top quark mass using charged particles in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 092006](#) | [arXiv:1603.06536](#) | [DOI:10.1103/PhysRevD.93.092006](#) | 2016-03-21

987. Search for neutral resonances decaying into a Z boson and a pair of b jets or τ leptons
 Khachatryan, Vardan et al.
[Phys.Lett.B 759 \(2016\) 369](#) | [arXiv:1603.02991](#) | [DOI:10.1016/j.physletb.2016.05.087](#) | 2016-03-09
988. Y(nS) polarizations versus particle multiplicity in pp collisions at $\sqrt{s} = 7$ TeV
 Khachatryan, Vardan et al.
[Phys.Lett.B 761 \(2016\) 31](#) | [arXiv:1603.02913](#) | [DOI:10.1016/j.physletb.2016.07.065](#) | 2016-03-09
989. Search for s channel single top quark production in pp collisions at $\sqrt{s}=7$ and 8 TeV
 Khachatryan, Vardan et al.
[JHEP 09 \(2016\) 027](#) | [arXiv:1603.02555](#) | [DOI:10.1007/JHEP09\(2016\)027](#) | 2016-03-08
990. Search for heavy Majorana neutrinos in $e^{\pm}e^{\pm}++$ jets and $e^{\pm}\mu^{\pm}++$ jets events in proton-proton collisions at $\sqrt{s}=8$ TeV
 Khachatryan, Vardan et al.
[JHEP 04 \(2016\) 169](#) | [arXiv:1603.02248](#) | [DOI:10.1007/JHEP04\(2016\)169](#) | 2016-03-07
991. Measurement of the t-tbar production cross section in the e-mu channel in proton-proton collisions at $\sqrt{s} = 7$ and 8 TeV
 Khachatryan, Vardan et al.
[JHEP 08 \(2016\) 029](#) | [arXiv:1603.02303](#) | [DOI:10.1007/JHEP08\(2016\)029](#) | 2016-03-07
992. Measurement of the differential cross section and charge asymmetry for inclusive $pp \rightarrow W^{\pm} + X$ production at $\sqrt{s} = 8$ TeV
 Khachatryan, Vardan et al.
[Eur.Phys.J.C 76 \(2016\) 469](#) | [arXiv:1603.01803](#) | [DOI:10.1140/epjc/s10052-016-4293-4](#) | 2016-03-06
993. Search for direct pair production of supersymmetric top quarks decaying to all-hadronic final states in pp collisions at $\sqrt{s} = 8$ TeV
 Khachatryan, Vardan et al.
[Eur.Phys.J.C 76 \(2016\) 460](#) | [arXiv:1603.00765](#) | [DOI:10.1140/epjc/s10052-016-4292-5](#) | 2016-03-02
994. Measurements of the tt production cross section in lepton+jets final states in pp collisions at 8 TeV and ratio of 8 to 7 TeV cross sections
 Khachatryan, Vardan et al.
[Eur.Phys.J.C 77 \(2017\) 15](#) | [arXiv:1602.09024](#) | [DOI:10.1140/epjc/s10052-016-4504-z](#) | 2016-02-29
995. Search for heavy resonances decaying to two Higgs bosons in final states containing four b quarks
 Khachatryan, Vardan et al.
[Eur.Phys.J.C 76 \(2016\) 371](#) | [arXiv:1602.08762](#) | [DOI:10.1140/epjc/s10052-016-4206-6](#) | 2016-02-28
996. Search for supersymmetry in electroweak production with photons and large missing transverse energy in pp collisions at $\sqrt{s} = 8$ TeV
 Khachatryan, Vardan et al.
[Phys.Lett.B 759 \(2016\) 479](#) | [arXiv:1602.08772](#) | [DOI:10.1016/j.physletb.2016.05.088](#) | 2016-02-28
997. Measurement of the $Z \gamma \rightarrow \nu \bar{\nu} \gamma$ production cross section in pp collisions at $\sqrt{s} = 8$ TeV and limits on anomalous ZZ γ and Z $\gamma\gamma$ trilinear gauge boson couplings
 Khachatryan, Vardan et al.
[Phys.Lett.B 760 \(2016\) 448](#) | [arXiv:1602.07152](#) | [DOI:10.1016/j.physletb.2016.06.080](#) | 2016-02-23
998. Search for supersymmetry in the multijet and missing transverse momentum final state in pp collisions at 13 TeV
 Khachatryan, Vardan et al.
[Phys.Lett.B 758 \(2016\) 152](#) | [arXiv:1602.06581](#) | [DOI:10.1016/j.physletb.2016.05.002](#) | 2016-02-21
999. Measurement of dijet azimuthal decorrelation in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 536](#) | [arXiv:1602.04384](#) | [DOI:10.1140/epjc/s10052-016-4346-8](#) | 2016-02-13

1000. Search for R-parity violating decays of a top squark in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 760 \(2016\) 178](#) | [arXiv:1602.04334](#) | [DOI:10.1016/j.physletb.2016.06.039](#) | 2016-02-13

1001. Combined search for anomalous pseudoscalar HVV couplings in $VH(H \rightarrow b\bar{b})$ production and $H \rightarrow VV$ decay

Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 672](#) | [arXiv:1602.04305](#) | [DOI:10.1016/j.physletb.2016.06.004](#) | 2016-02-13

1002. Search for supersymmetry in pp collisions at $\sqrt{s} = 8$ TeV in final states with boosted W bosons and b jets using razor variables

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 092009](#) | [arXiv:1602.02917](#) | [DOI:10.1103/PhysRevD.93.092009](#) | 2016-02-09

1003. Search for direct pair production of scalar top quarks in the single- and dilepton channels in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 07 \(2016\) 027](#) | [arXiv:1602.03169](#) | [DOI:10.1007/JHEP07\(2016\)027](#) | 2016-02-09

1004. Azimuthal decorrelation of jets widely separated in rapidity in pp collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[JHEP 08 \(2016\) 139](#) | [arXiv:1601.06713](#) | [DOI:10.1007/JHEP08\(2016\)139](#) | 2016-01-25

1005. Search for massive WH resonances decaying into the $\ell\nu b\bar{b}$ final state at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 237](#) | [arXiv:1601.06431](#) | [DOI:10.1140/epjc/s10052-016-4067-z](#) | 2016-01-24

1006. Forward-backward asymmetry of Drell-Yan lepton pairs in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 325](#) | [arXiv:1601.04768](#) | [DOI:10.1140/epjc/s10052-016-4156-z](#) | 2016-01-18

1007. Measurement of inclusive jet production and nuclear modifications in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 372](#) | [arXiv:1601.02001](#) | [DOI:10.1140/epjc/s10052-016-4205-7](#) | 2016-01-08

1008. Correlations between jets and charged particles in PbPb and pp collisions at $\sqrt{s_{NN}}=2.76$ TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2016\) 156](#) | [arXiv:1601.00079](#) | [DOI:10.1007/JHEP02\(2016\)156](#) | 2016-01-01

1009. Measurement of differential and integrated fiducial cross sections for Higgs boson production in the four-lepton decay channel in pp collisions at $\sqrt{s}=7$ and 8 TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2016\) 005](#) | [arXiv:1512.08377](#) | [DOI:10.1007/JHEP04\(2016\)005](#) | 2015-12-28

1010. Search for supersymmetry in events with soft leptons, low jet multiplicity, and missing transverse energy in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 9](#) | [arXiv:1512.08002](#) | [DOI:10.1016/j.physletb.2016.05.033](#) | 2015-12-25

1011. Study of Z boson production in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 36](#) | [arXiv:1512.06461](#) | [DOI:10.1016/j.physletb.2016.05.044](#) | 2015-12-20

1012. Measurement of the inclusive jet cross section in pp collisions at $\sqrt{s} = 2.76$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 265](#) | [arXiv:1512.06212](#) | [DOI:10.1140/epjc/s10052-016-4083-z](#) | 2015-12-19

1013. Search for narrow resonances decaying to dijets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 116 \(2016\) 071801](#) | [arXiv:1512.01224](#) | [DOI:10.1103/PhysRevLett.116.071801](#) | 2015-12-03

1014. Event generator tunes obtained from underlying event and multiparton scattering measurements

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 155](#) | [arXiv:1512.00815](#) | [DOI:10.1140/epjc/s10052-016-3988-x](#) | 2015-12-02

1015. Search for dark matter and unparticles produced in association with a Z boson in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 052011](#) | [arXiv:1511.09375](#) | [DOI:10.1103/PhysRevD.93.052011](#) | 2015-11-30

1016. Measurement of Spin Correlations in $t\bar{t}$ Production using the Matrix Element Method in the Muon+Jets Final State in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 758 \(2016\) 321](#) | [arXiv:1511.06170](#) | [DOI:10.1016/j.physletb.2016.05.005](#) | 2015-11-19

1017. Search for Anomalous Single Top Quark Production in Association with a Photon in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2016\) 035](#) | [arXiv:1511.03951](#) | [DOI:10.1007/JHEP04\(2016\)035](#) | 2015-11-12

1018. Search for a Low-Mass Pseudoscalar Higgs Boson Produced in Association with a $b\bar{b}$ Pair in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 758 \(2016\) 296](#) | [arXiv:1511.03610](#) | [DOI:10.1016/j.physletb.2016.05.003](#) | 2015-11-11

1019. Measurement of Top Quark Polarisation in T-Channel Single Top Quark Production

Khachatryan, Vardan et al.

[JHEP 04 \(2016\) 073](#) | [arXiv:1511.02138](#) | [DOI:10.1007/JHEP04\(2016\)073](#) | 2015-11-06

1020. Search for Excited Leptons in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2016\) 125](#) | [arXiv:1511.01407](#) | [DOI:10.1007/JHEP03\(2016\)125](#) | 2015-11-04

1021. Reconstruction and identification of τ lepton decays to hadrons and ν_τ at CMS

Khachatryan, Vardan et al.

[JINST 11 \(2016\) P01019](#) | [arXiv:1510.07488](#) | [DOI:10.1088/1748-0221/11/01/P01019](#) | 2015-10-26

1022. Search for a very light NMSSM Higgs boson produced in decays of the 125 GeV scalar boson and decaying into τ leptons in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2016\) 079](#) | [arXiv:1510.06534](#) | [DOI:10.1007/JHEP01\(2016\)079](#) | 2015-10-22

1023. Measurement of the top quark pair production cross section in proton-proton collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 116 \(2016\) 052002](#) | [arXiv:1510.05302](#) | [DOI:10.1103/PhysRevLett.116.052002](#) | 2015-10-18

1024. Search for a light charged Higgs boson decaying to cs in pp collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[JHEP 12 \(2015\) 178](#) | [arXiv:1510.04252](#) | [DOI:10.1007/JHEP12\(2015\)178](#) | 2015-10-14
1025. Transverse momentum spectra of inclusive b jets in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV
Khachatryan, Vardan et al.
[Phys.Lett.B 754 \(2016\) 59](#) | [arXiv:1510.03373](#) | [DOI:10.1016/j.physletb.2016.01.010](#) | 2015-10-12
1026. Measurement of long-range near-side two-particle angular correlations in pp collisions at $\sqrt{s} = 13$ TeV
Khachatryan, Vardan et al.
[Phys.Rev.Lett. 116 \(2016\) 172302](#) | [arXiv:1510.03068](#) | [DOI:10.1103/PhysRevLett.116.172302](#) | 2015-10-11
1027. Measurement of tt production with additional jet activity, including b quark jets, in the dilepton decay channel using pp collisions at $\sqrt{s} = 8$ TeV
Khachatryan, Vardan et al.
[Eur.Phys.J.C 76 \(2016\) 379](#) | [arXiv:1510.03072](#) | [DOI:10.1140/epjc/s10052-016-4105-x](#) | 2015-10-11
1028. Observation of top quark pairs produced in association with a vector boson in pp collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[JHEP 01 \(2016\) 096](#) | [arXiv:1510.01131](#) | [DOI:10.1007/JHEP01\(2016\)096](#) | 2015-10-05
1029. Searches for a heavy scalar boson H decaying to a pair of 125 GeV Higgs bosons hh or for a heavy pseudoscalar boson A decaying to Zh , in the final states with $h \rightarrow \tau\tau$
Khachatryan, Vardan et al.
[Phys.Lett.B 755 \(2016\) 217](#) | [arXiv:1510.01181](#) | [DOI:10.1016/j.physletb.2016.01.056](#) | 2015-10-05
1030. Measurement of transverse momentum relative to dijet systems in $PbPb$ and pp collisions at $\sqrt{s_{NN}}=2.76$ TeV
Khachatryan, Vardan et al.
[JHEP 01 \(2016\) 006](#) | [arXiv:1509.09029](#) | [DOI:10.1007/JHEP01\(2016\)006](#) | 2015-09-30
1031. Search for the production of an excited bottom quark decaying to tW in proton-proton collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[JHEP 01 \(2016\) 166](#) | [arXiv:1509.08141](#) | [DOI:10.1007/JHEP01\(2016\)166](#) | 2015-09-27
1032. Search for the associated production of a Higgs boson with a single top quark in proton-proton collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[JHEP 06 \(2016\) 177](#) | [arXiv:1509.08159](#) | [DOI:10.1007/JHEP06\(2016\)177](#) | 2015-09-27
1033. Measurement of the tt production cross section in the all-jets final state in pp collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[Eur.Phys.J.C 76 \(2016\) 128](#) | [arXiv:1509.06076](#) | [DOI:10.1140/epjc/s10052-016-3956-5](#) | 2015-09-20
1034. Search for $W' \rightarrow tb$ in proton-proton collisions at $\sqrt{s} = 8$ TeV
Khachatryan, Vardan et al.
[JHEP 02 \(2016\) 122](#) | [arXiv:1509.06051](#) | [DOI:10.1007/JHEP02\(2016\)122](#) | 2015-09-20
1035. Measurement of the top quark mass using proton-proton data at $\sqrt{s} = 7$ and 8 TeV
Khachatryan, Vardan et al.
[Phys.Rev.D 93 \(2016\) 072004](#) | [arXiv:1509.04044](#) | [DOI:10.1103/PhysRevD.93.072004](#) | 2015-09-14

1036. Search for vector-like charge $2/3$ T quarks in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 012003](#) | [arXiv:1509.04177](#) | [DOI:10.1103/PhysRevD.93.012003](#) | 2015-09-14

1037. Measurement of the inelastic cross section in proton-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 641](#) | [arXiv:1509.03893](#) | [DOI:10.1016/j.physletb.2016.06.027](#) | 2015-09-13

1038. Search for pair production of first and second generation leptoquarks in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 032004](#) | [arXiv:1509.03744](#) | [DOI:10.1103/PhysRevD.93.032004](#) | 2015-09-12

1039. Search for single production of scalar leptoquarks in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 032005](#) | [arXiv:1509.03750](#) | [DOI:10.1103/PhysRevD.93.032005](#) | 2015-09-12

1040. Search for a charged Higgs boson in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 11 \(2015\) 018](#) | [arXiv:1508.07774](#) | [DOI:10.1007/JHEP11\(2015\)018](#) | 2015-08-31

1041. Measurement of differential cross sections for Higgs boson production in the diphoton decay channel in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 13](#) | [arXiv:1508.07819](#) | [DOI:10.1140/epjc/s10052-015-3853-3](#) | 2015-08-31

1042. Search for supersymmetry in the vector-boson fusion topology in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 11 \(2015\) 189](#) | [arXiv:1508.07628](#) | [DOI:10.1007/JHEP11\(2015\)189](#) | 2015-08-30

1043. Study of B Meson Production in p+Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV Using Exclusive Hadronic Decays

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 116 \(2016\) 032301](#) | [arXiv:1508.06678](#) | [DOI:10.1103/PhysRevLett.116.032301](#) | 2015-08-26

1044. Search for W' decaying to tau lepton and neutrino in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 755 \(2016\) 196](#) | [arXiv:1508.04308](#) | [DOI:10.1016/j.physletb.2016.02.002](#) | 2015-08-18

1045. Measurement of the charge asymmetry in top quark pair production in pp collisions at $\sqrt{s} = 8$ TeV using a template method

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 034014](#) | [arXiv:1508.03862](#) | [DOI:10.1103/PhysRevD.93.034014](#) | 2015-08-16

1046. Search for neutral MSSM Higgs bosons decaying to $\mu^+ \mu^-$ in pp collisions at $\sqrt{s} = 7$ and 8 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 752 \(2016\) 221](#) | [arXiv:1508.01437](#) | [DOI:10.1016/j.physletb.2015.11.042](#) | 2015-08-06

1047. Search for supersymmetry in events with a photon, a lepton, and missing transverse momentum in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

- Phys.Lett.B 757 (2016) 6 | [arXiv:1508.01218](#) | [DOI:10.1016/j.physletb.2016.03.039](#) | 2015-08-05
1048. Angular analysis of the decay $B^0 \rightarrow K^{*0} \mu^+ \mu^-$ from pp collisions at $\sqrt{s} = 8$ TeV
Khachatryan, Vardan et al.
[Phys.Lett.B 753 \(2016\) 424](#) | [arXiv:1507.08126](#) | [DOI:10.1016/j.physletb.2015.12.020](#) | 2015-07-29
1049. Measurement of the CP-violating weak phase ϕ_s and the decay width difference $\Delta\Gamma_s$ using the $B_s^0 \rightarrow J/\psi\phi(1020)$ decay channel in pp collisions at $\sqrt{s} = 8$ TeV
Khachatryan, Vardan et al.
[Phys.Lett.B 757 \(2016\) 97](#) | [arXiv:1507.07527](#) | [DOI:10.1016/j.physletb.2016.03.046](#) | 2015-07-27
1050. Measurement of the underlying event activity using charged-particle jets in proton-proton collisions at $\sqrt{s} = 2.76$ TeV
Khachatryan, Vardan et al.
[JHEP 09 \(2015\) 137](#) | [arXiv:1507.07229](#) | [DOI:10.1007/JHEP09\(2015\)137](#) | 2015-07-26
1051. Search for pair-produced vectorlike B quarks in proton-proton collisions at $\sqrt{s} = 8$ TeV
Khachatryan, Vardan et al.
[Phys.Rev.D 93 \(2016\) 112009](#) | [arXiv:1507.07129](#) | [DOI:10.1103/PhysRevD.93.112009](#) | 2015-07-25
1052. Limits on the Higgs boson lifetime and width from its decay to four charged leptons
Khachatryan, Vardan et al.
[Phys.Rev.D 92 \(2015\) 072010](#) | [arXiv:1507.06656](#) | [DOI:10.1103/PhysRevD.92.072010](#) | 2015-07-23
1053. Pseudorapidity distribution of charged hadrons in proton-proton collisions at $\sqrt{s} = 13$ TeV
Khachatryan, Vardan et al.
[Phys.Lett.B 751 \(2015\) 143](#) | [arXiv:1507.05915](#) | [DOI:10.1016/j.physletb.2015.10.004](#) | 2015-07-21
1054. Inclusive and differential measurements of the tt charge asymmetry in pp collisions at $\sqrt{s} = 8$ TeV
Khachatryan, Vardan et al.
[Phys.Lett.B 757 \(2016\) 154](#) | [arXiv:1507.03119](#) | [DOI:10.1016/j.physletb.2016.03.060](#) | 2015-07-11
1055. Search for supersymmetry with photons in pp collisions at $\sqrt{s} = 8$ TeV
Khachatryan, Vardan et al.
[Phys.Rev.D 92 \(2015\) 072006](#) | [arXiv:1507.02898](#) | [DOI:10.1103/PhysRevD.92.072006](#) | 2015-07-10
1056. Search for a Higgs boson decaying into $\gamma^* \rightarrow \ell\ell\gamma$ with low dilepton mass in pp collisions at $\sqrt{s} = 8$ TeV
Khachatryan, Vardan et al.
[Phys.Lett.B 753 \(2016\) 341](#) | [arXiv:1507.03031](#) | [DOI:10.1016/j.physletb.2015.12.039](#) | 2015-07-10
1057. Production of leading charged particles and leading charged-particle jets at small transverse momenta in pp collisions at $\sqrt{s} = 8$ TeV
Khachatryan, Vardan et al.
[Phys.Rev.D 92 \(2015\) 112001](#) | [arXiv:1507.00233](#) | [DOI:10.1103/PhysRevD.92.112001](#) | 2015-07-01
1058. Search for exotic decays of a Higgs boson into undetectable particles and one or more photons
Khachatryan, Vardan et al.
[Phys.Lett.B 753 \(2016\) 363](#) | [arXiv:1507.00359](#) | [DOI:10.1016/j.physletb.2015.12.017](#) | 2015-07-01
1059. Search for neutral MSSM Higgs bosons decaying into a pair of bottom quarks
Khachatryan, Vardan et al.
[JHEP 11 \(2015\) 071](#) | [arXiv:1506.08329](#) | [DOI:10.1007/JHEP11\(2015\)071](#) | 2015-06-27

1060. Search for resonant $t\bar{t}$ production in proton-proton collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[Phys.Rev.D 93 \(2016\) 012001](#) | [arXiv:1506.03062](#) | [DOI:10.1103/PhysRevD.93.012001](#) | 2015-06-09
1061. Search for diphoton resonances in the mass range from 150 to 850 GeV in pp collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[Phys.Lett.B 750 \(2015\) 494](#) | [arXiv:1506.02301](#) | [DOI:10.1016/j.physletb.2015.09.062](#) | 2015-06-07
1062. Search for a massive resonance decaying into a Higgs boson and a W or Z boson in hadronic final states in proton-proton collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[JHEP 02 \(2016\) 145](#) | [arXiv:1506.01443](#) | [DOI:10.1007/JHEP02\(2016\)145](#) | 2015-06-03
1063. Search for the standard model Higgs boson produced through vector boson fusion and decaying to $b\bar{b}$
Khachatryan, Vardan et al.
[Phys.Rev.D 92 \(2015\) 032008](#) | [arXiv:1506.01010](#) | [DOI:10.1103/PhysRevD.92.032008](#) | 2015-06-02
1064. A search for pair production of new light bosons decaying into muons
Khachatryan, V. et al.
[Phys.Lett.B 752 \(2016\) 146](#) | [arXiv:1506.00424](#) | [DOI:10.1016/j.physletb.2015.10.067](#) | 2015-06-01
1065. Search for neutral color-octet weak-triplet scalar particles in proton-proton collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[JHEP 09 \(2015\) 201](#) | [arXiv:1505.08118](#) | [DOI:10.1007/JHEP09\(2015\)201](#) | 2015-05-29
1066. Comparison of the $Z/\gamma^* + \text{jets}$ to $\gamma + \text{jets}$ cross sections in pp collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[JHEP 10 \(2015\) 128](#) | [arXiv:1505.06520](#) | [DOI:10.1007/JHEP10\(2015\)128](#) | 2015-05-24
1067. Measurement of the differential cross section for top quark pair production in pp collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[Eur.Phys.J.C 75 \(2015\) 542](#) | [arXiv:1505.04480](#) | [DOI:10.1140/epjc/s10052-015-3709-x](#) | 2015-05-17
1068. Search for a pseudoscalar boson decaying into a Z boson and the 125 GeV Higgs boson in $\ell^+\ell^-b\bar{b}$ final states
Khachatryan, Vardan et al.
[Phys.Lett.B 748 \(2015\) 221](#) | [arXiv:1504.04710](#) | [DOI:10.1016/j.physletb.2015.07.010](#) | 2015-04-18
1069. Angular coefficients of Z bosons produced in pp collisions at $\sqrt{s}=8$ TeV and decaying to $\mu^+\mu^-$ as a function of transverse momentum and rapidity
Khachatryan, Vardan et al.
[Phys.Lett.B 750 \(2015\) 154](#) | [arXiv:1504.03512](#) | [DOI:10.1016/j.physletb.2015.08.061](#) | 2015-04-14
1070. Measurement of the Z boson differential cross section in transverse momentum and rapidity in proton-proton collisions at 8 TeV
Khachatryan, Vardan et al.
[Phys.Lett.B 749 \(2015\) 187](#) | [arXiv:1504.03511](#) | [DOI:10.1016/j.physletb.2015.07.065](#) | 2015-04-14
1071. Search for the production of dark matter in association with top-quark pairs in the single-lepton final state in proton-proton collisions at $\sqrt{s}=8$ TeV
Khachatryan, Vardan et al.
[JHEP 06 \(2015\) 121](#) | [arXiv:1504.03198](#) | [DOI:10.1007/JHEP06\(2015\)121](#) | 2015-04-13

1072. Search for a Higgs boson in the mass range from 145 to 1000 GeV decaying to a pair of W or Z bosons

Khachatryan, Vardan et al.

[JHEP 10 \(2015\) 144](#) | [arXiv:1504.00936](#) | [DOI:10.1007/JHEP10\(2015\)144](#) | 2015-04-03

1073. Search for Third-Generation Scalar Leptoquarks in the $t\tau$ Channel in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 07 \(2015\) 042](#) | [arXiv:1503.09049](#) | [DOI:10.1007/JHEP11\(2016\)056](#) | 2015-03-31

1074. Measurement of diffraction dissociation cross sections in pp collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 92 \(2015\) 012003](#) | [arXiv:1503.08689](#) | [DOI:10.1103/PhysRevD.92.012003](#) | 2015-03-30

1075. Searches for third-generation squark production in fully hadronic final states in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 06 \(2015\) 116](#) | [arXiv:1503.08037](#) | [DOI:10.1007/JHEP06\(2015\)116](#) | 2015-03-27

1076. Combined Measurement of the Higgs Boson Mass in pp Collisions at $\sqrt{s}=7$ and 8 TeV with the ATLAS and CMS Experiments

Aad, Georges et al.

[Phys.Rev.Lett. 114 \(2015\) 191803](#) | [arXiv:1503.07589](#) | [DOI:10.1103/PhysRevLett.114.191803](#) | 2015-03-26

1077. Study of W boson production in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 750 \(2015\) 565](#) | [arXiv:1503.05825](#) | [DOI:10.1016/j.physletb.2015.09.057](#) | 2015-03-19

1078. Measurements of the Z Z production cross sections in the $2l\,2\nu$ channel in proton-proton collisions at $\sqrt{s} = 7$ and 8 TeV and combined constraints on triple gauge couplings

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 511](#) | [arXiv:1503.05467](#) | [DOI:10.1140/epjc/s10052-015-3706-0](#) | 2015-03-18

1079. Search for resonant pair production of Higgs bosons decaying to two bottom quark-antiquark pairs in proton-proton collisions at 8 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 749 \(2015\) 560](#) | [arXiv:1503.04114](#) | [DOI:10.1016/j.physletb.2015.08.047](#) | 2015-03-13

1080. Search for vector-like T quarks decaying to top quarks and Higgs bosons in the all-hadronic channel using jet substructure

Khachatryan, Vardan et al.

[JHEP 06 \(2015\) 080](#) | [arXiv:1503.01952](#) | [DOI:10.1007/JHEP06\(2015\)080](#) | 2015-03-06

1081. Evidence for transverse momentum and pseudorapidity dependent event plane fluctuations in PbPb and pPb collisions

Khachatryan, Vardan et al.

[Phys.Rev.C 92 \(2015\) 034911](#) | [arXiv:1503.01692](#) | [DOI:10.1103/PhysRevC.92.034911](#) | 2015-03-05

1082. Study of Final-State Radiation in Decays of Z Bosons Produced in pp Collisions at 7 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 092012](#) | [arXiv:1502.07940](#) | [DOI:10.1103/PhysRevD.91.092012](#) | 2015-02-27

1083. Search for Lepton-Flavour-Violating Decays of the Higgs Boson

Khachatryan, Vardan et al.

[Phys.Lett.B 749 \(2015\) 337](#) | [arXiv:1502.07400](#) | [DOI:10.1016/j.physletb.2015.07.053](#) | 2015-02-25

1084. Measurement of the $Z\gamma$ Production Cross Section in pp Collisions at 8 TeV and Search for Anomalous Triple Gauge Boson Couplings

Khachatryan, Vardan et al.

[JHEP 04 \(2015\) 164](#) | [arXiv:1502.05664](#) | [DOI:10.1007/JHEP04\(2015\)164](#) | 2015-02-19

1085. Nuclear Effects on the Transverse Momentum Spectra of Charged Particles in pPb Collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 237](#) | [arXiv:1502.05387](#) | [DOI:10.1140/epjc/s10052-015-3435-4](#) | 2015-02-18

1086. Evidence for Collective Multiparticle Correlations in p-Pb Collisions

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 115 \(2015\) 012301](#) | [arXiv:1502.05382](#) | [DOI:10.1103/PhysRevLett.115.012301](#) | 2015-02-18

1087. Search for Narrow High-Mass Resonances in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV Decaying to a Z and a Higgs Boson

Khachatryan, Vardan et al.

[Phys.Lett.B 748 \(2015\) 255](#) | [arXiv:1502.04994](#) | [DOI:10.1016/j.physletb.2015.07.011](#) | 2015-02-17

1088. Distributions of Topological Observables in Inclusive Three- and Four-Jet Events in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 302](#) | [arXiv:1502.04785](#) | [DOI:10.1140/epjc/s10052-015-3491-9](#) | 2015-02-16

1089. Searches for Supersymmetry using the M_{T2} Variable in Hadronic Events Produced in pp Collisions at 8 TeV

Khachatryan, Vardan et al.

[JHEP 05 \(2015\) 078](#) | [arXiv:1502.04358](#) | [DOI:10.1007/JHEP05\(2015\)078](#) | 2015-02-15

1090. Measurement of J/ψ and $\psi(2S)$ Prompt Double-Differential Cross Sections in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 114 \(2015\) 191802](#) | [arXiv:1502.04155](#) | [DOI:10.1103/PhysRevLett.114.191802](#) | 2015-02-13

1091. Search for a Standard Model Higgs Boson Produced in Association with a Top-Quark Pair and Decaying to Bottom Quarks Using a Matrix Element Method

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 251](#) | [arXiv:1502.02485](#) | [DOI:10.1140/epjc/s10052-015-3454-1](#) | 2015-02-09

1092. Constraints on the pMSSM, AMSB model and on other models from the search for long-lived charged particles in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 325](#) | [arXiv:1502.02522](#) | [DOI:10.1140/epjc/s10052-015-3533-3](#) | 2015-02-09

1093. Performance of Photon Reconstruction and Identification with the CMS Detector in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JINST 10 \(2015\) P08010](#) | [arXiv:1502.02702](#) | [DOI:10.1088/1748-0221/10/08/P08010](#) | 2015-02-09

1094. Performance of Electron Reconstruction and Selection with the CMS Detector in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JINST 10 \(2015\) P06005](#) | [arXiv:1502.02701](#) | [DOI:10.1088/1748-0221/10/06/P06005](#) | 2015-02-09

1095. Search for Supersymmetry Using Razor Variables in Events with b-Tagged Jets in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 052018](#) | [arXiv:1502.00300](#) | [DOI:10.1103/PhysRevD.91.052018](#) | 2015-02-01

1096. Measurements of the $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ differential cross sections in pp collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 749 \(2015\) 14](#) | [arXiv:1501.07750](#) | [DOI:10.1016/j.physletb.2015.07.037](#) | 2015-01-30

1097. Measurement of the ratio $B(B^0_s \rightarrow J/\psi f_0(980)) / B(B^0_s \rightarrow J/\psi \phi(1020))$ in pp collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 756 \(2016\) 84](#) | [arXiv:1501.06089](#) | [DOI:10.1016/j.physletb.2016.02.047](#) | 2015-01-24

1098. Search for Decays of Stopped Long-Lived Particles Produced in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 151](#) | [arXiv:1501.05603](#) | [DOI:10.1140/epjc/s10052-015-3367-z](#) | 2015-01-22

1099. Search for heavy Majorana neutrinos in $\mu^+ \mu^- +$ jets events in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 748 \(2015\) 144](#) | [arXiv:1501.05566](#) | [DOI:10.1016/j.physletb.2015.06.070](#) | 2015-01-22

1100. Search for resonances and quantum black holes using dijet mass spectra in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 052009](#) | [arXiv:1501.04198](#) | [DOI:10.1103/PhysRevD.91.052009](#) | 2015-01-17

1101. Precise determination of the mass of the Higgs boson and tests of compatibility of its couplings with the standard model predictions using proton collisions at 7 and 8 TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 212](#) | [arXiv:1412.8662](#) | [DOI:10.1140/epjc/s10052-015-3351-7](#) | 2014-12-30

1102. Search for pair-produced resonances decaying to jet pairs in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 747 \(2015\) 98](#) | [arXiv:1412.7706](#) | [DOI:10.1016/j.physletb.2015.04.045](#) | 2014-12-24

1103. Search for physics beyond the standard model in dilepton mass spectra in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2015\) 025](#) | [arXiv:1412.6302](#) | [DOI:10.1007/JHEP04\(2015\)025](#) | 2014-12-19

1104. Searches for supersymmetry based on events with b jets and four W bosons in pp collisions at 8 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 745 \(2015\) 5](#) | [arXiv:1412.4109](#) | [DOI:10.1016/j.physletb.2015.04.002](#) | 2014-12-12

1105. Measurement of the inclusive 3-jet production differential cross section in proton-proton collisions at 7 TeV and determination of the strong coupling constant in the TeV range

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 186](#) | [arXiv:1412.1633](#) | [DOI:10.1140/epjc/s10052-015-3376-y](#) | 2014-12-04

1106. Measurements of differential and double-differential Drell-Yan cross sections in proton-proton collisions at 8 TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 147](#) | [arXiv:1412.1115](#) | [DOI:10.1140/epjc/s10052-015-3364-2](#) | 2014-12-02

1107. Search for stealth supersymmetry in events with jets, either photons or leptons, and low missing transverse momentum in pp collisions at 8 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 743 \(2015\) 503](#) | [arXiv:1411.7255](#) | [DOI:10.1016/j.physletb.2015.03.017](#) | 2014-11-26

1108. Search for long-lived particles that decay into final states containing two electrons or two muons in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 052012](#) | [arXiv:1411.6977](#) | [DOI:10.1103/PhysRevD.91.052012](#) | 2014-11-25

1109. Search for Long-Lived Neutral Particles Decaying to Quark-Antiquark Pairs in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 012007](#) | [arXiv:1411.6530](#) | [DOI:10.1103/PhysRevD.91.012007](#) | 2014-11-24

1110. Search for disappearing tracks in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, V. et al.

[JHEP 01 \(2015\) 096](#) | [arXiv:1411.6006](#) | [DOI:10.1007/JHEP01\(2015\)096](#) | 2014-11-21

1111. Measurement of the cross section ratio $\sigma_{t\bar{t}b\bar{b}} / \sigma_{t\bar{t}jj}$ in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 746 \(2015\) 132](#) | [arXiv:1411.5621](#) | [DOI:10.1016/j.physletb.2015.04.060](#) | 2014-11-20

1112. Constraints on the spin-parity and anomalous HVV couplings of the Higgs boson in proton collisions at 7 and 8 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 92 \(2015\) 012004](#) | [arXiv:1411.3441](#) | [DOI:10.1103/PhysRevD.92.012004](#) | 2014-11-12

1113. Search for quark contact interactions and extra spatial dimensions using dijet angular distributions in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 746 \(2015\) 79](#) | [arXiv:1411.2646](#) | [DOI:10.1016/j.physletb.2015.04.042](#) | 2014-11-10

1114. Performance of the CMS missing transverse momentum reconstruction in pp data at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JINST 10 \(2015\) P02006](#) | [arXiv:1411.0511](#) | [DOI:10.1088/1748-0221/10/02/P02006](#) | 2014-11-03

1115. Search for new phenomena in monophoton final states in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 755 \(2016\) 102](#) | [arXiv:1410.8812](#) | [DOI:10.1016/j.physletb.2016.01.057](#) | 2014-10-31

1116. Search for a standard model-like Higgs boson in the $\mu^+\mu^-$ and e^+e^- decay channels at the LHC

Khachatryan, Vardan et al.

[Phys.Lett.B 744 \(2015\) 184](#) | [arXiv:1410.6679](#) | [DOI:10.1016/j.physletb.2015.03.048](#) | 2014-10-24

1117. Constraints on parton distribution functions and extraction of the strong coupling constant from the inclusive jet cross section in pp collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 288](#) | [arXiv:1410.6765](#) | [DOI:10.1140/epjc/s10052-015-3499-1](#) | 2014-10-24

1118. Measurement of the ratio of the production cross sections times branching fractions of $B_c^{\pm} \rightarrow J/\psi \pi^{\pm}$ and $B^{\pm} \rightarrow J/\psi K^{\pm}$ and $\mathcal{B}(B_c^{\pm} \rightarrow J/\psi \pi^{\pm} \pi^{\pm} \pi^{\mp}) / \mathcal{B}(B_c^{\pm} \rightarrow J/\psi \pi^{\pm})$ in pp collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2015\) 063](#) | [arXiv:1410.5729](#) | [DOI:10.1007/JHEP01\(2015\)063](#) | 2014-10-21

1119. Study of Z production in PbPb and pp collisions at $\sqrt{s_{NN}}=2.76$ TeV in the dimuon and dielectron decay channels

Chatrchyan, Serguei et al.

[JHEP 03 \(2015\) 022](#) | [arXiv:1410.4825](#) | [DOI:10.1007/JHEP03\(2015\)022](#) | 2014-10-17

1120. Identification techniques for highly boosted W bosons that decay into hadrons

Khachatryan, Vardan et al.

[JHEP 12 \(2014\) 017](#) | [arXiv:1410.4227](#) | [DOI:10.1007/JHEP12\(2014\)017](#) | 2014-10-15

1121. Measurement of electroweak production of two jets in association with a Z boson in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 66](#) | [arXiv:1410.3153](#) | [DOI:10.1140/epjc/s10052-014-3232-5](#) | 2014-10-12

1122. Searches for heavy Higgs bosons in two-Higgs-doublet models and for t decay using multilepton and diphoton final states in pp collisions at 8 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 90 \(2014\) 112013](#) | [arXiv:1410.2751](#) | [DOI:10.1103/PhysRevD.90.112013](#) | 2014-10-10

1123. Measurement of Prompt $\psi(2S) \rightarrow J/\psi$ Yield Ratios in Pb-Pb and p-p Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 113 \(2014\) 262301](#) | [arXiv:1410.1804](#) | [DOI:10.1103/PhysRevLett.113.262301](#) | 2014-10-07

1124. Search for Monotop Signatures in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 114 \(2015\) 101801](#) | [arXiv:1410.1149](#) | [DOI:10.1103/PhysRevLett.114.101801](#) | 2014-10-05

1125. Measurement of the W boson helicity in events with a single reconstructed top quark in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2015\) 053](#) | [arXiv:1410.1154](#) | [DOI:10.1007/JHEP01\(2015\)053](#) | 2014-10-05

1126. Search for Standard Model Production of Four Top Quarks in the Lepton + Jets Channel in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 11 \(2014\) 154](#) | [arXiv:1409.7339](#) | [DOI:10.1007/JHEP11\(2014\)154](#) | 2014-09-25

1127. Measurement of the production cross section ratio $\sigma(Xb2(1P)) / \sigma(Xb1(1P))$ in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 743 \(2015\) 383](#) | [arXiv:1409.5761](#) | [DOI:10.1016/j.physletb.2015.02.048](#) | 2014-09-19

1128. Search for Displaced Supersymmetry in events with an electron and a muon with large impact parameters

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 114 \(2015\) 061801](#) | [arXiv:1409.4789](#) | [DOI:10.1103/PhysRevLett.114.061801](#) | 2014-09-16

1129. Long-range two-particle correlations of strange hadrons with charged particles in pPb and PbPb collisions at LHC energies

Khachatryan, Vardan et al.

[Phys.Lett.B 742 \(2015\) 200](#) | [arXiv:1409.3392](#) | [DOI:10.1016/j.physletb.2015.01.034](#) | 2014-09-11

1130. Search for dark matter, extra dimensions, and unparticles in monojet events in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 235](#) | [arXiv:1408.3583](#) | [DOI:10.1140/epjc/s10052-015-3451-4](#) | 2014-08-15

1131. Search for neutral MSSM Higgs bosons decaying to a pair of tau leptons in pp collisions

Khachatryan, Vardan et al.

[JHEP 10 \(2014\) 160](#) | [arXiv:1408.3316](#) | [DOI:10.1007/JHEP10\(2014\)160](#) | 2014-08-14

1132. Measurements of jet multiplicity and differential production cross sections of Z + jets events in proton-proton collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 052008](#) | [arXiv:1408.3104](#) | [DOI:10.1103/PhysRevD.91.052008](#) | 2014-08-13

1133. Search for physics beyond the standard model in final states with a lepton and missing transverse energy in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 092005](#) | [arXiv:1408.2745](#) | [DOI:10.1103/PhysRevD.91.092005](#) | 2014-08-12

1134. Search for the associated production of the Higgs boson with a top-quark pair

Khachatryan, Vardan et al.

[JHEP 09 \(2014\) 087](#) | [arXiv:1408.1682](#) | [DOI:10.1007/JHEP09\(2014\)087](#) | 2014-08-07

1135. Search for pair production of third-generation scalar leptoquarks and top squarks in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 739 \(2014\) 229](#) | [arXiv:1408.0806](#) | [DOI:10.1016/j.physletb.2014.10.063](#) | 2014-08-04

1136. Measurement of the $t\bar{t}$ Production Cross Section in pp Collisions at $\sqrt{s} = 8$ TeV in Dilepton Final States Containing One τ Lepton

Khachatryan, Vardan et al.

[Phys.Lett.B 739 \(2014\) 23](#) | [arXiv:1407.6643](#) | [DOI:10.1016/j.physletb.2014.10.032](#) | 2014-07-24

1137. Search for Heavy Neutrinos and W Bosons with Right-Handed Couplings in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 74 \(2014\) 3149](#) | [arXiv:1407.3683](#) | [DOI:10.1140/epjc/s10052-014-3149-z](#) | 2014-07-14

1138. Search for New Resonances Decaying via WZ to Leptons in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 740 \(2015\) 83](#) | [arXiv:1407.3476](#) | [DOI:10.1016/j.physletb.2014.11.026](#) | 2014-07-13

1139. Study of Hadronic Event-Shape Variables in Multijet Final States in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[JHEP 10 \(2014\) 087](#) | [arXiv:1407.2856](#) | [DOI:10.1007/JHEP10\(2014\)087](#) | 2014-07-10

1140. Observation of the Diphoton Decay of the Higgs Boson and Measurement of Its Properties

Khachatryan, Vardan et al.

[Eur.Phys.J.C 74 \(2014\) 3076](#) | [arXiv:1407.0558](#) | [DOI:10.1140/epjc/s10052-014-3076-z](#) | 2014-07-02

1141. Differential Cross Section Measurements for the Production of a W Boson in Association with Jets in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 741 \(2015\) 12](#) | [arXiv:1406.7533](#) | [DOI:10.1016/j.physletb.2014.12.003](#) | 2014-06-29

1142. Search for Excited Quarks in the γ + Jet Final State in Proton-Proton Collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 738 \(2014\) 274](#) | [arXiv:1406.5171](#) | [DOI:10.1016/j.physletb.2014.09.048](#) | 2014-06-19

1143. Measurement of Jet Fragmentation in PbPb and pp Collisions at $\sqrt{s_{NN}}= 2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.C 90 \(2014\) 024908](#) | [arXiv:1406.0932](#) | [DOI:10.1103/PhysRevC.90.024908](#) | 2014-06-03

1144. Measurement of Prompt J/ψ Pair Production in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[JHEP 09 \(2014\) 094](#) | [arXiv:1406.0484](#) | [DOI:10.1007/JHEP09\(2014\)094](#) | 2014-06-02

1145. Measurement of the Ratio of Inclusive Jet Cross Sections using the Anti- k_T Algorithm with Radius Parameters $R=0.5$ and 0.7 in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 90 \(2014\) 072006](#) | [arXiv:1406.0324](#) | [DOI:10.1103/PhysRevD.90.072006](#) | 2014-06-02

1146. Measurement of the $pp \rightarrow ZZ$ Production Cross Section and Constraints on Anomalous Triple Gauge Couplings in Four-Lepton Final States at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 740 \(2015\) 250](#) | [arXiv:1406.0113](#) | [DOI:10.1016/j.physletb.2016.04.010](#) | 2014-05-31

1147. Search for Jet Extinction in the Inclusive Jet- p_t Spectrum from Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 90 \(2014\) 032005](#) | [arXiv:1405.7653](#) | [DOI:10.1103/PhysRevD.90.032005](#) | 2014-05-29

1148. Measurement of Differential Cross Sections for the Production of a Pair of Isolated Photons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 3129](#) | [arXiv:1405.7225](#) | [DOI:10.1140/epjc/s10052-014-3129-3](#) | 2014-05-28

1149. Description and Performance of Track and Primary-Vertex Reconstruction with the CMS Tracker

Chatrchyan, Serguei et al.

[JINST 9 \(2014\) P10009](#) | [arXiv:1405.6569](#) | [DOI:10.1088/1748-0221/9/10/P10009](#) | 2014-05-26

1150. Search for Supersymmetry with Razor Variables in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 90 \(2014\) 112001](#) | [arXiv:1405.3961](#) | [DOI:10.1103/PhysRevD.90.112001](#) | 2014-05-15

1151. Search for Top-Squark Pairs Decaying into Higgs or Z Bosons in pp Collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 736 \(2014\) 371](#) | [arXiv:1405.3886](#) | [DOI:10.1016/j.physletb.2014.07.053](#) | 2014-05-15

1152. Search for Massive Resonances Decaying into Pairs of Boosted Bosons in Semi-Leptonic Final States at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 08 \(2014\) 174](#) | [arXiv:1405.3447](#) | [DOI:10.1007/JHEP08\(2014\)174](#) | 2014-05-14

1153. Constraints on the Higgs Boson Width from Off-Shell Production and Decay to Z-Boson Pairs

Khachatryan, Vardan et al.

[Phys.Lett.B 736 \(2014\) 64](#) | [arXiv:1405.3455](#) | [DOI:10.1016/j.physletb.2014.06.077](#) | 2014-05-14

1154. Search for Massive Resonances in Dijet Systems Containing Jets Tagged as W or Z Boson Decays in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 08 \(2014\) 173](#) | [arXiv:1405.1994](#) | [DOI:10.1007/JHEP08\(2014\)173](#) | 2014-05-08

1155. Measurement of Pseudorapidity Distributions of Charged Particles in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV by the CMS and TOTEM Experiments

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 3053](#) | [arXiv:1405.0722](#) | [DOI:10.1140/epjc/s10052-014-3053-6](#) | 2014-05-04

1156. Search for Anomalous Production of Events with Three or More Leptons in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 90 \(2014\) 032006](#) | [arXiv:1404.5801](#) | [DOI:10.1103/PhysRevD.90.032006](#) | 2014-04-23

1157. Search for WW γ and WZ γ Production and Constraints on Anomalous Quartic Gauge Couplings in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 90 \(2014\) 032008](#) | [arXiv:1404.4619](#) | [DOI:10.1103/PhysRevD.90.032008](#) | 2014-04-17

1158. Measurement of Jet Multiplicity Distributions in tt Production in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2015\) 3014](#) | [arXiv:1404.3171](#) | [DOI:10.1140/epjc/s10052-014-3014-0](#) | 2014-04-11

1159. Measurement of the Ratio $\mathcal{B}(t \rightarrow Wb)/\mathcal{B}(t \rightarrow Wq)$ in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 736 \(2014\) 33](#) | [arXiv:1404.2292](#) | [DOI:10.1016/j.physletb.2014.06.076](#) | 2014-04-08

1160. Search for Invisible Decays of Higgs Bosons in the Vector Boson Fusion and Associated ZH Production Modes

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 2980](#) | [arXiv:1404.1344](#) | [DOI:10.1140/epjc/s10052-014-2980-6](#) | 2014-04-04

1161. Measurement of the t-channel single-top-quark production cross section and of the $|V_{tb}|$ CKM matrix element in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 06 \(2014\) 090](#) | [arXiv:1403.7366](#) | [DOI:10.1007/JHEP06\(2014\)090](#) | 2014-03-28

1162. Measurement of WZ and ZZ Production in pp Collisions at $\sqrt{s} = 8$ TeV in Final States with b-Tagged Jets

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 2973](#) | [arXiv:1403.3047](#) | [DOI:10.1140/epjc/s10052-014-2973-5](#) | 2014-03-12

1163. Alignment of the CMS Tracker with LHC and Cosmic Ray Data

Chatrchyan, Serguei et al.

[JINST 9 \(2014\) P06009](#) | [arXiv:1403.2286](#) | [DOI:10.1088/1748-0221/9/06/P06009](#) | 2014-03-10

1164. Search for New Physics in the Multijet and Missing Transverse Momentum Final State in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2014\) 055](#) | [arXiv:1402.4770](#) | [DOI:10.1007/JHEP06\(2014\)055](#) | 2014-02-19

1165. Search for $W' \rightarrow tb$ Decays in the Lepton + Jets Final State in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[JHEP 05 \(2014\) 108](#) | [arXiv:1402.2176](#) | [DOI:10.1007/JHEP05\(2014\)108](#) | 2014-02-10

1166. Measurement of the Production Cross Sections for a Z Boson and One or More b Jets in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2014\) 120](#) | [arXiv:1402.1521](#) | [DOI:10.1007/JHEP06\(2014\)120](#) | 2014-02-06

1167. Measurement of Inclusive W and Z Boson Production Cross Sections in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 112 \(2014\) 191802](#) | [arXiv:1402.0923](#) | [DOI:10.1103/PhysRevLett.112.191802](#) | 2014-02-04

1168. Evidence for the Direct Decay of the 125 GeV Higgs Boson to Fermions

Chatrchyan, Serguei et al.

[Nature Phys. 10 \(2014\) 557](#) | [arXiv:1401.6527](#) | [DOI:10.1038/nphys3005](#) | 2014-01-25

1169. Determination of the Top-Quark Pole Mass and Strong Coupling Constant from the $t\bar{t}$ Production Cross Section in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 728 \(2014\) 496](#) | [arXiv:1307.1907](#) | [DOI:10.1016/j.physletb.2013.12.009](#) | 2014-01-20

1170. Evidence for the 125 GeV Higgs Boson Decaying to a Pair of τ Leptons

Chatrchyan, Serguei et al.

[JHEP 05 \(2014\) 104](#) | [arXiv:1401.5041](#) | [DOI:10.1007/JHEP05\(2014\)104](#) | 2014-01-20

1171. Studies of Dijet Transverse Momentum Balance and Pseudorapidity Distributions in pPb Collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 2951](#) | [arXiv:1401.4433](#) | [DOI:10.1140/epjc/s10052-014-2951-y](#) | 2014-01-17

1172. Observation of the Associated Production of a Single Top Quark and a W Boson in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 112 \(2014\) 231802](#) | [arXiv:1401.2942](#) | [DOI:10.1103/PhysRevLett.112.231802](#) | 2014-01-13

1173. Measurement of the $t\bar{t}$ production cross section in the dilepton channel in pp collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[JHEP 02 \(2014\) 024](#) | [arXiv:1312.7582](#) | [DOI:10.1007/JHEP02\(2014\)024](#) | 2013-12-29

1174. Measurement of the Production Cross Section for a W Boson and Two b Jets in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 735 \(2014\) 204](#) | [arXiv:1312.6608](#) | [DOI:10.1016/j.physletb.2014.06.041](#) | 2013-12-23

1175. Measurement of Four-Jet Production in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 89 \(2014\) 092010](#) | [arXiv:1312.6440](#) | [DOI:10.1103/PhysRevD.89.092010](#) | 2013-12-22

1176. Event Activity Dependence of $Y(nS)$ Production in $\sqrt{s_{NN}}=5.02$ TeV pPb and $\sqrt{s}=2.76$ TeV pp Collisions

Chatrchyan, Serguei et al.

[JHEP 04 \(2014\) 103](#) | [arXiv:1312.6300](#) | [DOI:10.1007/JHEP04\(2014\)103](#) | 2013-12-21

1177. Measurement of the Muon Charge Asymmetry in Inclusive $pp \rightarrow W+X$ Production at $\sqrt{s} = 7$ TeV and an Improved Determination of Light Parton Distribution Functions

Chatrchyan, Serguei et al.

[Phys.Rev.D 90 \(2014\) 032004](#) | [arXiv:1312.6283](#) | [DOI:10.1103/PhysRevD.90.032004](#) | 2013-12-21

1178. Study of Double Parton Scattering Using W + 2-Jet Events in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 03 \(2014\) 032](#) | [arXiv:1312.5729](#) | [DOI:10.1007/JHEP03\(2014\)032](#) | 2013-12-19

1179. Measurement of the Properties of a Higgs Boson in the Four-Lepton Final State

Chatrchyan, Serguei et al.

[Phys.Rev.D 89 \(2014\) 092007](#) | [arXiv:1312.5353](#) | [DOI:10.1103/PhysRevD.89.092007](#) | 2013-12-18

1180. Search for Flavor-Changing Neutral Currents in Top-Quark Decays $t \rightarrow Zq$ in pp Collisions at $\sqrt{s}=8$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 112 \(2014\) 171802](#) | [arXiv:1312.4194](#) | [DOI:10.1103/PhysRevLett.112.171802](#) | 2013-12-15

1181. Evidence of b-Jet Quenching in PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 113 \(2014\) 132301](#) | [arXiv:1312.4198](#) | [DOI:10.1103/PhysRevLett.113.132301](#) | 2013-12-15

1182. Search for Top Squark and Higgsino Production using Diphoton Higgs Boson Decays

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 112 \(2014\) 161802](#) | [arXiv:1312.3310](#) | [DOI:10.1103/PhysRevLett.112.161802](#) | 2013-12-11

1183. Search for Top-Quark Partners with Charge 5/3 in the Same-Sign Dilepton Final State

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 112 \(2014\) 171801](#) | [arXiv:1312.2391](#) | [DOI:10.1103/PhysRevLett.112.171801](#) | 2013-12-09

1184. Studies of Azimuthal Dihadron Correlations in Ultra-Central PbPb Collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Chatrchyan, Serguei et al.

[JHEP 02 \(2014\) 088](#) | [arXiv:1312.1845](#) | [DOI:10.1007/JHEP02\(2014\)088](#) | 2013-12-06

1185. Inclusive Search for a Vector-Like T Quark with Charge 2/3 in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 729 \(2014\) 149](#) | [arXiv:1311.7667](#) | [DOI:10.1016/j.physletb.2014.01.006](#) | 2013-11-29

1186. Measurement of the Triple-Differential Cross Section for Photon+Jets Production in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2014\) 009](#) | [arXiv:1311.6141](#) | [DOI:10.1007/JHEP06\(2014\)009](#) | 2013-11-24

1187. Probing Color Coherence Effects in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 2901](#) | [arXiv:1311.5815](#) | [DOI:10.1140/epjc/s10052-014-2901-8](#) | 2013-11-22

1188. Search for Pair Production of Excited Top Quarks in the Lepton + Jets Final State

Chatrchyan, Serguei et al.

[JHEP 06 \(2014\) 125](#) | [arXiv:1311.5357](#) | [DOI:10.1007/JHEP06\(2014\)125](#) | 2013-11-21

1189. Search for Supersymmetry in pp Collisions at $\sqrt{s}=8$ TeV in Events with a Single Lepton, Large Jet Multiplicity, and Multiple b Jets

Chatrchyan, Serguei et al.

[Phys.Lett.B 733 \(2014\) 328](#) | [arXiv:1311.4937](#) | [DOI:10.1016/j.physletb.2014.04.023](#) | 2013-11-19

1190. Searches for Light- and Heavy-Flavour Three-Jet Resonances in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 730 \(2014\) 193](#) | [arXiv:1311.1799](#) | [DOI:10.1016/j.physletb.2014.01.049](#) | 2013-11-07

1191. Measurement of Higher-Order Harmonic Azimuthal Anisotropy in PbPb Collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.C 89 \(2014\) 044906](#) | [arXiv:1310.8651](#) | [DOI:10.1103/PhysRevC.89.044906](#) | 2013-10-31

1192. Measurement of the Differential and Double-Differential Drell-Yan Cross Sections in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 12 \(2013\) 030](#) | [arXiv:1310.7291](#) | [DOI:10.1007/JHEP12\(2013\)030](#) | 2013-10-27

1193. Jet and Underlying Event Properties as a Function of Charged-Particle Multiplicity in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2674](#) | [arXiv:1310.4554](#) | [DOI:10.1140/epjc/s10052-013-2674-5](#) | 2013-10-16

1194. Search for the Standard Model Higgs Boson Produced in Association with a W or a Z Boson and Decaying to Bottom Quarks

Chatrchyan, Serguei et al.

[Phys.Rev.D 89 \(2014\) 012003](#) | [arXiv:1310.3687](#) | [DOI:10.1103/PhysRevD.89.012003](#) | 2013-10-14

1195. Rapidity Distributions in Exclusive Z + Jet and γ + Jet Events in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 88 \(2013\) 112009](#) | [arXiv:1310.3082](#) | [DOI:10.1103/PhysRevD.88.112009](#) | 2013-10-11

1196. Search for Baryon Number Violation in Top-Quark Decays

Chatrchyan, Serguei et al.

[Phys.Lett.B 731 \(2014\) 173](#) | [arXiv:1310.1618](#) | [DOI:10.1016/j.physletb.2014.02.033](#) | 2013-10-06

1197. Measurement of the Cross Section and Angular Correlations for Associated Production of a Z Boson with b Hadrons in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 12 \(2013\) 039](#) | [arXiv:1310.1349](#) | [DOI:10.1007/JHEP12\(2013\)039](#) | 2013-10-04

1198. Measurement of Associated W + Charm Production in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 02 \(2014\) 013](#) | [arXiv:1310.1138](#) | [DOI:10.1007/JHEP02\(2014\)013](#) | 2013-10-03

1199. Modification of Jet Shapes in PbPb Collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 730 \(2014\) 243](#) | [arXiv:1310.0878](#) | [DOI:10.1016/j.physletb.2014.01.042](#) | 2013-10-02

1200. Observation of a Peaking Structure in the $J/\psi\phi$ Mass Spectrum from $B^{\pm} \rightarrow J/\psi\phi K^{\pm}$ Decays

Chatrchyan, Serguei et al.

[Phys.Lett.B 734 \(2014\) 261](#) | [arXiv:1309.6920](#) | [DOI:10.1016/j.physletb.2014.05.055](#) | 2013-09-26

1201. Searches for New Physics using the $t\bar{t}$ Invariant Mass Distribution in pp Collisions at $\sqrt{s}=8$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 111 \(2013\) 211804](#) | [arXiv:1309.2030](#) | [DOI:10.1103/PhysRevLett.111.211804](#) | 2013-09-08

1202. Measurement of the Production Cross Section for $Z\gamma \rightarrow \nu\bar{\nu}\gamma$ in pp Collisions at $\sqrt{s} = 7$ TeV and Limits on $ZZ\gamma$ and $Z\gamma\gamma$ Triple Gauge Boson Couplings

Chatrchyan, Serguei et al.

[JHEP 10 \(2013\) 164](#) | [arXiv:1309.1117](#) | [DOI:10.1007/JHEP10\(2013\)164](#) | 2013-09-04

1203. Search for a New Bottomonium State Decaying to $\Upsilon(1S)\pi^+\pi^-$ in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 727 \(2013\) 57](#) | [arXiv:1309.0250](#) | [DOI:10.1016/j.physletb.2013.10.016](#) | 2013-09-01

1204. Measurement of the $W\gamma$ and $Z\gamma$ Inclusive Cross Sections in pp Collisions at $\sqrt{s}=7$ TeV and Limits on Anomalous Triple Gauge Boson Couplings

Chatrchyan, Serguei et al.

[Phys.Rev.D 89 \(2014\) 092005](#) | [arXiv:1308.6832](#) | [DOI:10.1103/PhysRevD.89.092005](#) | 2013-08-30

1205. Measurement of the W-Boson Helicity in Top-Quark decays from $t\bar{t}$ Production in Lepton + Jets Events in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 10 \(2013\) 167](#) | [arXiv:1308.3879](#) | [DOI:10.1007/JHEP10\(2013\)167](#) | 2013-08-18

1206. Angular Analysis and Branching Fraction Measurement of the Decay $B^0 \rightarrow K^{*0}\mu^+\mu^-$

Chatrchyan, Serguei et al.

[Phys.Lett.B 727 \(2013\) 77](#) | [arXiv:1308.3409](#) | [DOI:10.1016/j.physletb.2013.10.017](#) | 2013-08-15

1207. Measurement of the Prompt J/ψ and $\psi(2S)$ Polarizations in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 727 \(2013\) 381](#) | [arXiv:1307.6070](#) | [DOI:10.1016/j.physletb.2013.10.055](#) | 2013-07-23

1208. Search for a Higgs Boson Decaying into a Z and a Photon in pp Collisions at $\sqrt{s} = 7$ and 8 TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 726 \(2013\) 587](#) | [arXiv:1307.5515](#) | [DOI:10.1016/j.physletb.2013.09.057](#) | 2013-07-21

1209. Measurement of the $B^0_s \rightarrow \mu^+ \mu^-$ Branching Fraction and Search for $B^0 \rightarrow \mu^+ \mu^-$ with the CMS Experiment

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 111 \(2013\) 101804](#) | [arXiv:1307.5025](#) | [DOI:10.1103/PhysRevLett.111.101804](#) | 2013-07-18

1210. Measurement of the Top-Quark Mass in All-Jets $t\bar{t}$ Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 2758](#) | [arXiv:1307.4617](#) | [DOI:10.1140/epjc/s10052-014-2758-x](#) | 2013-07-17

1211. Study of the Production of Charged Pions, Kaons, and Protons in pPb Collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 2847](#) | [arXiv:1307.3442](#) | [DOI:10.1140/epjc/s10052-014-2847-x](#) | 2013-07-12

1212. The Performance of the CMS Muon Detector in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV at the LHC

Chatrchyan, Serguei et al.

[JINST 8 \(2013\) P11002](#) | [arXiv:1306.6905](#) | [DOI:10.1088/1748-0221/8/11/P11002](#) | 2013-06-28

1213. Search for Top Squarks in R-Parity-Violating Supersymmetry using Three or More Leptons and B-Tagged Jets

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 111 \(2013\) 221801](#) | [arXiv:1306.6643](#) | [DOI:10.1103/PhysRevLett.111.221801](#) | 2013-06-27

1214. Energy Calibration and Resolution of the CMS Electromagnetic Calorimeter in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JINST 8 \(2013\) P09009](#) | [arXiv:1306.2016](#) | [DOI:10.1088/1748-0221/8/09/P09009](#) | 2013-06-09

1215. Measurement of the Hadronic Activity in Events with a Z and Two Jets and Extraction of the Cross Section for the Electroweak Production of a Z with Two Jets in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 10 \(2013\) 062](#) | [arXiv:1305.7389](#) | [DOI:10.1007/JHEP10\(2013\)062](#) | 2013-05-31

1216. Measurement of Neutral Strange Particle Production in the Underlying Event in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 88 \(2013\) 052001](#) | [arXiv:1305.6016](#) | [DOI:10.1103/PhysRevD.88.052001](#) | 2013-05-26

1217. Study of Exclusive Two-Photon Production of W^+W^- in pp Collisions at $\sqrt{s} = 7$ TeV and Constraints on Anomalous Quartic Gauge Couplings

Chatrchyan, Serguei et al.

[JHEP 07 \(2013\) 116](#) | [arXiv:1305.5596](#) | [DOI:10.1007/JHEP07\(2013\)116](#) | 2013-05-23

1218. Search for Gluino Mediated Bottom- and Top-Squark Production in Multijet Final States in pp Collisions at 8 TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 725 \(2013\) 243](#) | [arXiv:1305.2390](#) | [DOI:10.1016/j.physletb.2013.06.058](#) | 2013-05-10

1219. Searches for Long-Lived Charged Particles in pp Collisions at $\sqrt{s}=7$ and 8 TeV

Chatrchyan, Serguei et al.

[JHEP 07 \(2013\) 122](#) | [arXiv:1305.0491](#) | [DOI:10.1007/JHEP07\(2013\)122](#) | 2013-05-02

1220. Multiplicity and Transverse Momentum Dependence of Two- and Four-Particle Correlations in pPb and PbPb Collisions

Chatrchyan, Serguei et al.

[Phys.Lett.B 724 \(2013\) 213](#) | [arXiv:1305.0609](#) | [DOI:10.1016/j.physletb.2013.06.028](#) | 2013-05-02

1221. Measurement of the Λ_b^0 Lifetime in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 07 \(2013\) 163](#) | [arXiv:1304.7495](#) | [DOI:10.1007/JHEP07\(2013\)163](#) | 2013-04-28

1222. Measurement of the Ratio of the Inclusive 3-Jet Cross Section to the Inclusive 2-Jet Cross Section in pp Collisions at $\sqrt{s} = 7$ TeV and First Determination of the Strong Coupling Constant in the TeV Range

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2604](#) | [arXiv:1304.7498](#) | [DOI:10.1140/epjc/s10052-013-2604-6](#) | 2013-04-28

1223. Measurement of Masses in the $t\bar{t}$ System by Kinematic Endpoints in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2494](#) | [arXiv:1304.5783](#) | [DOI:10.1140/epjc/s10052-013-2494-7](#) | 2013-04-21

1224. Search for a Standard-Model-Like Higgs Boson with a Mass in the Range 145 to 1000 GeV at the LHC

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2469](#) | [arXiv:1304.0213](#) | [DOI:10.1140/epjc/s10052-013-2469-8](#) | 2013-03-31

1225. Measurement of the $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ Cross Sections in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 727 \(2013\) 101](#) | [arXiv:1303.5900](#) | [DOI:10.1016/j.physletb.2013.10.033](#) | 2013-03-23

1226. Search for Microscopic Black Holes in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[JHEP 07 \(2013\) 178](#) | [arXiv:1303.5338](#) | [DOI:10.1007/JHEP07\(2013\)178](#) | 2013-03-21

1227. Observation of a New Boson with Mass Near 125 GeV in pp Collisions at $\sqrt{s} = 7$ and 8 TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2013\) 081](#) | [arXiv:1303.4571](#) | [DOI:10.1007/JHEP06\(2013\)081](#) | 2013-03-19

1228. Studies of Jet Mass in Dijet and W/Z + Jet Events

Chatrchyan, Serguei et al.

[JHEP 05 \(2013\) 090](#) | [arXiv:1303.4811](#) | [DOI:10.1007/JHEP05\(2013\)090](#) | 2013-03-19

1229. Measurement of associated production of vector bosons and top quark-antiquark pairs at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 110 \(2013\) 172002](#) | [arXiv:1303.3239](#) | [DOI:10.1103/PhysRevLett.110.172002](#) | 2013-03-13

1230. Search for Supersymmetry in Hadronic Final States with Missing Transverse Energy Using the Variables α_T and b-Quark Multiplicity in pp collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2568](#) | [arXiv:1303.2985](#) | [DOI:10.1140/epjc/s10052-013-2568-6](#) | 2013-03-12

1231. Search for the Standard Model Higgs Boson Produced in Association with a Top-Quark Pair in pp Collisions at the LHC

Chatrchyan, Serguei et al.

[JHEP 05 \(2013\) 145](#) | [arXiv:1303.0763](#) | [DOI:10.1007/JHEP05\(2013\)145](#) | 2013-03-04

1232. Search for Narrow Resonances Using the Dijet Mass Spectrum in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 114015](#) | [arXiv:1302.4794](#) | [DOI:10.1103/PhysRevD.87.114015](#) | 2013-02-19

1233. Measurement of the X(3872) Production Cross Section Via Decays to $J/\psi \pi^+ \pi^-$ in pp collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 04 \(2013\) 154](#) | [arXiv:1302.3968](#) | [DOI:10.1007/JHEP04\(2013\)154](#) | 2013-02-16

1234. Search for a Higgs Boson Decaying into a b-Quark Pair and Produced in Association with b Quarks in Proton-Proton Collisions at 7 TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 722 \(2013\) 207](#) | [arXiv:1302.2892](#) | [DOI:10.1016/j.physletb.2013.04.017](#) | 2013-02-12

1235. Search for New Physics in Final States with a Lepton and Missing Transverse Energy in pp Collisions at the LHC

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 072005](#) | [arXiv:1302.2812](#) | [DOI:10.1103/PhysRevD.87.072005](#) | 2013-02-12

1236. Study of the Underlying Event at Forward Rapidity in pp Collisions at $\sqrt{s} = 0.9, 2.76$, and 7 TeV

Chatrchyan, Serguei et al.

[JHEP 04 \(2013\) 072](#) | [arXiv:1302.2394](#) | [DOI:10.1007/JHEP04\(2013\)072](#) | 2013-02-10

1237. Searches for Higgs Bosons in pp Collisions at $\sqrt{s} = 7$ and 8 TeV in the Context of Four-Generation and Fermiophobic Models

Chatrchyan, Serguei et al.

[Phys.Lett.B 725 \(2013\) 36](#) | [arXiv:1302.1764](#) | [DOI:10.1016/j.physletb.2013.06.043](#) | 2013-02-07

1238. Search for Pair-Produced Dijet Resonances in Four-Jet Final States in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 110 \(2013\) 141802](#) | [arXiv:1302.0531](#) | [DOI:10.1103/PhysRevLett.110.141802](#) | 2013-02-03

1239. Measurement of the $t\bar{t}$ Production Cross Section in the All-Jet Final State in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 05 \(2013\) 065](#) | [arXiv:1302.0508](#) | [DOI:10.1007/JHEP05\(2013\)065](#) | 2013-02-03

1240. Measurement of the $t\bar{t}$ Production Cross Section in the τ +Jets Channel in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2386](#) | [arXiv:1301.5755](#) | [DOI:10.1140/epjc/s10052-013-2386-x](#) | 2013-01-24

1241. Search for Contact Interactions Using the Inclusive Jet p_T Spectrum in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 052017](#) | [arXiv:1301.5023](#) | [DOI:10.1103/PhysRevD.87.052017](#) | 2013-01-21

1242. Search for Supersymmetry in Events with Opposite-Sign Dileptons and Missing Transverse Energy Using an Artificial Neural Network

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 072001](#) | [arXiv:1301.0916](#) | [DOI:10.1103/PhysRevD.87.072001](#) | 2013-01

1243. Event Shapes and Azimuthal Correlations in Z + Jets Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 722 \(2013\) 238](#) | [arXiv:1301.1646](#) | [DOI:10.1016/j.physletb.2013.04.025](#) | 2013-01

1244. Search for Physics Beyond the Standard Model in Events with τ Leptons, Jets, and Large Transverse Momentum Imbalance in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2493](#) | [arXiv:1301.3792](#) | [DOI:10.1140/epjc/s10052-013-2493-8](#) | 2013-01

1245. Interpretation of Searches for Supersymmetry with Simplified Models

Chatrchyan, Serguei et al.

[Phys.Rev.D 88 \(2013\) 052017](#) | [arXiv:1301.2175](#) | [DOI:10.1103/PhysRevD.88.052017](#) | 2013-01

1246. Search for Long-Lived Particles Decaying to Photons and Missing Energy in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 722 \(2013\) 273](#) | [arXiv:1212.1838](#) | [DOI:10.1016/j.physletb.2013.04.027](#) | 2012-12

1247. Search for Heavy Resonances in the W/Z-Tagged Dijet Mass Spectrum in pp Collisions at 7 TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 723 \(2013\) 280](#) | [arXiv:1212.1910](#) | [DOI:10.1016/j.physletb.2013.05.040](#) | 2012-12

1248. Measurements of Differential Jet Cross Sections in Proton-Proton Collisions at $\sqrt{s}=7$ TeV with the CMS Detector

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 112002](#) | [arXiv:1212.6660](#) | [DOI:10.1103/PhysRevD.87.112002](#) | 2012-12

1249. Measurement of the $t\bar{t}$ Production Cross Section in pp Collisions at $\sqrt{s}=7$ TeV with Lepton + Jets Final States

Chatrchyan, Serguei et al.

Phys.Lett.B 720 (2013) 83 | [arXiv:1212.6682](#) | [DOI:10.1016/j.physletb.2013.02.021](#) | 2012-12

1250. Search for Contact Interactions in $\mu^+\mu^-$ Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Phys.Rev.D 87 (2013) 032001 | [arXiv:1212.4563](#) | [DOI:10.1103/PhysRevD.87.032001](#) | 2012-12

1251. Search for Heavy Narrow Dilepton Resonances in pp Collisions at $\sqrt{s}=7$ TeV and $\sqrt{s}=8$ TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 720 (2013) 63 | [arXiv:1212.6175](#) | [DOI:10.1016/j.physletb.2013.02.003](#) | 2012-12

1252. Study of the Mass and Spin-Parity of the Higgs Boson Candidate Via Its Decays to Z Boson Pairs

Chatrchyan, Serguei et al.

Phys.Rev.Lett. 110 (2013) 081803 | [arXiv:1212.6639](#) | [DOI:10.1103/PhysRevLett.110.081803](#) | 2012-12

1253. Search for Supersymmetry in pp Collisions at $\sqrt{s}=7$ TeV in Events with a Single Lepton, Jets, and Missing Transverse Momentum

Chatrchyan, Serguei et al.

Eur.Phys.J.C 73 (2013) 2404 | [arXiv:1212.6428](#) | [DOI:10.1140/epjc/s10052-013-2404-z](#) | 2012-12

1254. Inclusive Search for Supersymmetry Using the Razor Variables in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Phys.Rev.Lett. 111 (2013) 081802 | [arXiv:1212.6961](#) | [DOI:10.1103/PhysRevLett.111.081802](#) | 2012-12

1255. Search in Leptonic Channels for Heavy Resonances Decaying to Long-Lived Neutral Particles

Chatrchyan, Serguei et al.

JHEP 02 (2013) 085 | [arXiv:1211.2472](#) | [DOI:10.1007/JHEP02\(2013\)085](#) | 2012-11

1256. Measurement of the ZZ Production Cross Section and Search for Anomalous Couplings in $2\ell\ell'$ Final States in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

JHEP 01 (2013) 063 | [arXiv:1211.4890](#) | [DOI:10.1007/JHEP01\(2013\)063](#) | 2012-11

1257. Identification of b-Quark Jets with the CMS Experiment

Chatrchyan, Serguei et al.

JINST 8 (2013) P04013 | [arXiv:1211.4462](#) | [DOI:10.1088/1748-0221/8/04/P04013](#) | 2012-11

1258. Measurement of Differential Top-Quark Pair Production Cross Sections in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Eur.Phys.J.C 73 (2013) 2339 | [arXiv:1211.2220](#) | [DOI:10.1140/epjc/s10052-013-2339-4](#) | 2012-11

1259. Search for Exotic Resonances Decaying into WZ/ZZ in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

JHEP 02 (2013) 036 | [arXiv:1211.5779](#) | [DOI:10.1007/JHEP02\(2013\)036](#) | 2012-11

1260. Search for Supersymmetry in Final States with a Single Lepton, b-Quark Jets, and Missing Transverse Energy in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Phys.Rev.D 87 (2013) 052006 | [arXiv:1211.3143](#) | [DOI:10.1103/PhysRevD.87.052006](#) | 2012-11

1261. Search for New Physics in Events with Photons, Jets, and Missing Transverse Energy in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

JHEP 03 (2013) 111 | [arXiv:1211.4784](#) | [DOI:10.1007/JHEP03\(2013\)111](#) | 2012-11

1262. Search for Z' Resonances Decaying to $t\bar{t}$ in Dilepton + Jets Final States in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 072002](#) | [arXiv:1211.3338](#) | [DOI:10.1103/PhysRevD.87.072002](#) | 2012-11

1263. Observation of Z Decays to Four Leptons with the CMS Detector at the LHC

Chatrchyan, Serguei et al.

[JHEP 12 \(2012\) 034](#) | [arXiv:1210.3844](#) | [DOI:10.1007/JHEP12\(2012\)034](#) | 2012-10

1264. Search for Third-Generation Leptoquarks and Scalar Bottom Quarks in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 12 \(2012\) 055](#) | [arXiv:1210.5627](#) | [DOI:10.1007/JHEP12\(2012\)055](#) | 2012-10

1265. Measurement of the Relative Prompt Production Rate of χ_{c2} and χ_{c1} in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 2251](#) | [arXiv:1210.0875](#) | [DOI:10.1140/epjc/s10052-012-2251-3](#) | 2012-10

1266. Search for Heavy Lepton Partners of Neutrinos in Proton-Proton Collisions in the Context of the Type III Seesaw Mechanism

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2012\) 348](#) | [arXiv:1210.1797](#) | [DOI:10.1016/j.physletb.2012.10.070](#) | 2012-10

1267. Search for Heavy Neutrinos and W_R Bosons with Right-Handed Couplings in a Left-Right Symmetric Model in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 261802](#) | [arXiv:1210.2402](#) | [DOI:10.1103/PhysRevLett.109.261802](#) | 2012-10

1268. Search for Excited Leptons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 720 \(2013\) 309](#) | [arXiv:1210.2422](#) | [DOI:10.1016/j.physletb.2013.02.031](#) | 2012-10

1269. Search for Pair Production of Third-Generation Leptoquarks and Top Squarks in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 110 \(2013\) 081801](#) | [arXiv:1210.5629](#) | [DOI:10.1103/PhysRevLett.110.081801](#) | 2012-10

1270. Measurement of the Inelastic Proton-Proton Cross Section at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 722 \(2013\) 5](#) | [arXiv:1210.6718](#) | [DOI:10.1016/j.physletb.2013.03.024](#) | 2012-10

1271. Search for Heavy Quarks Decaying into a Top Quark and a W or Z Boson using Lepton + Jets Events in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 01 \(2013\) 154](#) | [arXiv:1210.7471](#) | [DOI:10.1007/JHEP01\(2013\)154](#) | 2012-10

1272. Observation of Long-Range Near-Side Angular Correlations in Proton-Lead Collisions at the LHC

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2013\) 795](#) | [arXiv:1210.5482](#) | [DOI:10.1016/j.physletb.2012.11.025](#) | 2012-10

1273. Search for a Non-Standard-Model Higgs Boson Decaying to a Pair of New Light Bosons in Four-Muon Final States

Chatrchyan, Serguei et al.

[Phys.Lett.B 726 \(2013\) 564](#) | [arXiv:1210.7619](#) | [DOI:10.1016/j.physletb.2013.09.009](#) | 2012-10

1274. Search for Supersymmetry in Final States with Missing Transverse Energy and 0, 1, 2, or at Least 3 b-Quark Jets in 7 TeV pp Collisions using the Variable α_T

Chatrchyan, Serguei et al.

[JHEP 01 \(2013\) 077](#) | [arXiv:1210.8115](#) | [DOI:10.1007/JHEP01\(2013\)077](#) | 2012-10

1275. Search for Anomalous Production of Highly Boosted Z Bosons Decaying to $\mu^+ \mu^-$ in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 722 \(2013\) 28](#) | [arXiv:1210.0867](#) | [DOI:10.1016/j.physletb.2013.03.037](#) | 2012-10

1276. Search for Supersymmetry in Events with Photons and Low Missing Transverse Energy in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 719 \(2013\) 42](#) | [arXiv:1210.2052](#) | [DOI:10.1016/j.physletb.2012.12.055](#) | 2012-10

1277. Measurement of the Sum of W W and WZ Production with W+Dijet Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2283](#) | [arXiv:1210.7544](#) | [DOI:10.1140/epjc/s10052-013-2283-3](#) | 2012-10

1278. Search for Fractionally Charged Particles in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, S. et al.

[Phys.Rev.D 87 \(2013\) 092008](#) | [arXiv:1210.2311](#) | [DOI:10.1103/PhysRevD.87.092008](#) | 2012-10

1279. Search for Narrow Resonances and Quantum Black Holes in Inclusive and b-Tagged Dijet Mass Spectra from pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 01 \(2013\) 013](#) | [arXiv:1210.2387](#) | [DOI:10.1007/JHEP01\(2013\)013](#) | 2012-10

1280. Search for the Standard Model Higgs Boson Produced in Association with W and Z Bosons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 11 \(2012\) 088](#) | [arXiv:1209.3937](#) | [DOI:10.1007/JHEP11\(2012\)088](#) | 2012-09

1281. Search for Exclusive or Semi-Exclusive Photon Pair Production and Observation of Exclusive and Semi-Exclusive Electron Pair Production in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 11 \(2012\) 080](#) | [arXiv:1209.1666](#) | [DOI:10.1007/JHEP11\(2012\)080](#) | 2012-09

1282. Search for Resonant $t\bar{t}$ Production in Lepton+Jets Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 12 \(2012\) 015](#) | [arXiv:1209.4397](#) | [DOI:10.1007/JHEP12\(2012\)015](#) | 2012-09

1283. Search for Pair Produced Fourth-Generation Up-Type Quarks in pp Collisions at $\sqrt{s}=7$ TeV with a Lepton in the Final State

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2012\) 307](#) | [arXiv:1209.0471](#) | [DOI:10.1016/j.physletb.2012.10.038](#) | 2012-09

1284. Measurement of the Top-Quark Mass in $t\bar{t}$ Events with Lepton+Jets Final States in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 12 \(2012\) 105](#) | [arXiv:1209.2319](#) | [DOI:10.1007/JHEP12\(2012\)105](#) | 2012-09

1285. Measurement of the Top-Quark Mass in $t\bar{t}$ Events with Dilepton Final States in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

- Eur.Phys.J.C 72 (2012) 2202 | [arXiv:1209.2393](#) | [DOI:10.1140/epjc/s10052-012-2202-z](#) | 2012-09
1286. Combined Search for the Quarks of a Sequential Fourth Generation
Chatrchyan, Serguei et al.
[Phys.Rev.D 86 \(2012\) 112003](#) | [arXiv:1209.1062](#) | [DOI:10.1103/PhysRevD.86.112003](#) | 2012-09
1287. Measurement of the Single-Top-Quark t-Channel Cross Section in pp Collisions at $\sqrt{s}=7$ TeV
Chatrchyan, Serguei et al.
[JHEP 12 \(2012\) 035](#) | [arXiv:1209.4533](#) | [DOI:10.1007/JHEP12\(2012\)035](#) | 2012-09
1288. Measurement of the Y(1S), Y(2S) and Y(3S) Polarizations in pp Collisions at $\sqrt{s}=7$ TeV
Chatrchyan, Serguei et al.
[Phys.Rev.Lett. 110 \(2013\) 081802](#) | [arXiv:1209.2922](#) | [DOI:10.1103/PhysRevLett.110.081802](#) | 2012-09
1289. Search for a Narrow Spin-2 Resonance Decaying to a Pair of Z Vector Bosons in the Semileptonic Final State
Chatrchyan, Serguei et al.
[Phys.Lett.B 718 \(2013\) 1208](#) | [arXiv:1209.3807](#) | [DOI:10.1016/j.physletb.2012.11.063](#) | 2012-09
1290. Observation of a Diffractive Contribution to Dijet Production in Proton-Proton Collisions at $\sqrt{s}=7$ TeV
Chatrchyan, Serguei et al.
[Phys.Rev.D 87 \(2013\) 012006](#) | [arXiv:1209.1805](#) | [DOI:10.1103/PhysRevD.87.012006](#) | 2012-09
1291. Evidence for Associated Production of a Single Top Quark and W Boson in pp Collisions at $\sqrt{s} = 7$ TeV
Chatrchyan, Serguei et al.
[Phys.Rev.Lett. 110 \(2013\) 022003](#) | [arXiv:1209.3489](#) | [DOI:10.1103/PhysRevLett.110.022003](#) | 2012-09
1292. Study of the Dijet Mass Spectrum in $pp \rightarrow W + \text{Jets}$ Events at $\sqrt{s}=7$ TeV
Chatrchyan, Serguei et al.
[Phys.Rev.Lett. 109 \(2012\) 251801](#) | [arXiv:1208.3477](#) | [DOI:10.1103/PhysRevLett.109.251801](#) | 2012-08
1293. Measurement of the $t\bar{t}$ Production Cross Section in the Dilepton Channel in pp Collisions at $\sqrt{s}=7$ TeV
Chatrchyan, Serguei et al.
[JHEP 11 \(2012\) 067](#) | [arXiv:1208.2671](#) | [DOI:10.1007/JHEP11\(2012\)067](#) | 2012-08
1294. Observation of Sequential Upsilon Suppression in PbPb Collisions
Chatrchyan, Serguei et al.
[Phys.Rev.Lett. 109 \(2012\) 222301](#) | [arXiv:1208.2826](#) | [DOI:10.1103/PhysRevLett.109.222301](#) | 2012-08
1295. Search for Supersymmetry in Events with b-Quark Jets and Missing Transverse Energy in pp Collisions at 7 TeV
Chatrchyan, Serguei et al.
[Phys.Rev.D 86 \(2012\) 072010](#) | [arXiv:1208.4859](#) | [DOI:10.1103/PhysRevD.86.072010](#) | 2012-08
1296. Search for Three-Jet Resonances in pp Collisions at $\sqrt{s}=7$ TeV
Chatrchyan, Serguei et al.
[Phys.Lett.B 718 \(2012\) 329](#) | [arXiv:1208.2931](#) | [DOI:10.1016/j.physletb.2012.10.048](#) | 2012-08
1297. Measurement of the Azimuthal Anisotropy of Neutral Pions in PbPb cCollisions at $\sqrt{s_{NN}}=2.76$ TeV
Chatrchyan, Serguei et al.
[Phys.Rev.Lett. 110 \(2013\) 042301](#) | [arXiv:1208.2470](#) | [DOI:10.1103/PhysRevLett.110.042301](#) | 2012-08

1298. Search for a W' Boson Decaying to a Bottom Quark and a Top Quark in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 718 (2013) 1229 | [arXiv:1208.0956](#) | [DOI:10.1016/j.physletb.2012.12.008](#) | 2012-08

1299. Search for Flavor Changing Neutral Currents in Top Quark Decays in pp Collisions at 7 TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 718 (2013) 1252 | [arXiv:1208.0957](#) | [DOI:10.1016/j.physletb.2012.12.045](#) | 2012-08

1300. A New Boson with a Mass of 125 GeV Observed with the CMS Experiment at the Large Hadron Collider

Chatrchyan, Serguei et al.

Science 338 (2012) 1569 | [DOI:10.1126/science.1230816](#) | 2012

1301. A Search for a Doubly-Charged Higgs Boson in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Eur.Phys.J.C 72 (2012) 2189 | [arXiv:1207.2666](#) | [DOI:10.1140/epjc/s10052-012-2189-5](#) | 2012-07

1302. Search for New Physics with Long-Lived Particles Decaying to Photons and Missing Energy in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

JHEP 11 (2012) 172 | [arXiv:1207.0627](#) | [DOI:10.1007/JHEP11\(2012\)172](#) | 2012-07

1303. Search for Pair Production of First- and Second-Generation Scalar Leptoquarks in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Phys.Rev.D 86 (2012) 052013 | [arXiv:1207.5406](#) | [DOI:10.1103/PhysRevD.86.052013](#) | 2012-07

1304. Search for Supersymmetry in Hadronic Final States using MT_2 in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

JHEP 10 (2012) 018 | [arXiv:1207.1798](#) | [DOI:10.1007/JHEP10\(2012\)018](#) | 2012-07

1305. Inclusive and Differential Measurements of the $t\bar{t}$ Charge Asymmetry in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 717 (2012) 129 | [arXiv:1207.0065](#) | [DOI:10.1016/j.physletb.2012.09.028](#) | 2012-07

1306. Measurement of the Underlying Event Activity in pp Collisions at $\sqrt{s}=0.9$ and 7 TeV with the Novel Jet-Area/Median Approach

Chatrchyan, Serguei et al.

JHEP 08 (2012) 130 | [arXiv:1207.2392](#) | [DOI:10.1007/JHEP08\(2012\)130](#) | 2012-07

1307. Search for Stopped Long-Lived Particles Produced in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

JHEP 08 (2012) 026 | [arXiv:1207.0106](#) | [DOI:10.1007/JHEP08\(2012\)026](#) | 2012-07

1308. Search for heavy Majorana Neutrinos in $\mu^+\mu^\pm + \text{Jets}$ and $e^+e^\pm + \text{Jets}$ Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 717 (2012) 109 | [arXiv:1207.6079](#) | [DOI:10.1016/j.physletb.2012.09.012](#) | 2012-07

1309. Search for a Fermiophobic Higgs Boson in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

JHEP 09 (2012) 111 | [arXiv:1207.1130](#) | [DOI:10.1007/JHEP09\(2012\)111](#) | 2012-07

1310. Study of the Inclusive Production of Charged Pions, Kaons, and Protons in pp Collisions at $\sqrt{s}=0.9, 2.76$, and 7 TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 2164](#) | [arXiv:1207.4724](#) | [DOI:10.1140/epjc/s10052-012-2164-1](#) | 2012-07

1311. Search for New Physics in the Multijet and Missing Transverse Momentum Final State in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 171803](#) | [arXiv:1207.1898](#) | [DOI:10.1103/PhysRevLett.109.171803](#) | 2012-07

1312. Forward-Backward Asymmetry of Drell-Yan Lepton Pairs in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2013\) 752](#) | [arXiv:1207.3973](#) | [DOI:10.1016/j.physletb.2012.10.082](#) | 2012-07

1313. Search for Narrow Resonances in Dilepton Mass Spectra in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 714 \(2012\) 158](#) | [arXiv:1206.1849](#) | [DOI:10.1016/j.physletb.2012.06.051](#) | 2012-06

1314. Search for a Light Pseudoscalar Higgs Boson in the Dimuon Decay Channel in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 121801](#) | [arXiv:1206.6326](#) | [DOI:10.1103/PhysRevLett.109.121801](#) | 2012-06

1315. Search for a $W'^{\pm}$ or ρ_{TC} Decaying into WZ in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 141801](#) | [arXiv:1206.0433](#) | [DOI:10.1103/PhysRevLett.109.141801](#) | 2012-06

1316. Search for Charge-Asymmetric Production of W' Bosons in $t\bar{t}$ + Jet Events from pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 717 \(2012\) 351](#) | [arXiv:1206.3921](#) | [DOI:10.1016/j.physletb.2012.09.048](#) | 2012-06

1317. Measurement of the Electron Charge Asymmetry in Inclusive W Production in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 111806](#) | [arXiv:1206.2598](#) | [DOI:10.1103/PhysRevLett.109.111806](#) | 2012-06

1318. Search for High-Mass Resonances Decaying into τ -Lepton Pairs in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 716 \(2012\) 82](#) | [arXiv:1206.1725](#) | [DOI:10.1016/j.physletb.2012.07.062](#) | 2012-06

1319. Performance of CMS Muon Reconstruction in pp Collision Events at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JINST 7 \(2012\) P10002](#) | [arXiv:1206.4071](#) | [DOI:10.1088/1748-0221/7/10/P10002](#) | 2012-06

1320. Search for Dark Matter and Large Extra Dimensions in Monojet Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 09 \(2012\) 094](#) | [arXiv:1206.5663](#) | [DOI:10.1007/JHEP09\(2012\)094](#) | 2012-06

1321. Study of W Boson Production in PbPb and pp Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 715 \(2012\) 66](#) | [arXiv:1205.6334](#) | [DOI:10.1016/j.physletb.2012.07.025](#) | 2012-05

1322. Search for a Light Charged Higgs Boson in Top Quark Decays in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 07 \(2012\) 143](#) | [arXiv:1205.5736](#) | [DOI:10.1007/JHEP07\(2012\)143](#) | 2012-05

1323. Measurement of the Pseudorapidity and Centrality Dependence of the Transverse Energy Density in PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 152303](#) | [arXiv:1205.2488](#) | [DOI:10.1103/PhysRevLett.109.152303](#) | 2012-05

1324. Measurement of Jet Fragmentation into Charged Particles in pp and PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[JHEP 10 \(2012\) 087](#) | [arXiv:1205.5872](#) | [DOI:10.1007/JHEP10\(2012\)087](#) | 2012-05

1325. Search for Heavy Long-Lived Charged Particles in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 713 \(2012\) 408](#) | [arXiv:1205.0272](#) | [DOI:10.1016/j.physletb.2012.06.023](#) | 2012-05

1326. Measurement of the Λ_b Cross Section and the Λ_b to Λ_b Ratio with $J/\Psi \Lambda$ Decays in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 714 \(2012\) 136](#) | [arXiv:1205.0594](#) | [DOI:10.1016/j.physletb.2012.05.063](#) | 2012-05

1327. Studies of Jet Quenching using Isolated-Photon + Jet Correlations in PbPb and pp Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2013\) 773](#) | [arXiv:1205.0206](#) | [DOI:10.1016/j.physletb.2012.11.003](#) | 2012-05

1328. Search for Anomalous $t\bar{t}$ Production in the Highly-Boosted All-Hadronic Final State

Chatrchyan, Serguei et al.

[JHEP 09 \(2012\) 029](#) | [arXiv:1204.2488](#) | [DOI:10.1007/JHEP09\(2012\)029](#) | 2012-04

1329. Measurement of the Z/γ^*+b -jet Cross Section in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2012\) 126](#) | [arXiv:1204.1643](#) | [DOI:10.1007/JHEP06\(2012\)126](#) | 2012-04

1330. Observation of a new Ξ_b Baryon

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 108 \(2012\) 252002](#) | [arXiv:1204.5955](#) | [DOI:10.1103/PhysRevLett.108.252002](#) | 2012-04

1331. Measurement of the Mass Difference between Top and Antitop Quarks

Chatrchyan, Serguei et al.

[JHEP 06 \(2012\) 109](#) | [arXiv:1204.2807](#) | [DOI:10.1007/JHEP06\(2012\)109](#) | 2012-04

1332. Search for Heavy Bottom-like Quarks in 4.9 fb^{-1} of pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 05 \(2012\) 123](#) | [arXiv:1204.1088](#) | [DOI:10.1007/JHEP05\(2012\)123](#) | 2012-04

1333. Search for Dark Matter and Large Extra Dimensions in pp Collisions Yielding a Photon and Missing Transverse Energy

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 108 \(2012\) 261803](#) | [arXiv:1204.0821](#) | [DOI:10.1103/PhysRevLett.108.261803](#) | 2012-04

1334. Search for Leptonic Decays of $W^\pm$ Bosons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 08 \(2012\) 023](#) | [arXiv:1204.4764](#) | [DOI:10.1007/JHEP08\(2012\)023](#) | 2012-04

1335. Measurement of the Underlying Event in the Drell-Yan Process in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 2080](#) | [arXiv:1204.1411](#) | [DOI:10.1140/epjc/s10052-012-2080-4](#) | 2012-04

1336. Shape, Transverse Size, and Charged Hadron Multiplicity of Jets in pp Collisions at 7 TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2012\) 160](#) | [arXiv:1204.3170](#) | [DOI:10.1007/JHEP06\(2012\)160](#) | 2012-04

1337. Search for Anomalous Production of Multilepton Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2012\) 169](#) | [arXiv:1204.5341](#) | [DOI:10.1007/JHEP06\(2012\)169](#) | 2012-04

1338. Azimuthal Anisotropy of Charged Particles at High Transverse Momenta in PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 022301](#) | [arXiv:1204.1850](#) | [DOI:10.1103/PhysRevLett.109.022301](#) | 2012-04

1339. Ratios of Dijet Production Cross Sections as a Function of the Absolute Difference in Rapidity between Jets in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 2216](#) | [arXiv:1204.0696](#) | [DOI:10.1140/epjc/s10052-012-2216-6](#) | 2012-04

1340. Measurement of the Elliptic Anisotropy of Charged Particles Produced in PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.C 87 \(2013\) 014902](#) | [arXiv:1204.1409](#) | [DOI:10.1103/PhysRevC.87.014902](#) | 2012-04

1341. Measurement of the Top Quark Pair Production Cross Section in pp Collisions at $\sqrt{s}=7$ TeV in Dilepton Final States Containing a τ

Chatrchyan, Serguei et al.

[Phys.Rev.D 85 \(2012\) 112007](#) | [arXiv:1203.6810](#) | [DOI:10.1103/PhysRevD.85.112007](#) | 2012-03

1342. Search for $B^0_s \rightarrow \mu^+ \mu^-$ and $B^0 \rightarrow \mu^+ \mu^-$ decays

Chatrchyan, Serguei et al.

[JHEP 04 \(2012\) 033](#) | [arXiv:1203.3976](#) | [DOI:10.1007/JHEP04\(2012\)033](#) | 2012-03

1343. Measurement of the Inclusive Cross Section $\sigma(pp \rightarrow b\bar{b} X \rightarrow \mu\mu X')$ at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2012\) 110](#) | [arXiv:1203.3458](#) | [DOI:10.1007/JHEP06\(2012\)110](#) | 2012-03

1344. Study of High-pT Charged Particle Suppression in PbPb Compared to pp Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 1945](#) | [arXiv:1202.2554](#) | [DOI:10.1140/epjc/s10052-012-1945-x](#) | 2012-02

1345. Search for Neutral Higgs Bosons Decaying to Tau Pairs in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 713 \(2012\) 68](#) | [arXiv:1202.4083](#) | [DOI:10.1016/j.physletb.2012.05.028](#) | 2012-02

1346. Search for Microscopic Black Holes in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 04 \(2012\) 061](#) | [arXiv:1202.6396](#) | [DOI:10.1007/JHEP04\(2012\)061](#) | 2012-02

1347. Search for the Standard Model Higgs Boson Decaying into Two Photons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 710 \(2012\) 403](#) | [arXiv:1202.1487](#) | [DOI:10.1016/j.physletb.2012.03.003](#) | 2012-02

1348. Search for the Standard Model Higgs Boson in the $H \rightarrow ZZ \rightarrow \ell^+ \ell^- \tau^+ \tau^-$ Decay Channel in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 03 \(2012\) 081](#) | [arXiv:1202.3617](#) | [DOI:10.1007/JHEP03\(2012\)081](#) | 2012-02

1349. Inclusive b-Jet Production in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 04 \(2012\) 084](#) | [arXiv:1202.4617](#) | [DOI:10.1007/JHEP04\(2012\)084](#) | 2012-02

1350. Measurement of the Inclusive Production Cross Sections for Forward Jets and for Dijet Events with One Forward and One Central Jet in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2012\) 036](#) | [arXiv:1202.0704](#) | [DOI:10.1007/JHEP06\(2012\)036](#) | 2012-02

1351. Search for a Higgs Boson in the Decay Channel $H \rightarrow ZZ^* \rightarrow q\bar{q}\ell^+ \ell^-$ in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 04 \(2012\) 036](#) | [arXiv:1202.1416](#) | [DOI:10.1007/JHEP04\(2012\)036](#) | 2012-02

1352. Search for Quark Compositeness in Dijet Angular Distributions from pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 05 \(2012\) 055](#) | [arXiv:1202.5535](#) | [DOI:10.1007/JHEP05\(2012\)055](#) | 2012-02

1353. Jet Momentum Dependence of Jet Quenching in PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 712 \(2012\) 176](#) | [arXiv:1202.5022](#) | [DOI:10.1016/j.physletb.2012.04.058](#) | 2012-02

1354. Search for the Standard Model Higgs Boson in the Decay Channel $H \rightarrow ZZ \rightarrow 4$ Leptons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 108 \(2012\) 111804](#) | [arXiv:1202.1997](#) | [DOI:10.1103/PhysRevLett.108.111804](#) | 2012-02

1355. Search for the Standard Model Higgs Boson Decaying to Bottom Quarks in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 710 \(2012\) 284](#) | [arXiv:1202.4195](#) | [DOI:10.1016/j.physletb.2012.02.085](#) | 2012-02

1356. Search for Large Extra Dimensions in Dimuon and Dielectron Events in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 711 \(2012\) 15](#) | [arXiv:1202.3827](#) | [DOI:10.1016/j.physletb.2012.03.029](#) | 2012-02

1357. Combined Results of Searches for the Standard Model Higgs Boson in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 710 \(2012\) 26](#) | [arXiv:1202.1488](#) | [DOI:10.1016/j.physletb.2012.02.064](#) | 2012-02

1358. Suppression of Non-Prompt J/ψ , Prompt J/ψ , and $Y(1S)$ in PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[JHEP 05 \(2012\) 063](#) | [arXiv:1201.5069](#) | [DOI:10.1007/JHEP05\(2012\)063](#) | 2012-01

1359. Measurement of Isolated Photon Production in pp and PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 710 \(2012\) 256](#) | [arXiv:1201.3093](#) | [DOI:10.1016/j.physletb.2012.02.077](#) | 2012-01

1360. Centrality Dependence of Dihadron Correlations and Azimuthal anisotropy Harmonics in PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 2012](#) | [arXiv:1201.3158](#) | [DOI:10.1140/epjc/s10052-012-2012-3](#) | 2012-01

1361. Search for signatures of extra dimensions in the diphoton mass spectrum at the Large Hadron Collider

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 108 \(2012\) 111801](#) | [arXiv:1112.0688](#) | [DOI:10.1103/PhysRevLett.108.111801](#) | 2011-12

1362. Measurement of the charge asymmetry in top-quark pair production in proton-proton collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 709 \(2012\) 28](#) | [arXiv:1112.5100](#) | [DOI:10.1016/j.physletb.2012.01.078](#) | 2011-12

1363. J/ψ and ψ_{2S} production in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 02 \(2012\) 011](#) | [arXiv:1111.1557](#) | [DOI:10.1007/JHEP02\(2012\)011](#) | 2011-11

1364. Exclusive photon-photon production of muon pairs in proton-proton collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 01 \(2012\) 052](#) | [arXiv:1111.5536](#) | [DOI:10.1007/JHEP01\(2012\)052](#) | 2011-11

1365. Measurement of the weak mixing angle with the Drell-Yan process in proton-proton collisions at the LHC

Chatrchyan, Serguei et al.

[Phys.Rev.D 84 \(2011\) 112002](#) | [arXiv:1110.2682](#) | [DOI:10.1103/PhysRevD.84.112002](#) | 2011-10

1366. Measurement of energy flow at large pseudorapidities in pp collisions at $\sqrt{s} = 0.9$ and 7 TeV

Chatrchyan, Serguei et al.

[JHEP 11 \(2011\) 148](#) | [arXiv:1110.0211](#) | [DOI:10.1007/JHEP11\(2011\)148](#) | 2011-10

1367. Jet Production Rates in Association with W and Z Bosons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 01 \(2012\) 010](#) | [arXiv:1110.3226](#) | [DOI:10.1007/JHEP01\(2012\)010](#) | 2011-10

1368. Forward Energy Flow, Central Charged-Particle Multiplicities, and Pseudorapidity Gaps in W and Z Boson Events from pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 1839](#) | [arXiv:1110.0181](#) | [DOI:10.1140/epjc/s10052-011-1839-3](#) | 2011-10

1369. Measurement of the Rapidity and Transverse Momentum Distributions of Z Bosons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 85 \(2012\) 032002](#) | [arXiv:1110.4973](#) | [DOI:10.1103/PhysRevD.85.032002](#) | 2011-10

1370. Measurement of the Production Cross Section for Pairs of Isolated Photons in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 01 \(2012\) 133](#) | [arXiv:1110.6461](#) | [DOI:10.1007/JHEP01\(2012\)133](#) | 2011-10

1371. Search for Supersymmetry at the LHC in Events with Jets and Missing Transverse Energy

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 221804](#) | [arXiv:1109.2352](#) | [DOI:10.1103/PhysRevLett.107.221804](#) | 2011-09

1372. Search for a Vector-like Quark with Charge 2/3 in $t + Z$ Events from pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 271802](#) | [arXiv:1109.4985](#) | [DOI:10.1103/PhysRevLett.107.271802](#) | 2011-09

1373. Performance of tau-lepton reconstruction and identification in CMS

Chatrchyan, Serguei et al.

[JINST 7 \(2012\) P01001](#) | [arXiv:1109.6034](#) | [DOI:10.1088/1748-0221/7/01/P01001](#) | 2011-09

1374. Measurement of the Drell-Yan Cross Section in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 10 \(2011\) 007](#) | [arXiv:1108.0566](#) | [DOI:10.1007/JHEP10\(2011\)007](#) | 2011-08

1375. Measurement of the Differential Cross Section for Isolated Prompt Photon Production in pp Collisions at 7 TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 84 \(2011\) 052011](#) | [arXiv:1108.2044](#) | [DOI:10.1103/PhysRevD.84.052011](#) | 2011-08

1376. Measurement of the $t\bar{t}$ Production Cross Section in pp Collisions at 7 TeV in Lepton + Jets Events Using b-quark Jet Identification

Chatrchyan, Serguei et al.

[Phys.Rev.D 84 \(2011\) 092004](#) | [arXiv:1108.3773](#) | [DOI:10.1103/PhysRevD.84.092004](#) | 2011-08

1377. Measurement of the Inclusive W and Z Production Cross Sections in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 10 \(2011\) 132](#) | [arXiv:1107.4789](#) | [DOI:10.1007/JHEP10\(2011\)132](#) | 2011-07

1378. Search for supersymmetry in pp collisions at $\sqrt{s}=7$ TeV in events with a single lepton, jets, and missing transverse momentum

Chatrchyan, Serguei et al.

[JHEP 08 \(2011\) 156](#) | [arXiv:1107.1870](#) | [DOI:10.1007/JHEP08\(2011\)156](#) | 2011-07

1379. Dependence on pseudorapidity and centrality of charged hadron production in PbPb collisions at a nucleon-nucleon centre-of-mass energy of 2.76 TeV

Chatrchyan, Serguei et al.

[JHEP 08 \(2011\) 141](#) | [arXiv:1107.4800](#) | [DOI:10.1007/JHEP08\(2011\)141](#) | 2011-07

1380. Search for Resonances in the Dijet Mass Spectrum from 7 TeV pp Collisions at CMS

Chatrchyan, Serguei et al.

[Phys.Lett.B 704 \(2011\) 123](#) | [arXiv:1107.4771](#) | [DOI:10.1016/j.physletb.2011.09.015](#) | 2011-07

1381. A search for excited leptons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 704 \(2011\) 143](#) | [arXiv:1107.1773](#) | [DOI:10.1016/j.physletb.2011.09.021](#) | 2011-07

1382. Search for Three-Jet Resonances in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 101801](#) | [arXiv:1107.3084](#) | [DOI:10.1103/PhysRevLett.107.101801](#) | 2011-07

1383. Measurement of the Underlying Event Activity at the LHC with $\sqrt{s}=7$ TeV and Comparison with $\sqrt{s}=0.9$ TeV

Chatrchyan, Serguei et al.

[JHEP 09 \(2011\) 109](#) | [arXiv:1107.0330](#) | [DOI:10.1007/JHEP09\(2011\)109](#) | 2011-07

1384. Determination of Jet Energy Calibration and Transverse Momentum Resolution in CMS

Chatrchyan, Serguei et al.

[JINST 6 \(2011\) P11002](#) | [arXiv:1107.4277](#) | [DOI:10.1088/1748-0221/6/11/P11002](#) | 2011-07

1385. Search for B(s) and B to dimuon decays in pp collisions at 7 TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 191802](#) | [arXiv:1107.5834](#) | [DOI:10.1103/PhysRevLett.107.191802](#) | 2011-07

1386. Inclusive search for squarks and gluinos in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 85 \(2012\) 012004](#) | [arXiv:1107.1279](#) | [DOI:10.1103/PhysRevD.85.012004](#) | 2011-07

1387. Measurement of the Inclusive Jet Cross Section in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 132001](#) | [arXiv:1106.0208](#) | [DOI:10.1103/PhysRevLett.107.132001](#) | 2011-06

1388. Missing transverse energy performance of the CMS detector

Chatrchyan, Serguei et al.

[JINST 6 \(2011\) P09001](#) | [arXiv:1106.5048](#) | [DOI:10.1088/1748-0221/6/09/P09001](#) | 2011-06

1389. Measurement of the Top-antitop Production Cross Section in pp Collisions at $\sqrt{s}=7$ TeV using the Kinematic Properties of Events with Leptons and Jets

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 71 \(2011\) 1721](#) | [arXiv:1106.0902](#) | [DOI:10.1140/epjc/s10052-011-1721-3](#) | 2011-06

1390. Search for New Physics with a Mono-Jet and Missing Transverse Energy in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 201804](#) | [arXiv:1106.4775](#) | [DOI:10.1103/PhysRevLett.107.201804](#) | 2011-06

1391. Search for physics beyond the standard model using multilepton signatures in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 704 \(2011\) 411](#) | [arXiv:1106.0933](#) | [DOI:10.1016/j.physletb.2011.09.047](#) | 2011-06

1392. Measurement of the Strange B Meson Production Cross Section with J/Psi ϕ Decays in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 84 \(2011\) 052008](#) | [arXiv:1106.4048](#) | [DOI:10.1103/PhysRevD.84.052008](#) | 2011-06

1393. Measurement of the ratio of the 3-jet to 2-jet cross sections in pp collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 702 \(2011\) 336](#) | [arXiv:1106.0647](#) | [DOI:10.1016/j.physletb.2011.07.067](#) | 2011-06

1394. Measurement of the t-channel single top quark production cross section in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 091802](#) | [arXiv:1106.3052](#) | [DOI:10.1103/PhysRevLett.107.091802](#) | 2011-06

1395. Search for Supersymmetry in Events with b Jets and Missing Transverse Momentum at the LHC

Chatrchyan, Serguei et al.

[JHEP 07 \(2011\) 113](#) | [arXiv:1106.3272](#) | [DOI:10.1007/JHEP07\(2011\)113](#) | 2011-06

1396. Search for Light Resonances Decaying into Pairs of Muons as a Signal of New Physics

Chatrchyan, Serguei et al.

[JHEP 07 \(2011\) 098](#) | [arXiv:1106.2375](#) | [DOI:10.1007/JHEP07\(2011\)098](#) | 2011-06

1397. Search for New Physics with Jets and Missing Transverse Momentum in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 08 \(2011\) 155](#) | [arXiv:1106.4503](#) | [DOI:10.1007/JHEP08\(2011\)155](#) | 2011-06

1398. Indications of suppression of excited Υ states in PbPb collisions at $\sqrt{S_{NN}} = 2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 052302](#) | [arXiv:1105.4894](#) | [DOI:10.1103/PhysRevLett.107.052302](#) | 2011-05

1399. Long-range and short-range dihadron angular correlations in central PbPb collisions at a nucleon-nucleon center of mass energy of 2.76 TeV

Chatrchyan, Serguei et al.

[JHEP 07 \(2011\) 076](#) | [arXiv:1105.2438](#) | [DOI:10.1007/JHEP07\(2011\)076](#) | 2011-05

1400. Measurement of $W\gamma$ and $Z\gamma$ production in pp collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 701 \(2011\) 535](#) | [arXiv:1105.2758](#) | [DOI:10.1016/j.physletb.2011.06.034](#) | 2011-05

1401. Search for supersymmetry in events with a lepton, a photon, and large missing transverse energy in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2011\) 093](#) | [arXiv:1105.3152](#) | [DOI:10.1007/JHEP06\(2011\)093](#) | 2011-05

1402. Search for First Generation Scalar Leptoquarks in the $e\nu jj$ channel in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 703 \(2011\) 246](#) | [arXiv:1105.5237](#) | [DOI:10.1016/j.physletb.2011.07.089](#) | 2011-05

1403. Measurement of the Polarization of W Bosons with Large Transverse Momenta in W+Jets Events at the LHC

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 021802](#) | [arXiv:1104.3829](#) | [DOI:10.1103/PhysRevLett.107.021802](#) | 2011-04

1404. Measurement of the Inclusive Z Cross Section via Decays to Tau Pairs in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 08 \(2011\) 117](#) | [arXiv:1104.1617](#) | [DOI:10.1007/JHEP08\(2011\)117](#) | 2011-04

1405. Search for Neutral MSSM Higgs Bosons Decaying to Tau Pairs in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 106 \(2011\) 231801](#) | [arXiv:1104.1619](#) | [DOI:10.1103/PhysRevLett.106.231801](#) | 2011-04

1406. Charged particle transverse momentum spectra in pp collisions at $\sqrt{s} = 0.9$ and 7 TeV

Chatrchyan, Serguei et al.

[JHEP 08 \(2011\) 086](#) | [arXiv:1104.3547](#) | [DOI:10.1007/JHEP08\(2011\)086](#) | 2011-04

1407. Measurement of the differential dijet production cross section in proton-proton collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 700 \(2011\) 187](#) | [arXiv:1104.1693](#) | [DOI:10.1016/j.physletb.2011.05.027](#) | 2011-04

1408. Measurement of the B^0 production cross section in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 106 \(2011\) 252001](#) | [arXiv:1104.2892](#) | [DOI:10.1103/PhysRevLett.106.252001](#) | 2011-04

1409. Search for Supersymmetry in pp Collisions at $\sqrt{s} = 7$ TeV in Events with Two Photons and Missing Transverse Energy

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 106 \(2011\) 211802](#) | [arXiv:1103.0953](#) | [DOI:10.1103/PhysRevLett.106.211802](#) | 2011-03

1410. Measurement of the lepton charge asymmetry in inclusive W production in pp collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 04 \(2011\) 050](#) | [arXiv:1103.3470](#) | [DOI:10.1007/JHEP04\(2011\)050](#) | 2011-03

1411. Search for a $W^\pm$ boson decaying to a muon and a neutrino in pp collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 701 \(2011\) 160](#) | [arXiv:1103.0030](#) | [DOI:10.1016/j.physletb.2011.05.048](#) | 2011-03

1412. Search for Large Extra Dimensions in the Diphoton Final State at the Large Hadron Collider

Chatrchyan, Serguei et al.

[JHEP 05 \(2011\) 085](#) | [arXiv:1103.4279](#) | [DOI:10.1007/JHEP05\(2011\)085](#) | 2011-03

1413. Search for Resonances in the Dilepton Mass Distribution in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 05 \(2011\) 093](#) | [arXiv:1103.0981](#) | [DOI:10.1007/JHEP05\(2011\)093](#) | 2011-03

1414. Measurement of $W^+ W^-$ production and search for the Higgs boson in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 699 \(2011\) 25](#) | [arXiv:1102.5429](#) | [DOI:10.1016/j.physletb.2011.03.056](#) | 2011-02

1415. Measurement of Dijet Angular Distributions and Search for Quark Compositeness in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 106 \(2011\) 201804](#) | [arXiv:1102.2020](#) | [DOI:10.1103/PhysRevLett.106.201804](#) | 2011-02

1416. Observation and studies of jet quenching in PbPb collisions at nucleon-nucleon center-of-mass energy = 2.76 TeV

Chatrchyan, Serguei et al.

[Phys.Rev.C 84 \(2011\) 024906](#) | [arXiv:1102.1957](#) | [DOI:10.1103/PhysRevC.84.024906](#) | 2011-02

1417. Strange Particle Production in pp Collisions at $\sqrt{s}=0.9$ and 7 TeV

Khachatryan, Vardan et al.

[JHEP 05 \(2011\) 064](#) | [arXiv:1102.4282](#) | [DOI:10.1007/JHEP05\(2011\)064](#) | 2011-02

1418. Search for a Heavy Bottom-like Quark in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 701 \(2011\) 204](#) | [arXiv:1102.4746](#) | [DOI:10.1016/j.physletb.2011.05.074](#) | 2011-02

1419. First Measurement of Hadronic Event Shapes in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 699 \(2011\) 48](#) | [arXiv:1102.0068](#) | [DOI:10.1016/j.physletb.2011.03.060](#) | 2011-02

1420. Measurement of BB Angular Correlations based on Secondary Vertex Reconstruction at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2011\) 136](#) | [arXiv:1102.3194](#) | [DOI:10.1007/JHEP03\(2011\)136](#) | 2011-02

1421. Study of Z boson production in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 106 \(2011\) 212301](#) | [arXiv:1102.5435](#) | [DOI:10.1103/PhysRevLett.106.212301](#) | 2011-02

1422. Search for Heavy Stable Charged Particles in pp collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2011\) 024](#) | [arXiv:1101.1645](#) | [DOI:10.1007/JHEP03\(2011\)024](#) | 2011-01

1423. Dijet Azimuthal Decorrelations in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 106 \(2011\) 122003](#) | [arXiv:1101.5029](#) | [DOI:10.1103/PhysRevLett.106.122003](#) | 2011-01

1424. Measurement of Bose-Einstein Correlations in pp Collisions at $\sqrt{s}=0.9$ and 7 TeV

Khachatryan, Vardan et al.

[JHEP 05 \(2011\) 029](#) | [arXiv:1101.3518](#) | [DOI:10.1007/JHEP05\(2011\)029](#) | 2011-01

1425. Measurement of the B^+ Production Cross Section in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 106 \(2011\) 112001](#) | [arXiv:1101.0131](#) | [DOI:10.1103/PhysRevLett.106.112001](#) | 2011-01

1426. Search for Supersymmetry in pp Collisions at 7 TeV in Events with Jets and Missing Transverse Energy

Khachatryan, Vardan et al.

[Phys.Lett.B 698 \(2011\) 196](#) | [arXiv:1101.1628](#) | [DOI:10.1016/j.physletb.2011.03.021](#) | 2011-01

1427. Inclusive b-Hadron Production Cross Section with Muons in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2011\) 090](#) | [arXiv:1101.3512](#) | [DOI:10.1007/JHEP03\(2011\)090](#) | 2011-01

1428. Search for Pair Production of First-Generation Scalar Leptoquarks in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 106 \(2011\) 201802](#) | [arXiv:1012.4031](#) | [DOI:10.1103/PhysRevLett.106.201802](#) | 2010-12

1429. Search for a Heavy Gauge Boson W' in the Final State with an Electron and Large Missing Transverse Energy in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 698 \(2011\) 21](#) | [arXiv:1012.5945](#) | [DOI:10.1016/j.physletb.2011.02.048](#) | 2010-12

1430. Search for Microscopic Black Hole Signatures at the Large Hadron Collider

Khachatryan, Vardan et al.

[Phys.Lett.B 697 \(2011\) 434](#) | [arXiv:1012.3375](#) | [DOI:10.1016/j.physletb.2011.02.032](#) | 2010-12

1431. Search for Pair Production of Second-Generation Scalar Leptoquarks in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 106 \(2011\) 201803](#) | [arXiv:1012.4033](#) | [DOI:10.1103/PhysRevLett.106.201803](#) | 2010-12

1432. Measurement of the Isolated Prompt Photon Production Cross Section in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 106 \(2011\) 082001](#) | [arXiv:1012.0799](#) | [DOI:10.1103/PhysRevLett.106.082001](#) | 2010-12

1433. Measurements of Inclusive W and Z Cross Sections in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2011\) 080](#) | [arXiv:1012.2466](#) | [DOI:10.1007/JHEP01\(2011\)080](#) | 2010-12

1434. Upsilon Production Cross-Section in pp Collisions at $\sqrt{s}=7$ TeV
Khachatryan, Vardan et al.
[Phys.Rev.D 83 \(2011\) 112004](#) | [arXiv:1012.5545](#) | [DOI:10.1103/PhysRevD.83.112004](#) | 2010-12
1435. Search for Stopped Gluinos in pp collisions at $\sqrt{s}=7$ TeV
Khachatryan, Vardan et al.
[Phys.Rev.Lett. 106 \(2011\) 011801](#) | [arXiv:1011.5861](#) | [DOI:10.1103/PhysRevLett.106.011801](#) | 2010-11
1436. Prompt and Non-Prompt J/ ψ Production in pp Collisions at $\sqrt{s}=7$ TeV
Khachatryan, Vardan et al.
[Eur.Phys.J.C 71 \(2011\) 1575](#) | [arXiv:1011.4193](#) | [DOI:10.1140/epjc/s10052-011-1575-8](#) | 2010-11
1437. Charged Particle Multiplicities in pp Interactions at $\sqrt{s}=0.9, 2.36$, and 7 TeV
Khachatryan, Vardan et al.
[JHEP 01 \(2011\) 079](#) | [arXiv:1011.5531](#) | [DOI:10.1007/JHEP01\(2011\)079](#) | 2010-11
1438. Search for Quark Compositeness with the Dijet Centrality Ratio in pp Collisions at $\sqrt{s}=7$ TeV
Khachatryan, Vardan et al.
[Phys.Rev.Lett. 105 \(2010\) 262001](#) | [arXiv:1010.4439](#) | [DOI:10.1103/PhysRevLett.105.262001](#) | 2010-10
1439. Search for Dijet Resonances in 7 TeV pp Collisions at CMS
Khachatryan, Vardan et al.
[Phys.Rev.Lett. 105 \(2010\) 211801](#) | [arXiv:1010.0203](#) | [DOI:10.1103/PhysRevLett.105.211801](#) | 2010-10
1440. Observation of Long-Range Near-Side Angular Correlations in Proton-Proton Collisions at the LHC
Khachatryan, Vardan et al.
[JHEP 09 \(2010\) 091](#) | [arXiv:1009.4122](#) | [DOI:10.1007/JHEP09\(2010\)091](#) | 2010-09
1441. First Measurement of the Underlying Event Activity at the LHC with $\sqrt{s} = 0.9$ TeV
Khachatryan, Vardan et al.
[Eur.Phys.J.C 70 \(2010\) 555](#) | [arXiv:1006.2083](#) | [DOI:10.1140/epjc/s10052-010-1453-9](#) | 2010
1442. CMS Tracking Performance Results from Early LHC Operation
Khachatryan, Vardan et al.
[Eur.Phys.J.C 70 \(2010\) 1165](#) | [arXiv:1007.1988](#) | [DOI:10.1140/epjc/s10052-010-1491-3](#) | 2010
1443. First Measurement of Bose-Einstein Correlations in Proton-Proton Collisions at $\sqrt{s}=0.9$ and 2.36 TeV at the LHC
Khachatryan, Vardan et al.
[Phys.Rev.Lett. 105 \(2010\) 032001](#) | [arXiv:1005.3294](#) | [DOI:10.1103/PhysRevLett.105.032001](#) | 2010-05
1444. Transverse-momentum and pseudorapidity distributions of charged hadrons in pp collisions at $\sqrt{s}=7$ TeV
Khachatryan, Vardan et al.
[Phys.Rev.Lett. 105 \(2010\) 022002](#) | [arXiv:1005.3299](#) | [DOI:10.1103/PhysRevLett.105.022002](#) | 2010-05
1445. Measurement of the Charge Ratio of Atmospheric Muons with the CMS Detector
Khachatryan, Vardan et al.
[Phys.Lett.B 692 \(2010\) 83](#) | [arXiv:1005.5332](#) | [DOI:10.1016/j.physletb.2010.07.033](#) | 2010-05
1446. Transverse Momentum and Pseudorapidity Distributions of Charged Hadrons in pp Collisions at $\sqrt{s} = 0.9$ and 2.36 TeV
Khachatryan, Vardan et al.
[JHEP 02 \(2010\) 041](#) | [arXiv:1002.0621](#) | [DOI:10.1007/JHEP02\(2010\)041](#) | 2010-02
1447. Performance of the CMS Cathode Strip Chambers with Cosmic Rays

Chatrchyan, S et al.

[JINST 5 \(2010\) T03018](#) | [arXiv:0911.4992](#) | [DOI:10.1088/1748-0221/5/03/T03018](#) | 2009-11

1448. Commissioning and Performance of the CMS Pixel Tracker with Cosmic Ray Muons

Chatrchyan, S et al.

[JINST 5 \(2010\) T03007](#) | [arXiv:0911.5434](#) | [DOI:10.1088/1748-0221/5/03/T03007](#) | 2009-11

1449. Performance of CMS Muon Reconstruction in Cosmic-Ray Events

Chatrchyan, S et al.

[JINST 5 \(2010\) T03022](#) | [arXiv:0911.4994](#) | [DOI:10.1088/1748-0221/5/03/T03022](#) | 2009-11

1450. Aligning the CMS Muon Chambers with the Muon Alignment System during an Extended Cosmic Ray Run

Chatrchyan, S et al.

[JINST 5 \(2010\) T03019](#) | [arXiv:0911.4770](#) | [DOI:10.1088/1748-0221/5/03/T03019](#) | 2009-11

1451. Performance of the CMS Hadron Calorimeter with Cosmic Ray Muons and LHC Beam Data

Chatrchyan, S et al.

[JINST 5 \(2010\) T03012](#) | [arXiv:0911.4991](#) | [DOI:10.1088/1748-0221/5/03/T03012](#) | 2009-11

1452. Commissioning of the CMS High-Level Trigger with Cosmic Rays

Chatrchyan, S et al.

[JINST 5 \(2010\) T03005](#) | [arXiv:0911.4889](#) | [DOI:10.1088/1748-0221/5/03/T03005](#) | 2009-11

1453. Performance of the CMS Drift-Tube Local Trigger with Cosmic Rays

Chatrchyan, S et al.

[JINST 5 \(2010\) T03003](#) | [arXiv:0911.4893](#) | [DOI:10.1088/1748-0221/5/03/T03003](#) | 2009-11

1454. Commissioning of the CMS Experiment and the Cosmic Run at Four Tesla

Chatrchyan, S et al.

[JINST 5 \(2010\) T03001](#) | [arXiv:0911.4845](#) | [DOI:10.1088/1748-0221/5/03/T03001](#) | 2009-11

1455. Performance of the CMS Drift Tube Chambers with Cosmic Rays

Chatrchyan, S et al.

[JINST 5 \(2010\) T03015](#) | [arXiv:0911.4855](#) | [DOI:10.1088/1748-0221/5/03/T03015](#) | 2009-11

1456. Alignment of the CMS Muon System with Cosmic-Ray and Beam-Halo Muons

Chatrchyan, S et al.

[JINST 5 \(2010\) T03020](#) | [arXiv:0911.4022](#) | [DOI:10.1088/1748-0221/5/03/T03020](#) | 2009-11

1457. Identification and Filtering of Uncharacteristic Noise in the CMS Hadron Calorimeter

Chatrchyan, S et al.

[JINST 5 \(2010\) T03014](#) | [arXiv:0911.4881](#) | [DOI:10.1088/1748-0221/5/03/T03014](#) | 2009-11

1458. Commissioning and Performance of the CMS Silicon Strip Tracker with Cosmic Ray Muons

Chatrchyan, S et al.

[JINST 5 \(2010\) T03008](#) | [arXiv:0911.4996](#) | [DOI:10.1088/1748-0221/5/03/T03008](#) | 2009-11

1459. Performance of CMS Hadron Calorimeter Timing and Synchronization using Test Beam, Cosmic Ray, and LHC Beam Data

Chatrchyan, S et al.

[JINST 5 \(2010\) T03013](#) | [arXiv:0911.4877](#) | [DOI:10.1088/1748-0221/5/03/T03013](#) | 2009-11

1460. Time Reconstruction and Performance of the CMS Electromagnetic Calorimeter

Chatrchyan, S et al.

[JINST 5 \(2010\) T03011](#) | [arXiv:0911.4044](#) | [DOI:10.1088/1748-0221/5/03/T03011](#) | 2009-11

1461. Performance Study of the CMS Barrel Resistive Plate Chambers with Cosmic Rays
Chatrchyan, S et al.
[JINST 5 \(2010\) T03017](#) | [arXiv:0911.4045](#) | [DOI:10.1088/1748-0221/5/03/T03017](#) | 2009-11
1462. Fine Synchronization of the CMS Muon Drift-Tube Local Trigger using Cosmic Rays
Chatrchyan, S et al.
[JINST 5 \(2010\) T03004](#) | [arXiv:0911.4904](#) | [DOI:10.1088/1748-0221/5/03/T03004](#) | 2009-11
1463. Calibration of the CMS Drift Tube Chambers and Measurement of the Drift Velocity with Cosmic Rays
Chatrchyan, S et al.
[JINST 5 \(2010\) T03016](#) | [arXiv:0911.4895](#) | [DOI:10.1088/1748-0221/5/03/T03016](#) | 2009-11
1464. Measurement of the Muon Stopping Power in Lead Tungstate
Chatrchyan, S et al.
[JINST 5 \(2010\) P03007](#) | [arXiv:0911.5397](#) | [DOI:10.1088/1748-0221/5/03/P03007](#) | 2009-11
1465. Performance of the CMS Level-1 Trigger during Commissioning with Cosmic Ray Muons
Chatrchyan, S et al.
[JINST 5 \(2010\) T03002](#) | [arXiv:0911.5422](#) | [DOI:10.1088/1748-0221/5/03/T03002](#) | 2009-11
1466. CMS Data Processing Workflows during an Extended Cosmic Ray Run
Chatrchyan, S et al.
[JINST 5 \(2010\) T03006](#) | [arXiv:0911.4842](#) | [DOI:10.1088/1748-0221/5/03/T03006](#) | 2009-11
1467. Performance and Operation of the CMS Electromagnetic Calorimeter
Chatrchyan, S et al.
[JINST 5 \(2010\) T03010](#) | [arXiv:0910.3423](#) | [DOI:10.1088/1748-0221/5/03/T03010](#) | 2009-10
1468. Alignment of the CMS Silicon Tracker during Commissioning with Cosmic Rays
Chatrchyan, S et al.
[JINST 5 \(2010\) T03009](#) | [arXiv:0910.2505](#) | [DOI:10.1088/1748-0221/5/03/T03009](#) | 2009-10
1469. Precise Mapping of the Magnetic Field in the CMS Barrel Yoke using Cosmic Rays
Chatrchyan, S et al.
[JINST 5 \(2010\) T03021](#) | [arXiv:0910.5530](#) | [DOI:10.1088/1748-0221/5/03/T03021](#) | 2009-10
1470. Alignment of the CMS Silicon Strip Tracker during stand-alone Commissioning
Adam, W. et al.
[JINST 4 \(2009\) T07001](#) | [arXiv:0904.1220](#) | [DOI:10.1088/1748-0221/4/07/T07001](#) | 2009-04
1471. Performance studies of the CMS Strip Tracker before installation
Adam, W. et al.
[JINST 4 \(2009\) P06009](#) | [arXiv:0901.4316](#) | [DOI:10.1088/1748-0221/4/06/P06009](#) | 2009-01
1472. Track Reconstruction with Cosmic Ray Data at the Tracker Integration Facility
Adam, Wolfgang et al.
2008-09-17
1473. The CMS Experiment at the CERN LHC
Chatrchyan, S. et al.
[JINST 3 \(2008\) S08004](#) | [DOI:10.1088/1748-0221/3/08/S08004](#) | 2008-08
1474. The CMS tracker operation and performance at the Magnet Test and Cosmic Challenge
Adam, W. et al.
[JINST 3 \(2008\) P07006](#) | [DOI:10.1088/1748-0221/3/07/P07006](#) | 2008

1475. CMS data and workflow management system

Fanfani, A. et al.

(2008) 441 | DOI:10.1142/9789812819093_0076 | 2008

1476. CMS technical design report, volume II: Physics performance

Bayatian, G.L. et al.

J.Phys.G 34 (2007) 995 | DOI:10.1088/0954-3899/34/6/S01 | 2007

1477. Tracker operation and performance at the Magnet Test and Cosmic Challenge

Abbateo, D. et al.

2007-06

1478. CMS physics technical design report: Addendum on high density QCD with heavy ions

d'Enterria, David G. et al.

J.Phys.G 34 (2007) 2307 | DOI:10.1088/0954-3899/34/11/008 | 2007-03

1479. CMS expression of interest in the SLHC

Nash, J. et al.

2007-03

1480. The CMS Simulation Software

Abdouline, S. et al.

(2006) 1655 | DOI:10.1109/NSSMIC.2006.354216 | 2006

1481. Measurement of neutral current cross sections at high Bjorken-x with the ZEUS detector at HERA

Chekanov, S. et al.

Eur.Phys.J.C 49 (2007) 523 | arXiv:hep-ex/0608014 | DOI:10.1140/epjc/s10052-006-0164-8 | 2006-08

1482. Measurement of azimuthal asymmetries in neutral current deep inelastic scattering at HERA

Chekanov, S. et al.

Eur.Phys.J.C 51 (2007) 289 | arXiv:hep-ex/0608053 | DOI:10.1140/epjc/s10052-007-0310-y | 2006-08

1483. Measurement of prompt photons with associated jets in photoproduction at HERA

Chekanov, S. et al.

Eur.Phys.J.C 49 (2007) 511 | arXiv:hep-ex/0608028 | DOI:10.1140/epjc/s10052-006-0134-1 | 2006-08

1484. Inclusive-jet and dijet cross-sections in deep inelastic scattering at HERA

Chekanov, S. et al.

Nucl.Phys.B 765 (2007) 1 | arXiv:hep-ex/0608048 | DOI:10.1016/j.nuclphysb.2006.09.018 | 2006-08

1485. CMS Physics

Bayatian, G. L. et al.

DOI:10.2172/2510878 | 2006

1486. Event shapes in deep inelastic scattering at HERA

Chekanov, S. et al.

Nucl.Phys.B 767 (2007) 1 | arXiv:hep-ex/0604032 | DOI:10.1016/j.nuclphysb.2006.05.016 | 2006-04

1487. Measurement of high- Q^2 deep inelastic scattering cross sections with a longitudinally polarised positron beam at HERA

Chekanov, S. et al.

Phys.Lett.B 637 (2006) 210 | arXiv:hep-ex/0602026 | DOI:10.1016/j.physletb.2006.04.047 | 2006-02

1488. The CMS object-oriented simulation

Abdouline, S. et al.

(2005) 995 | DOI:10.1109/NSSMIC.2005.1596421 | 2005

1489. Measurement of charm fragmentation ratios and fractions in photoproduction at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 44 \(2005\) 351](#) | [arXiv:hep-ex/0508019](#) | [DOI:10.1140/epjc/s2005-02397-3](#) | 2005-08

1490. Inclusive jet cross sections and dijet correlations in D^{*+} photoproduction at HERA

Chekanov, S. et al.

[Nucl.Phys.B 729 \(2005\) 492](#) | [arXiv:hep-ex/0507089](#) | [DOI:10.1016/j.nuclphysb.2005.09.021](#) | 2005-07

1491. Measurement of inelastic J/ψ production in deep inelastic scattering at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 44 \(2005\) 13](#) | [arXiv:hep-ex/0505008](#) | [DOI:10.1140/epjc/s2005-02346-2](#) | 2005-05

1492. Exclusive electroproduction of phi mesons at HERA

Chekanov, S. et al.

[Nucl.Phys.B 718 \(2005\) 3](#) | [arXiv:hep-ex/0504010](#) | [DOI:10.1016/j.nuclphysb.2005.04.009](#) | 2005-04

1493. An NLO QCD analysis of inclusive cross-section and jet-production data from the zeus experiment

Chekanov, S. et al.

[Eur.Phys.J.C 42 \(2005\) 1](#) | [arXiv:hep-ph/0503274](#) | [DOI:10.1140/epjc/s2005-02293-x](#) | 2005-03

1494. Multijet production in neutral current deep inelastic scattering at HERA and determination of $\alpha(s)$

Chekanov, S. et al.

[Eur.Phys.J.C 44 \(2005\) 183](#) | [arXiv:hep-ex/0502007](#) | [DOI:10.1140/epjc/s2005-02347-1](#) | 2005-02

1495. Search for lepton-flavor violation at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 44 \(2005\) 463](#) | [arXiv:hep-ex/0501070](#) | [DOI:10.1140/epjc/s2005-02399-1](#) | 2005-01

1496. Search for pentaquarks decaying to Ξ - π in deep inelastic scattering at HERA

Chekanov, S. et al.

[Phys.Lett.B 610 \(2005\) 212](#) | [arXiv:hep-ex/0501069](#) | [DOI:10.1016/j.physletb.2005.02.016](#) | 2005-01

1497. Study of deep inelastic inclusive and diffractive scattering with the ZEUS forward plug calorimeter

Chekanov, S. et al.

[Nucl.Phys.B 713 \(2005\) 3](#) | [arXiv:hep-ex/0501060](#) | [DOI:10.1016/j.nuclphysb.2005.02.001](#) | 2005-01

1498. Forward jet production in deep inelastic ep scattering and low-x parton dynamics at HERA

Chekanov, S. et al.

[Phys.Lett.B 632 \(2006\) 13](#) | [arXiv:hep-ex/0502029](#) | [DOI:10.1016/j.physletb.2005.09.066](#) | 2005-01

1499. Search for a narrow charmed baryonic state decaying to D^{*+} - p in ep collisions at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 38 \(2004\) 29](#) | [arXiv:hep-ex/0409033](#) | [DOI:10.1140/epjc/s2004-02042-9](#) | 2004-09

1500. Dissociation of virtual photons in events with a leading proton at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 38 \(2004\) 43](#) | [arXiv:hep-ex/0408009](#) | [DOI:10.1140/epjc/s2004-02047-4](#) | 2004-08

1501. Diffractive charm and jet production

Gutsche, O. et al.

728 | 2004-08

1502. Substructure dependence of jet cross sections at HERA and determination of $\alpha(s)$
Chekanov, S. et al.
[Nucl.Phys.B 700 \(2004\) 3](#) | [arXiv:hep-ex/0405065](#) | [DOI:10.1016/j.nuclphysb.2004.08.049](#) | 2004-05
1503. Measurement of beauty production in deep inelastic scattering at HERA
Chekanov, S. et al.
[Phys.Lett.B 599 \(2004\) 173](#) | [arXiv:hep-ex/0405069](#) | [DOI:10.1016/j.physletb.2004.08.048](#) | 2004-05
1504. Evidence for a narrow baryonic state decaying to $K_0(S)$ p and $K_0(S)$ anti-p in deep inelastic scattering at HERA
Chekanov, S. et al.
[Phys.Lett.B 591 \(2004\) 7](#) | [arXiv:hep-ex/0403051](#) | [DOI:10.1016/j.physletb.2004.04.024](#) | 2004-03
1505. The Dependence of dijet production on photon virtuality in ep collisions at HERA
Chekanov, S. et al.
[Eur.Phys.J.C 35 \(2004\) 487](#) | [arXiv:hep-ex/0404033](#) | [DOI:10.1140/epjc/s2004-01885-2](#) | 2004-03
1506. Exclusive electroproduction of J/psi mesons at HERA
Chekanov, S. et al.
[Nucl.Phys.B 695 \(2004\) 3](#) | [arXiv:hep-ex/0404008](#) | [DOI:10.1016/j.nuclphysb.2004.06.034](#) | 2004-03
1507. Study of the pion trajectory in the photoproduction of leading neutrons at HERA
Chekanov, S. et al.
[Phys.Lett.B 610 \(2005\) 199](#) | [arXiv:hep-ex/0404002](#) | [DOI:10.1016/j.physletb.2005.01.101](#) | 2004-03
1508. Observation of isolated high $E(T)$ photons in deep inelastic scattering
Chekanov, S. et al.
[Phys.Lett.B 595 \(2004\) 86](#) | [arXiv:hep-ex/0402019](#) | [DOI:10.1016/j.physletb.2004.05.033](#) | 2004-02
1509. Photoproduction of D^{*+-} mesons associated with a leading neutron
Chekanov, S. et al.
[Phys.Lett.B 590 \(2004\) 143](#) | [arXiv:hep-ex/0401017](#) | [DOI:10.1016/j.physletb.2004.03.076](#) | 2004-01
1510. Search for contact interactions, large extra dimensions and finite quark radius in e p collisions at HERA
Chekanov, S. et al.
[Phys.Lett.B 591 \(2004\) 23](#) | [arXiv:hep-ex/0401009](#) | [DOI:10.1016/j.physletb.2004.03.081](#) | 2003-12
1511. Search for QCD instanton induced events in deep inelastic ep scattering at HERA
Chekanov, S. et al.
[Eur.Phys.J.C 34 \(2004\) 255](#) | [arXiv:hep-ex/0312048](#) | [DOI:10.1140/epjc/s2004-01735-3](#) | 2003-12
1512. High Q^2 neutral current cross-sections in $e^+ p$ deep inelastic scattering at $s^{1/2} = 318$ -GeV
Chekanov, S. et al.
[Phys.Rev.D 70 \(2004\) 052001](#) | [arXiv:hep-ex/0401003](#) | [DOI:10.1103/PhysRevD.70.052001](#) | 2003-12
1513. Beauty photoproduction measured using decays into muons in dijet events in e p collisions at $s^{1/2} = 318$ -GeV
Chekanov, S. et al.
[Phys.Rev.D 70 \(2004\) 012008](#) | [arXiv:hep-ex/0312057](#) | [DOI:10.1103/PhysRevD.70.012008](#) | 2003-12
1514. Isolated tau leptons in events with large missing transverse momentum at HERA
Chekanov, S. et al.
[Phys.Lett.B 583 \(2004\) 41](#) | [arXiv:hep-ex/0311028](#) | [DOI:10.1016/j.physletb.2003.12.054](#) | 2003-11
1515. Search for single-top production in e p collisions at HERA. (Addendum)
Chekanov, S. et al.

2003-11

1516. Bose-Einstein correlations in one and two-dimensions in deep inelastic scattering

Chekanov, S. et al.

[Phys.Lett.B 583 \(2004\) 231](#) | [arXiv:hep-ex/0311030](#) | [DOI:10.1016/j.physletb.2003.12.068](#) | 2003-10

1517. Measurement of D^{*+} production in deep inelastic e^+p scattering at HERA

Chekanov, S. et al.

[Phys.Rev.D 69 \(2004\) 012004](#) | [arXiv:hep-ex/0308068](#) | [DOI:10.1103/PhysRevD.69.012004](#) | 2003-08

1518. Observation of $K_0^*(800)$ resonances in deep inelastic scattering at HERA

Chekanov, S. et al.

[Phys.Lett.B 578 \(2004\) 33](#) | [arXiv:hep-ex/0308006](#) | [DOI:10.1016/j.physletb.2003.10.049](#) | 2003-08

1519. Measurement of high Q^2 charged current cross-sections in e^+p deep inelastic scattering at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 32 \(2003\) 1](#) | [arXiv:hep-ex/0307043](#) | [DOI:10.1140/epjc/s2003-01363-5](#) | 2003-07

1520. Measurement of the open charm contribution to the diffractive proton structure function

Chekanov, S. et al.

[Nucl.Phys.B 672 \(2003\) 3](#) | [arXiv:hep-ex/0307068](#) | [DOI:10.1016/j.nuclphysb.2003.09.001](#) | 2003-07

1521. Jet production in charged current deep inelastic e^+p scattering at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 31 \(2003\) 149](#) | [arXiv:hep-ex/0306018](#) | [DOI:10.1140/epjc/s2003-01358-2](#) | 2003-06

1522. Measurement of deeply virtual Compton scattering at HERA

Chekanov, S. et al.

[Phys.Lett.B 573 \(2003\) 46](#) | [arXiv:hep-ex/0305028](#) | [DOI:10.1016/j.physletb.2003.08.048](#) | 2003-04

1523. A Search for resonance decays to lepton + jet at HERA and limits on leptoquarks

Chekanov, S. et al.

[Phys.Rev.D 68 \(2003\) 052004](#) | [arXiv:hep-ex/0304008](#) | [DOI:10.1103/PhysRevD.68.052004](#) | 2003-03

1524. Dijet angular distributions in photoproduction of charm at HERA

Chekanov, S. et al.

[Phys.Lett.B 565 \(2003\) 87](#) | [arXiv:hep-ex/0302025](#) | [DOI:10.1016/S0370-2693\(03\)00752-4](#) | 2003-02

1525. Search for single top production in ep collisions at HERA

Chekanov, S. et al.

[Phys.Lett.B 559 \(2003\) 153](#) | [arXiv:hep-ex/0302010](#) | [DOI:10.1016/S0370-2693\(03\)00333-2](#) | 2003-02

1526. Measurement of subjet multiplicities in neutral current deep inelastic scattering at HERA and determination of $\alpha(s)$

Chekanov, S. et al.

[Phys.Lett.B 558 \(2003\) 41](#) | [arXiv:hep-ex/0212030](#) | [DOI:10.1016/S0370-2693\(03\)00216-8](#) | 2002-12

1527. Scaling violations and determination of $\alpha(s)$ from jet production in γp interactions at HERA

Chekanov, S. et al.

[Phys.Lett.B 560 \(2003\) 7](#) | [arXiv:hep-ex/0212064](#) | [DOI:10.1016/S0370-2693\(03\)00380-0](#) | 2002-12

1528. Observation of the strange sea in the proton via inclusive ϕ meson production in neutral current deep inelastic scattering at HERA

Chekanov, S. et al.

[Phys.Lett.B 553 \(2003\) 141](#) | [arXiv:hep-ex/0211025](#) | [DOI:10.1016/S0370-2693\(02\)03206-9](#) | 2002-11

1529. Measurement of event shapes in deep inelastic scattering at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 27 \(2003\) 531](#) | [arXiv:hep-ex/0211040](#) | [DOI:10.1140/epjc/s2003-01148-x](#) | 2002-11

1530. Study of the azimuthal asymmetry of jets in neutral current deep inelastic scattering at HERA

Chekanov, S. et al.

[Phys.Lett.B 551 \(2003\) 226](#) | [arXiv:hep-ex/0210064](#) | [DOI:10.1016/S0370-2693\(02\)03072-1](#) | 2002-10

1531. Measurements of inelastic J / psi and psi-prime photoproduction at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 27 \(2003\) 173](#) | [arXiv:hep-ex/0211011](#) | [DOI:10.1140/epjc/s2002-01130-2](#) | 2002-10

1532. Leading proton production in e+ p collisions at HERA

Chekanov, S. et al.

[Nucl.Phys.B 658 \(2003\) 3](#) | [arXiv:hep-ex/0210029](#) | [DOI:10.1016/S0550-3213\(03\)00152-4](#) | 2002-09

1533. Inclusive jet cross-sections in the Breit frame in neutral current deep inelastic scattering at HERA and determination of alpha(s)

Chekanov, S. et al.

[Phys.Lett.B 547 \(2002\) 164](#) | [arXiv:hep-ex/0208037](#) | [DOI:10.1016/S0370-2693\(02\)02763-6](#) | 2002-08

1534. Measurement of high Q^2 e- p neutral current cross-sections at HERA and the extraction of xF(3)

Chekanov, S. et al.

[Eur.Phys.J.C 28 \(2003\) 175](#) | [arXiv:hep-ex/0208040](#) | [DOI:10.1140/epjc/s2003-01163-y](#) | 2002-08

1535. A ZEUS next-to-leading-order QCD analysis of data on deep inelastic scattering

Chekanov, S. et al.

[Phys.Rev.D 67 \(2003\) 012007](#) | [arXiv:hep-ex/0208023](#) | [DOI:10.1103/PhysRevD.67.012007](#) | 2002-08

1536. Measurement of diffractive production of D^{*+} -(2010) mesons in deep inelastic scattering at HERA

Chekanov, S. et al.

[Phys.Lett.B 545 \(2002\) 244](#) | [arXiv:hep-ex/0206020](#) | [DOI:10.1016/S0370-2693\(02\)02595-9](#) | 2002-06

1537. Leading neutron production in e+ p collisions at HERA

Chekanov, S. et al.

[Nucl.Phys.B 637 \(2002\) 3](#) | [arXiv:hep-ex/0205076](#) | [DOI:10.1016/S0550-3213\(02\)00439-X](#) | 2002-05

1538. Measurement of high Q^2 charged current cross-sections in e^+p deep inelastic scattering at HERA

Chekanov, S. et al.

[Phys.Lett.B 539 \(2002\) 197](#) | [arXiv:hep-ex/0205091](#) | [DOI:10.1016/S0370-2693\(02\)02093-2](#) | 2002-05

1539. Measurement of proton dissociative diffractive photoproduction of vector mesons at large momentum transfer at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 26 \(2003\) 389](#) | [arXiv:hep-ex/0205081](#) | [DOI:10.1140/epjc/s2002-01079-0](#) | 2002-05

1540. Measurement of the Q^2 and energy dependence of diffractive interactions at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 25 \(2002\) 169](#) | [arXiv:hep-ex/0203039](#) | [DOI:10.1007/s10052-002-1003-1](#) | 2002-03

1541. MONOLITH: A massive magnetized iron detector for neutrino oscillation studies

Agafonova, N.Y. et al.

2000-08